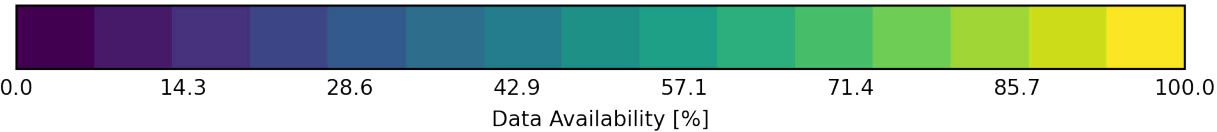


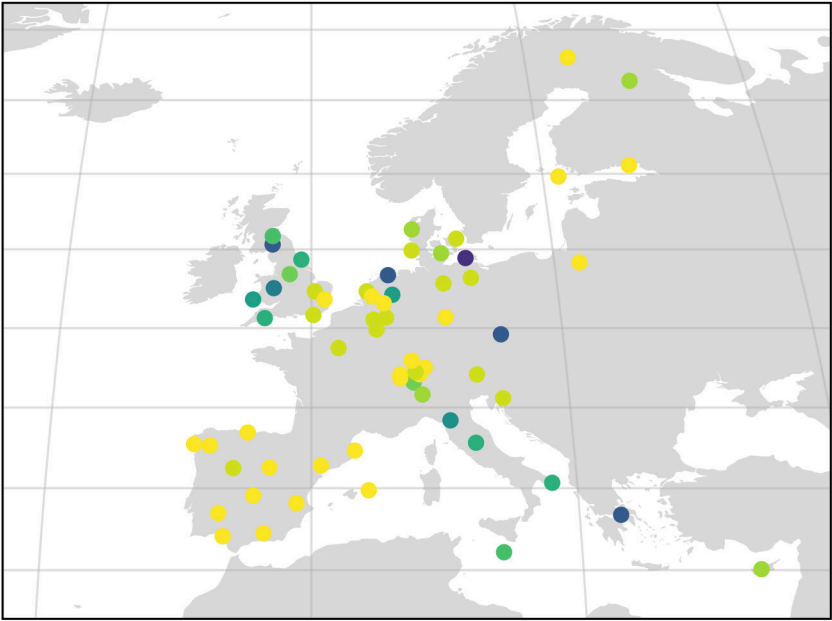
EVALUATION REPORT

Network: EBAS
Species: O3, NO2
Temporal resolution: Hourly
Date range: 01/01/2018 - 01/04/2018
Models: MONARCH
Subsections: LOCAL·All, LOCAL·France, LOCAL·Germany, LOCAL
·Spain, LOCAL·UK

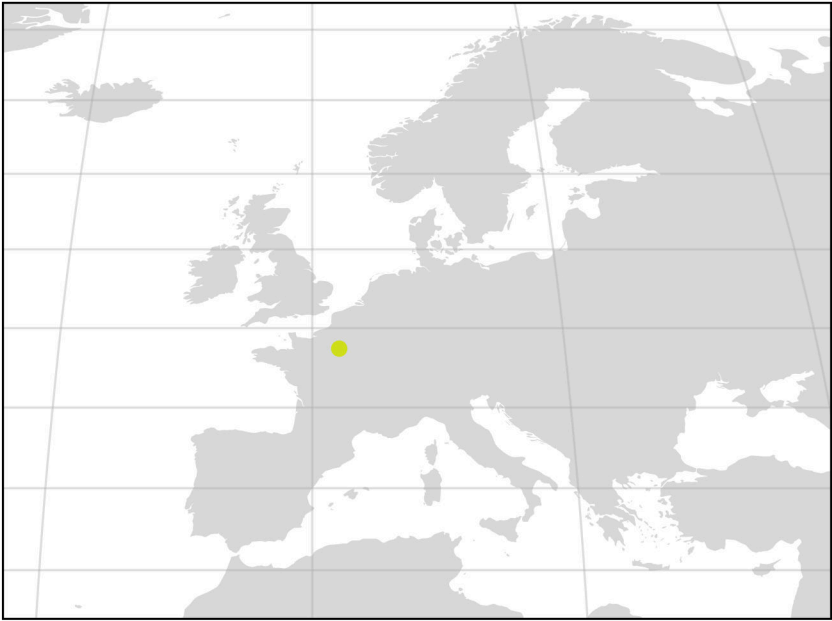
Map Data Availability (Summary)
EBAS|sconco3



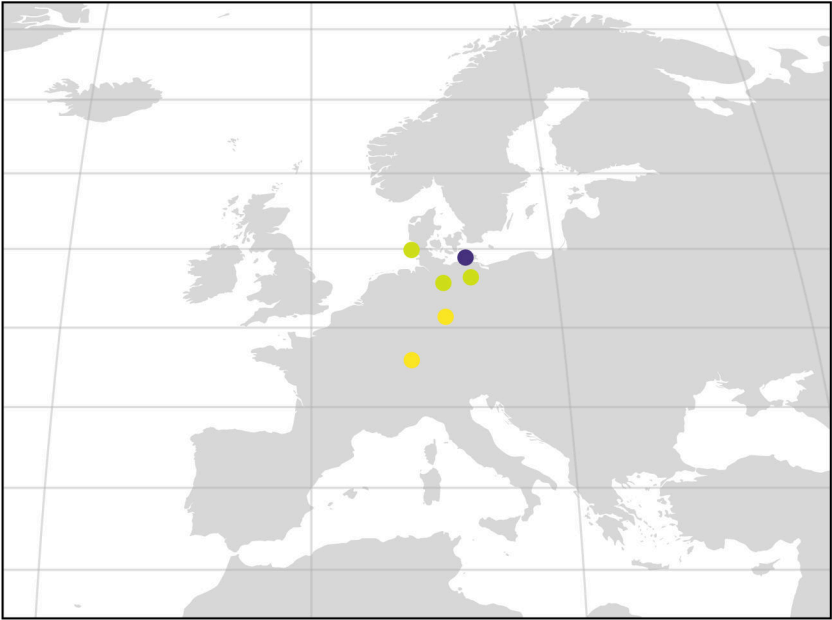
observations
LOCAL·All (62 stations)



observations
LOCAL·France (1 station)



observations
LOCAL·Germany (6 stations)

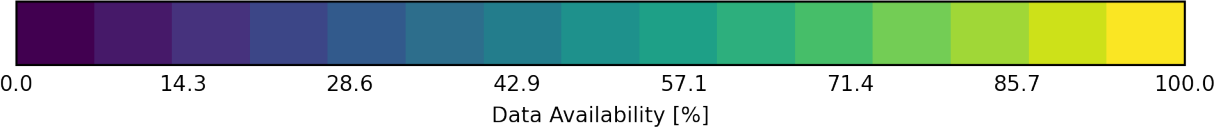


observations
LOCAL·Spain (13 stations)

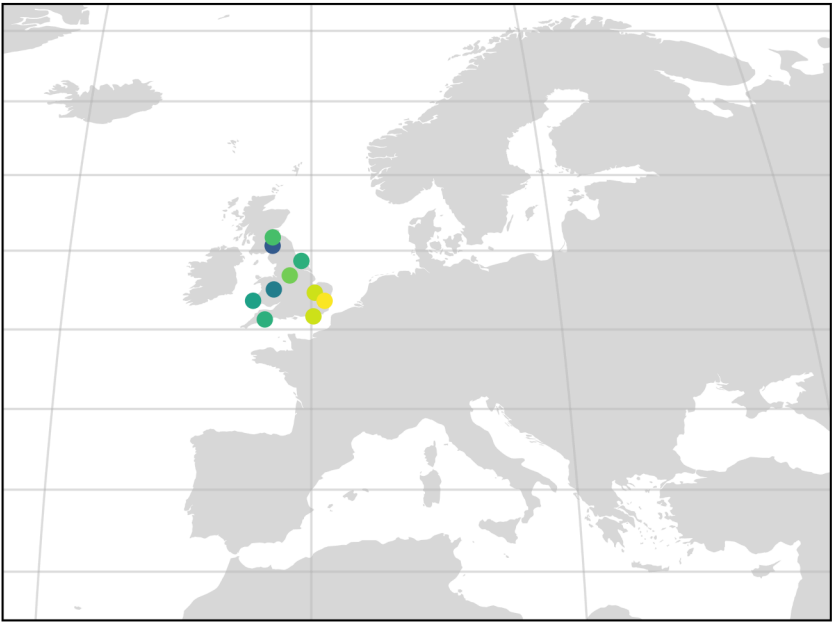


Map Data Availability (Summary)

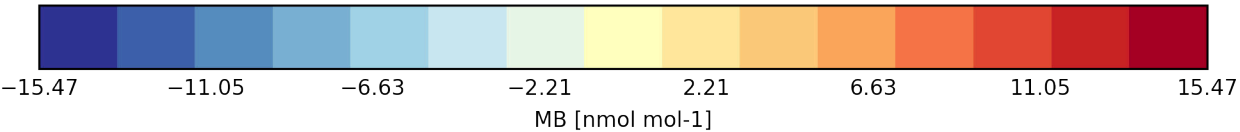
EBAS|sconco3



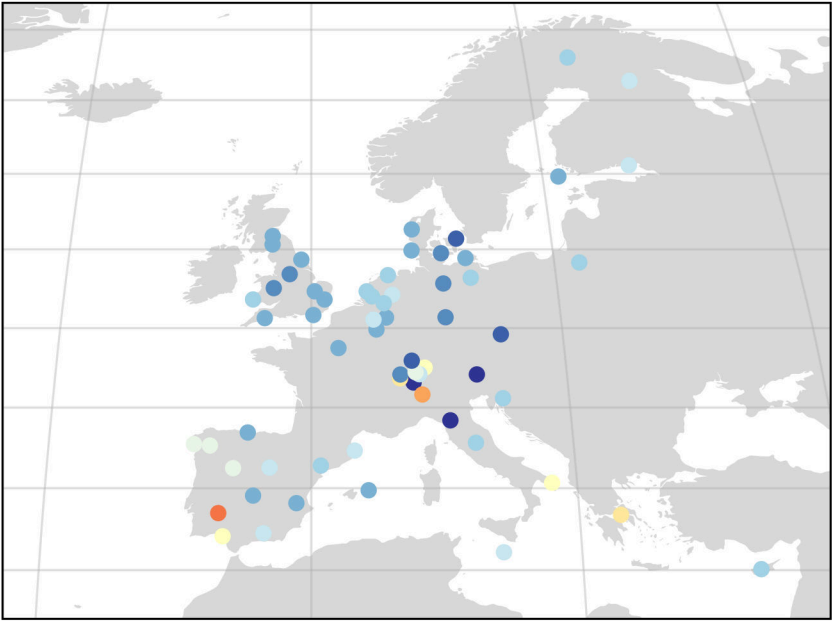
observations
LOCAL-UK (10 stations)



Map MB (Summary)
EBAS|sconco3



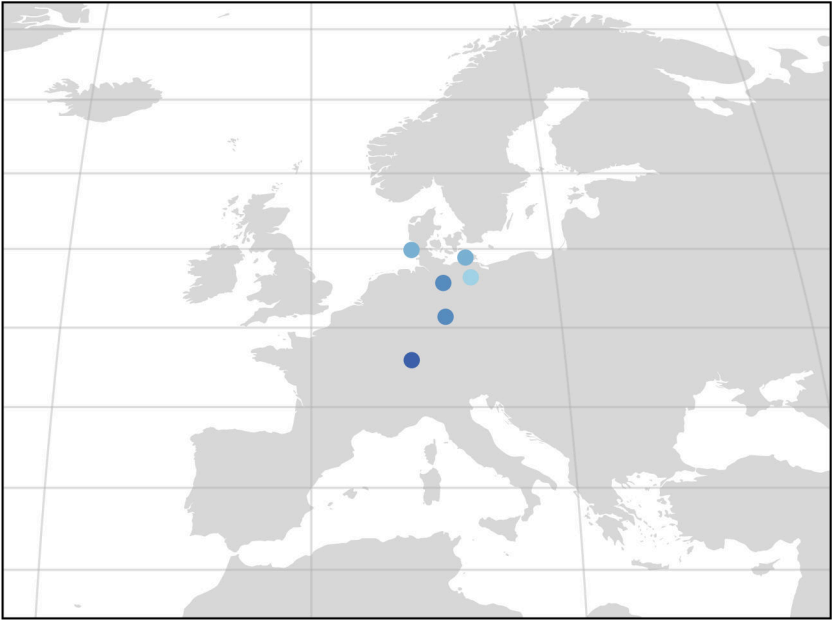
MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



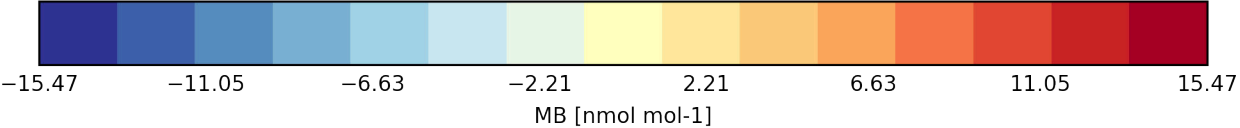
MONARCH
LOCAL·Germany (6 stations)



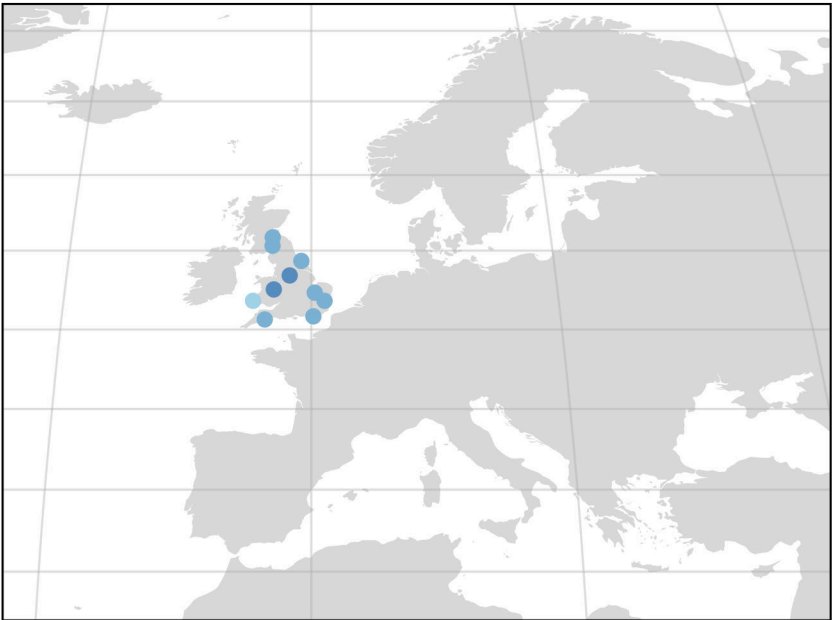
MONARCH
LOCAL·Spain (13 stations)



Map MB (Summary)
EBAS|sconco3

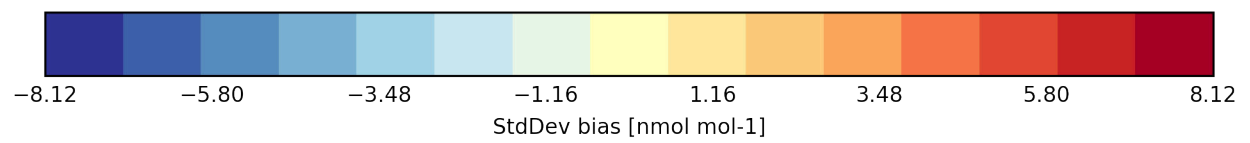


MONARCH
LOCAL·UK (10 stations)

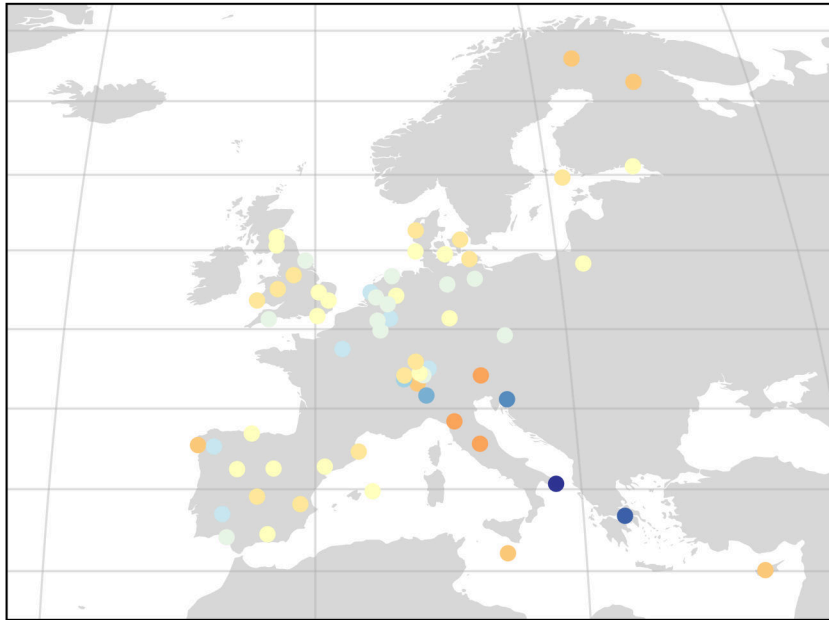


Map StdDev bias (Summary)

EBAS|sconco3



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



MONARCH
LOCAL·Germany (6 stations)

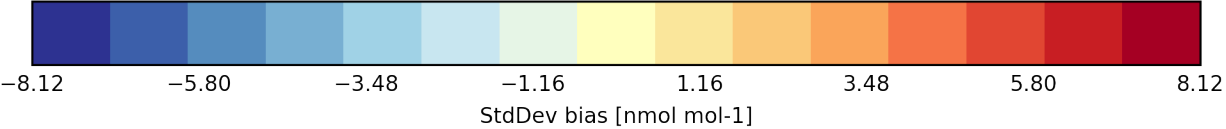


MONARCH
LOCAL·Spain (13 stations)

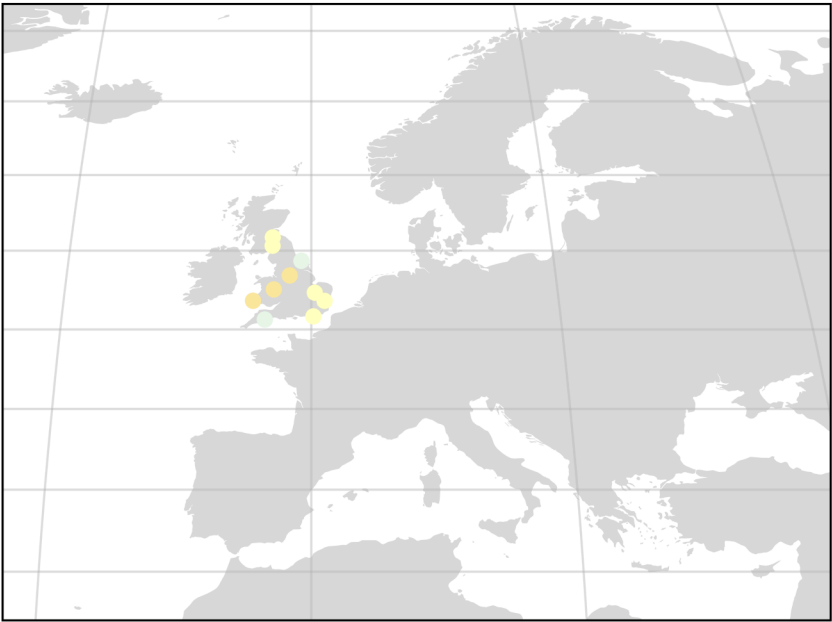


Map StdDev bias (Summary)

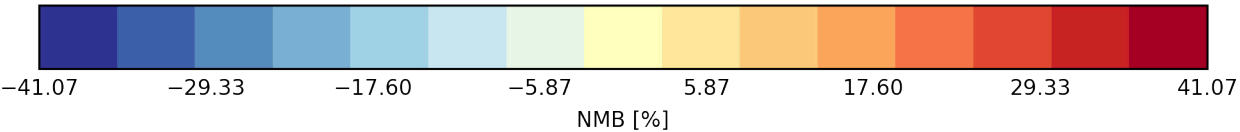
EBAS|sconco3



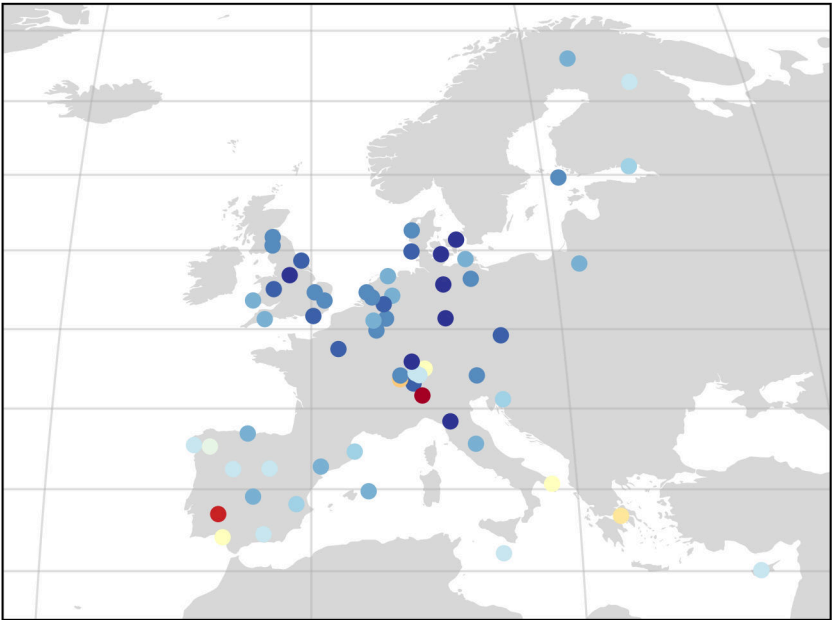
MONARCH
LOCAL·UK (10 stations)



Map NMB (Summary)
EBAS|sconco3



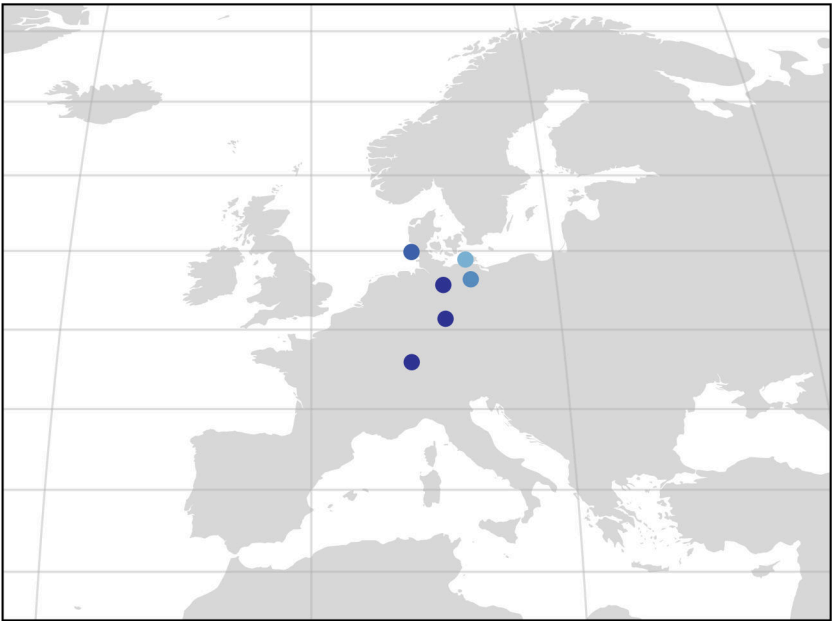
MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



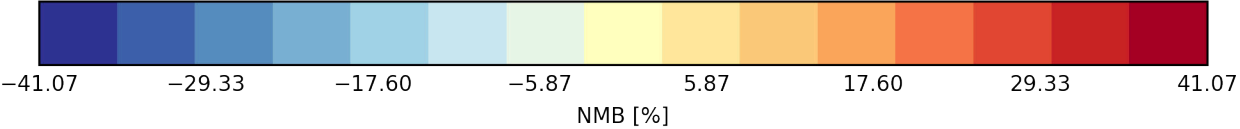
MONARCH
LOCAL·Germany (6 stations)



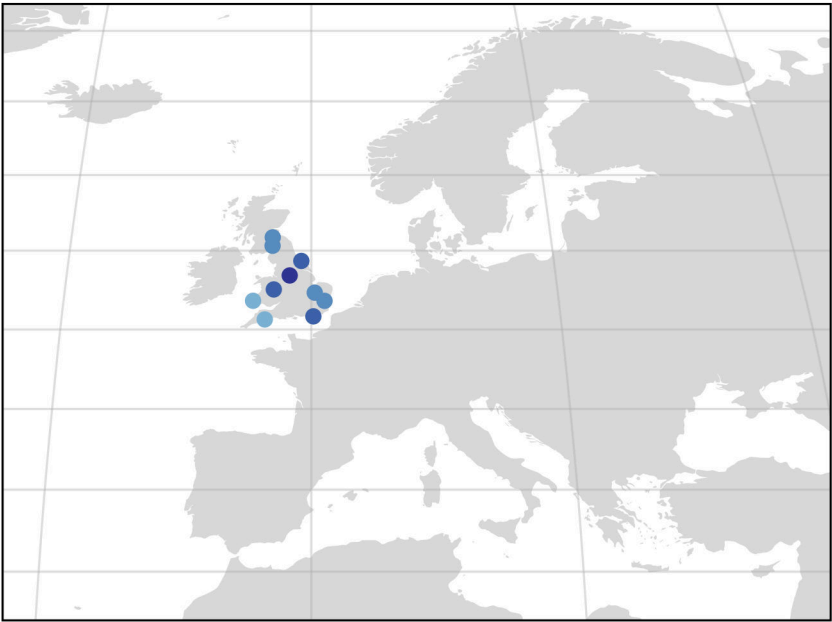
MONARCH
LOCAL·Spain (13 stations)



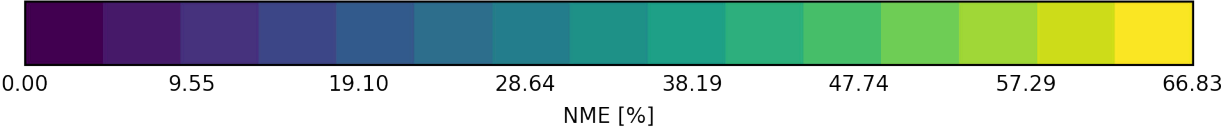
Map NMB (Summary)
EBAS|sconco3



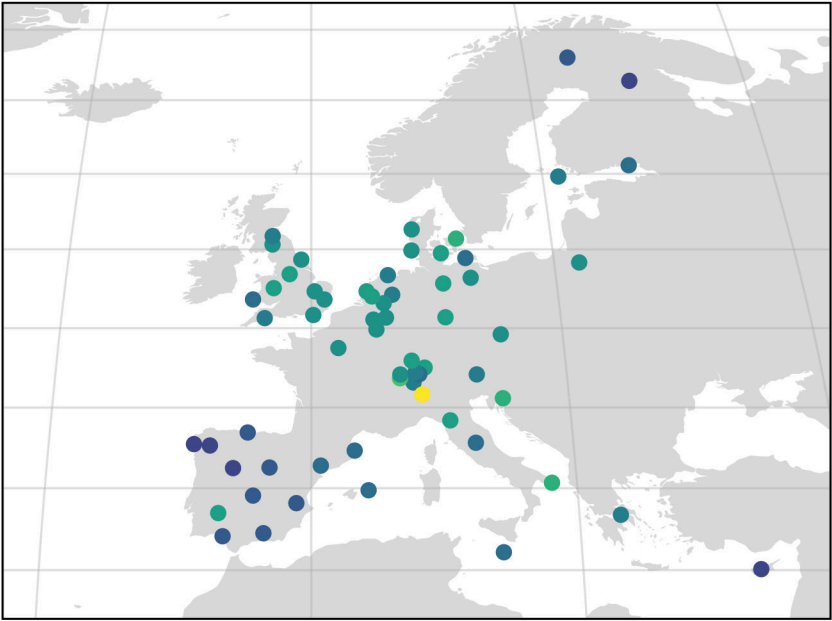
MONARCH
LOCAL·UK (10 stations)



Map NME (Summary)
EBAS|sconco3



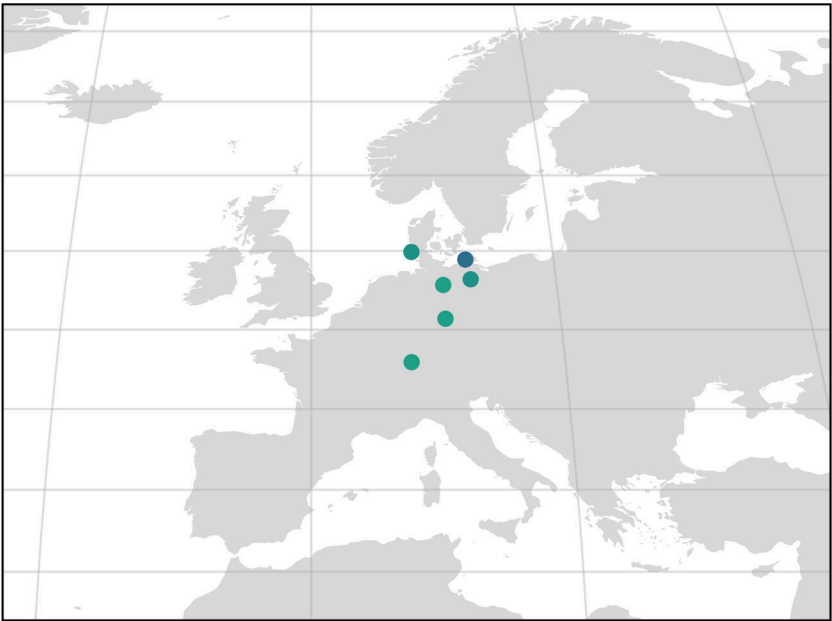
MONARCH
LOCAL·All (62 stations)



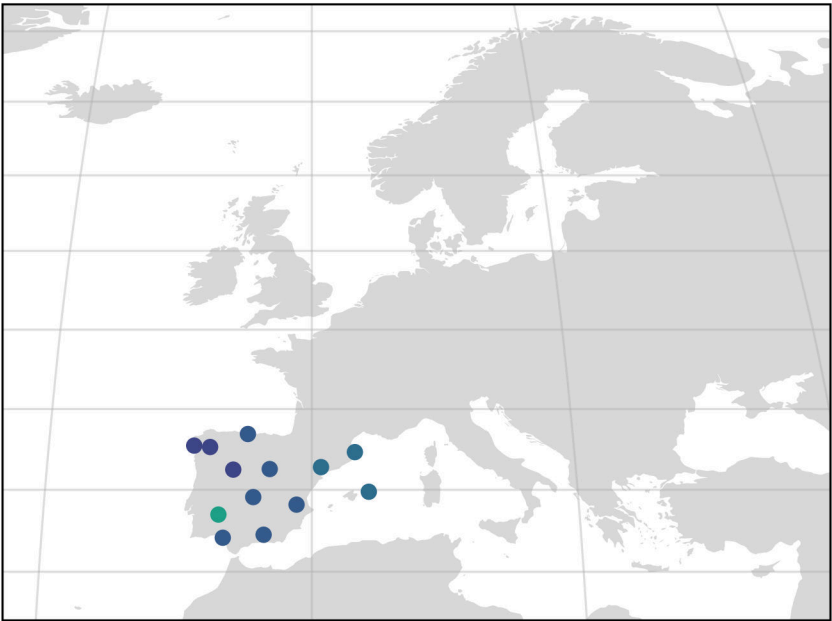
MONARCH
LOCAL·France (1 station)



MONARCH
LOCAL·Germany (6 stations)

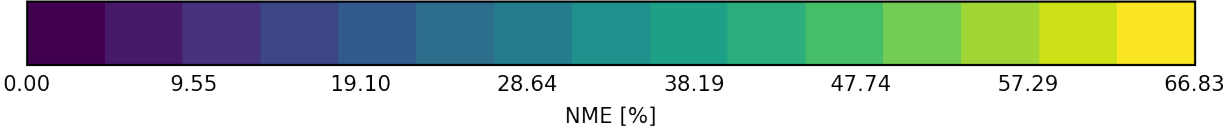


MONARCH
LOCAL·Spain (13 stations)

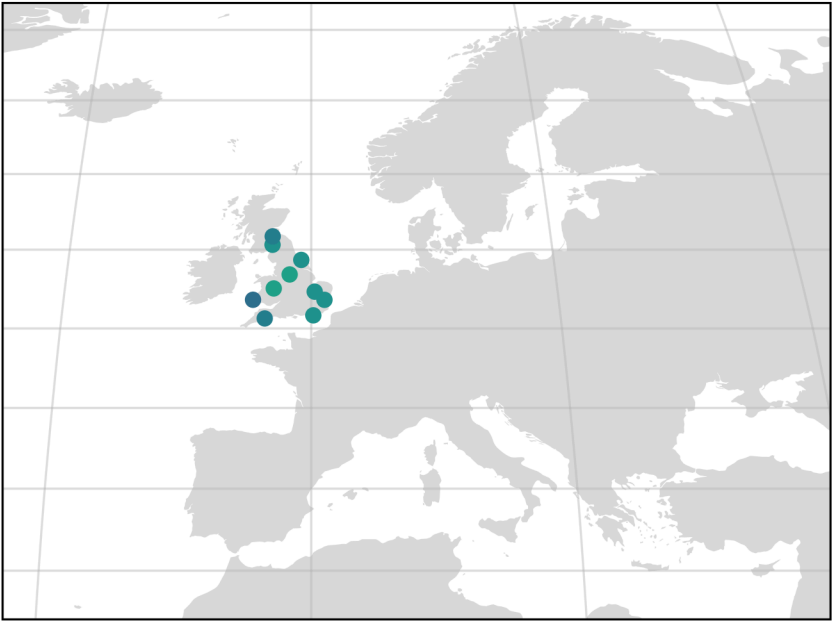


Map NME (Summary)

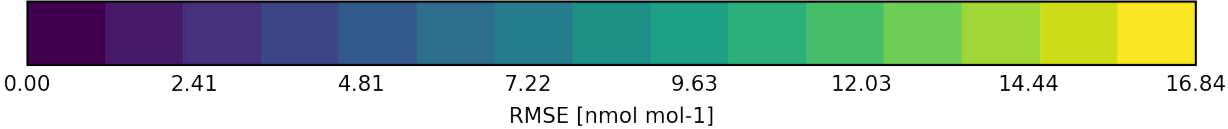
EBAS|sconco3



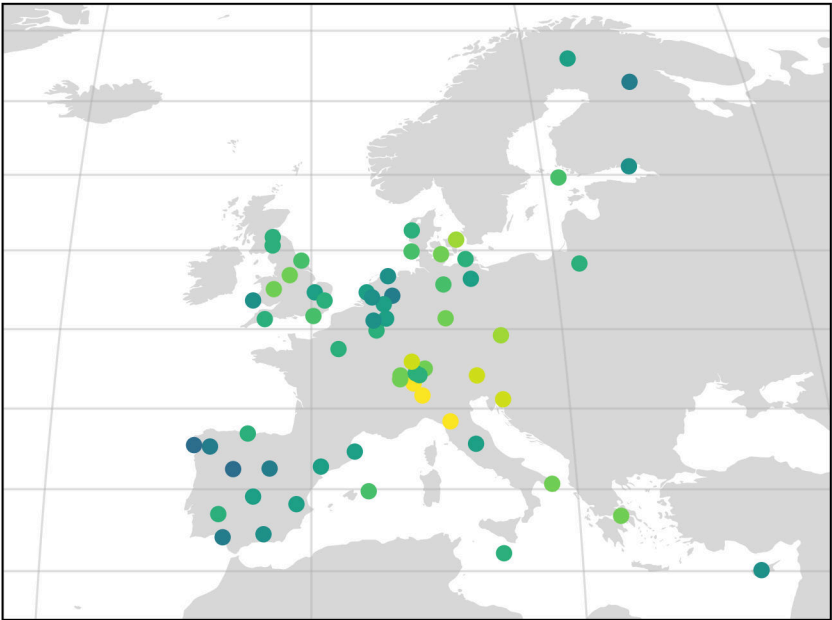
MONARCH
LOCAL·UK (10 stations)



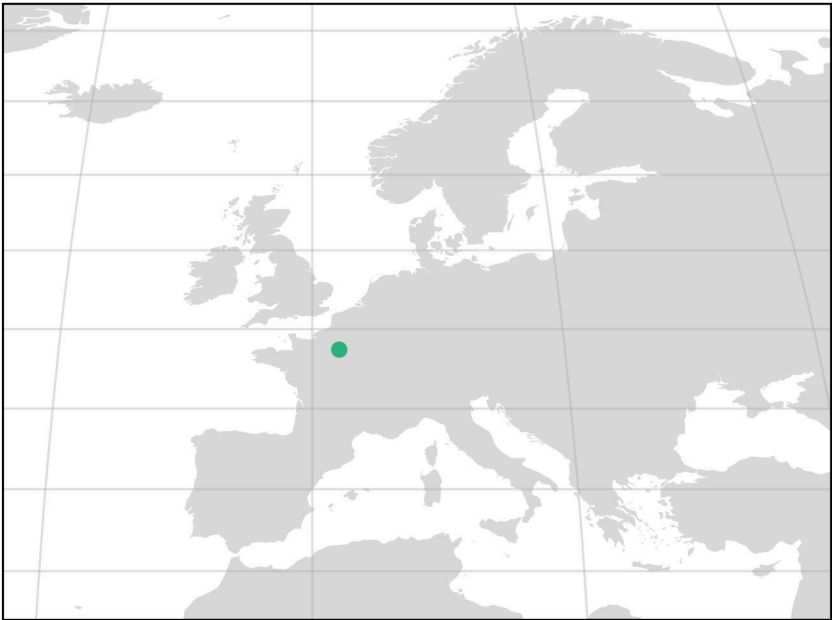
Map RMSE (Summary)
EBAS|sconco3



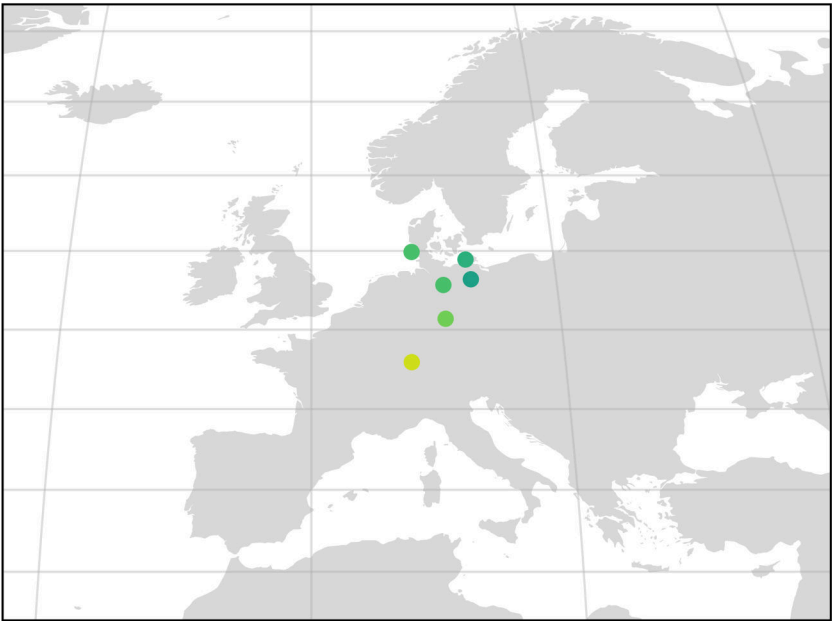
MONARCH
LOCAL·All (62 stations)



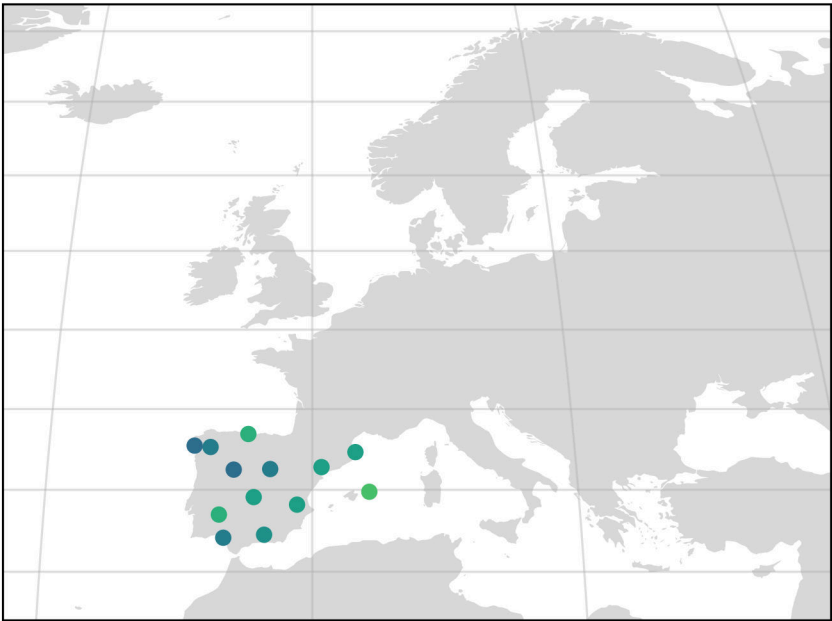
MONARCH
LOCAL·France (1 station)



MONARCH
LOCAL·Germany (6 stations)

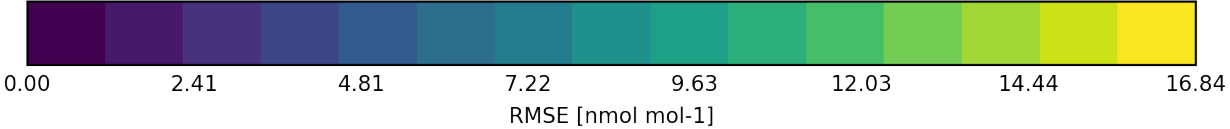


MONARCH
LOCAL·Spain (13 stations)

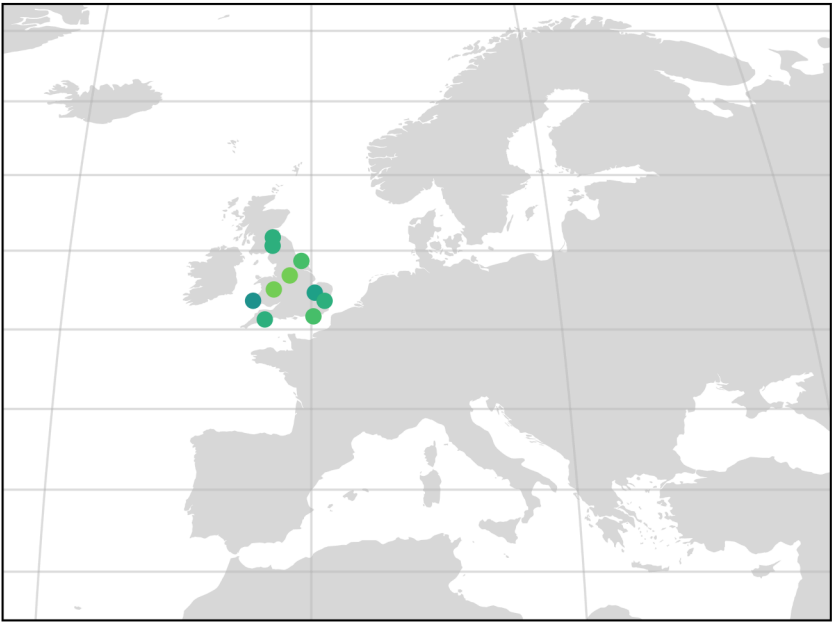


Map RMSE (Summary)

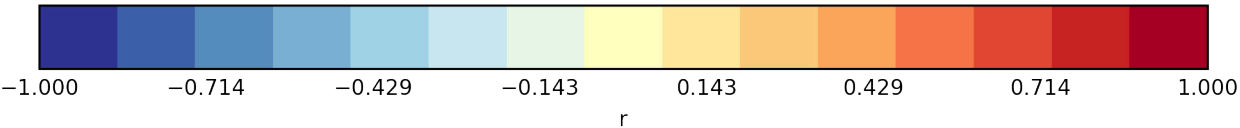
EBAS|sconco3



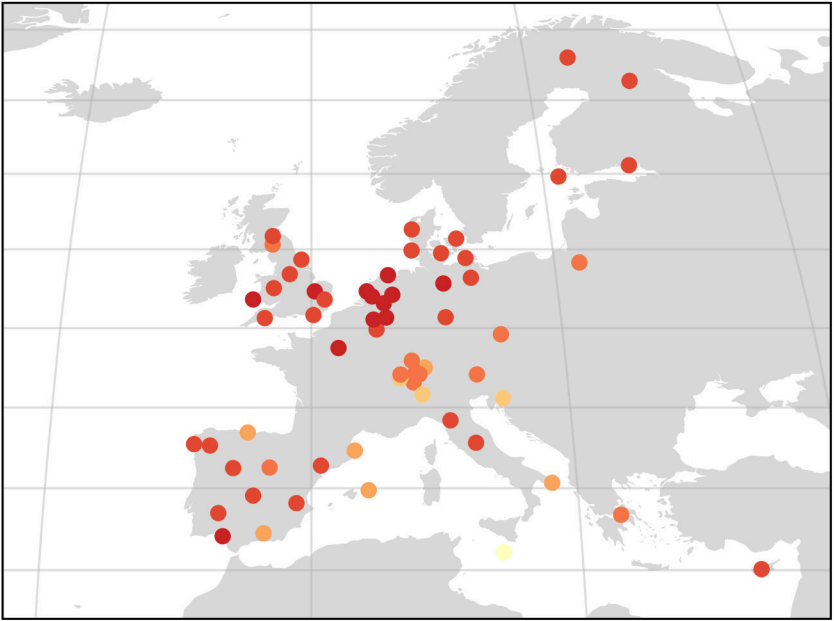
MONARCH
LOCAL-UK (10 stations)



Map r (Summary)
EBAS|sconco3



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



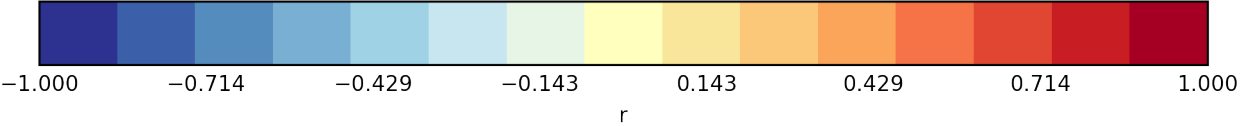
MONARCH
LOCAL·Germany (6 stations)



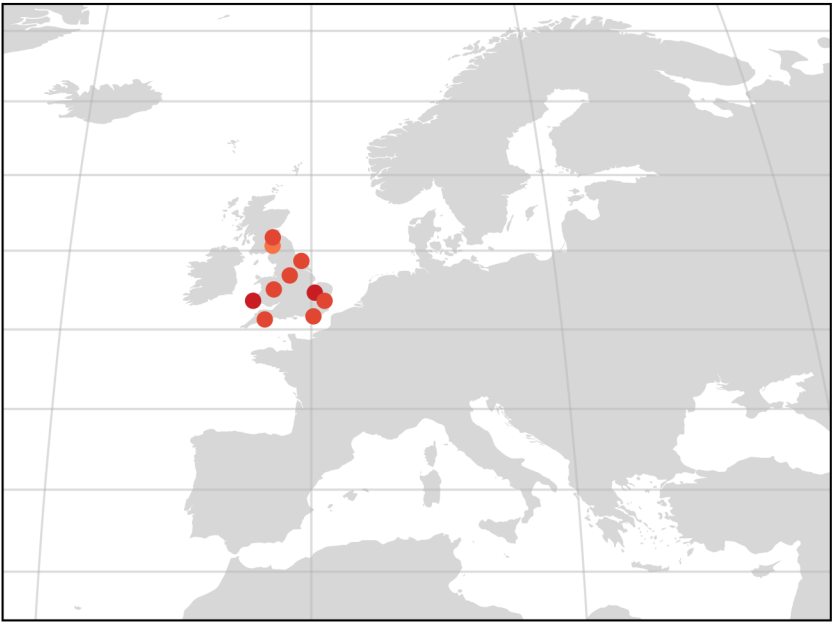
MONARCH
LOCAL·Spain (13 stations)

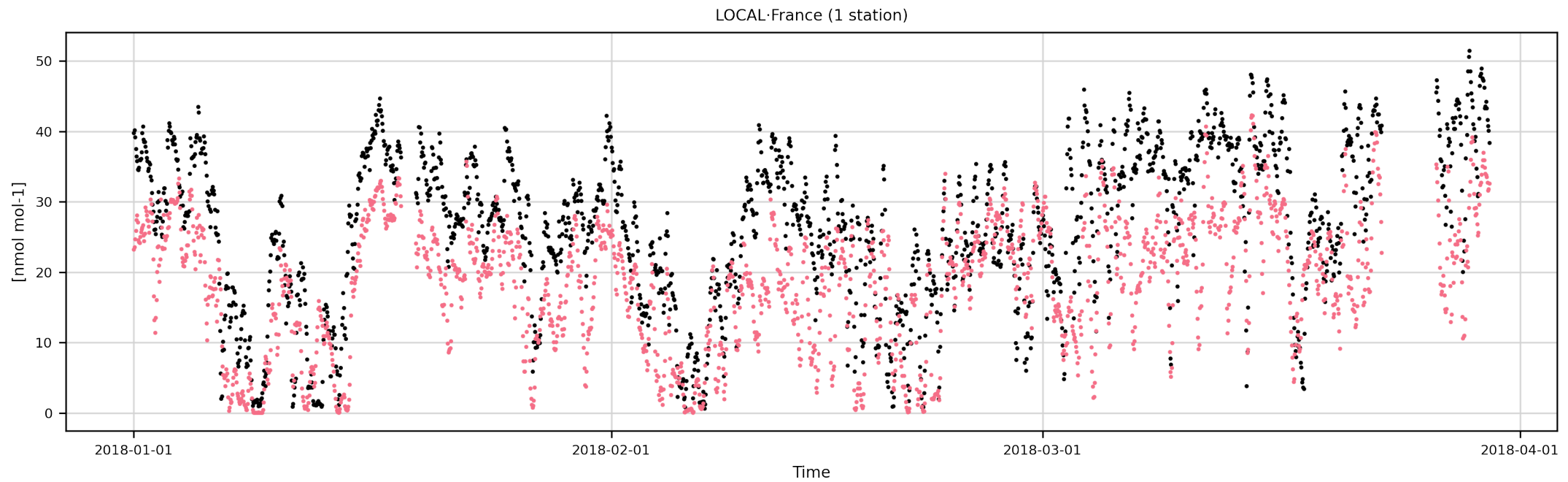
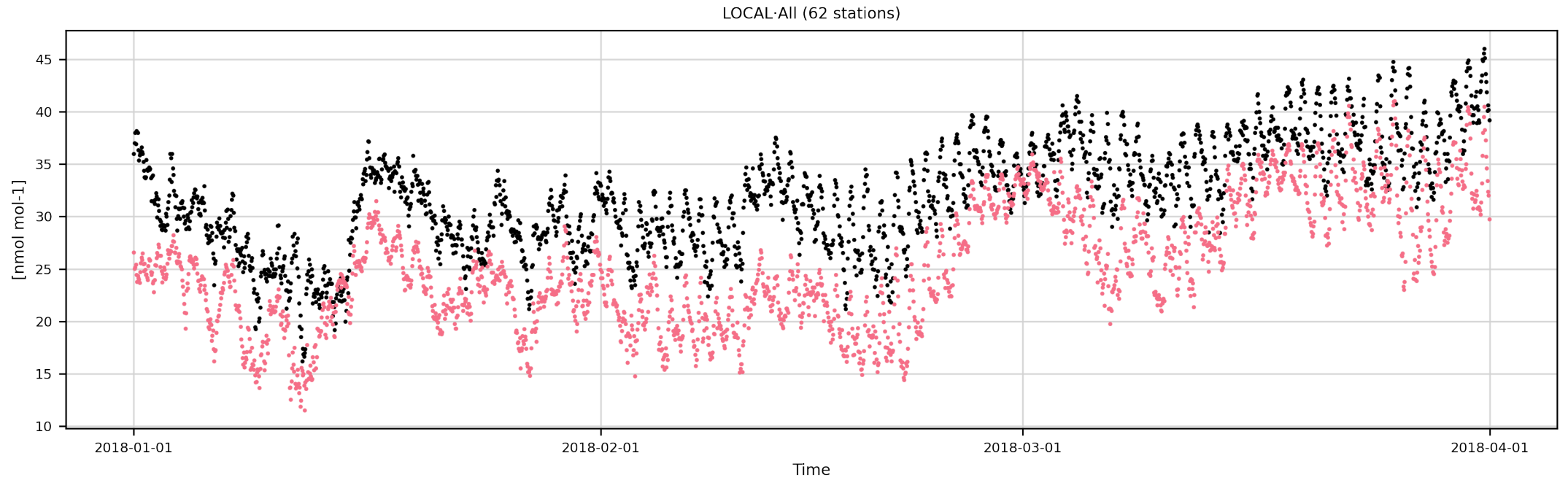


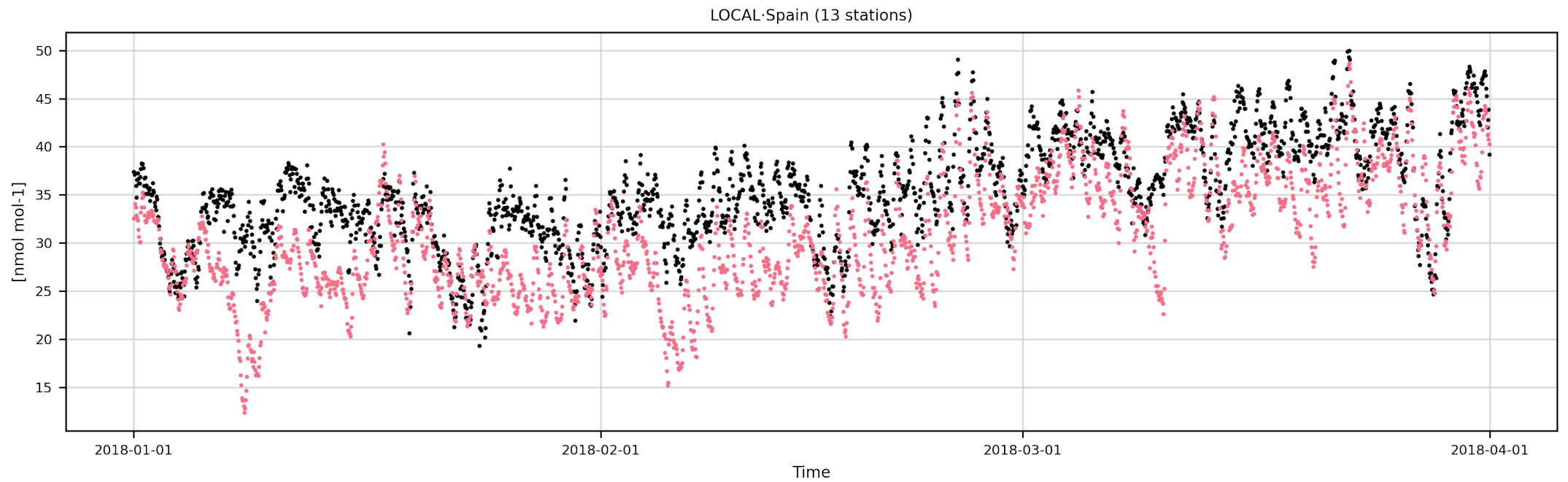
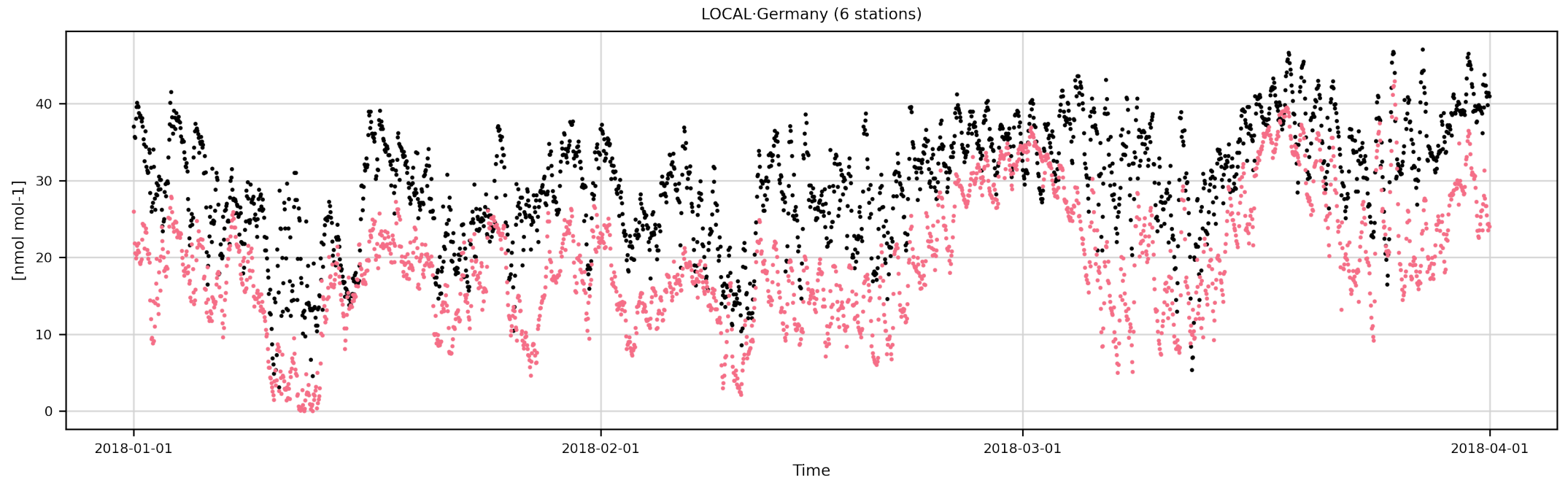
Map r (Summary)
EBAS|sconco3

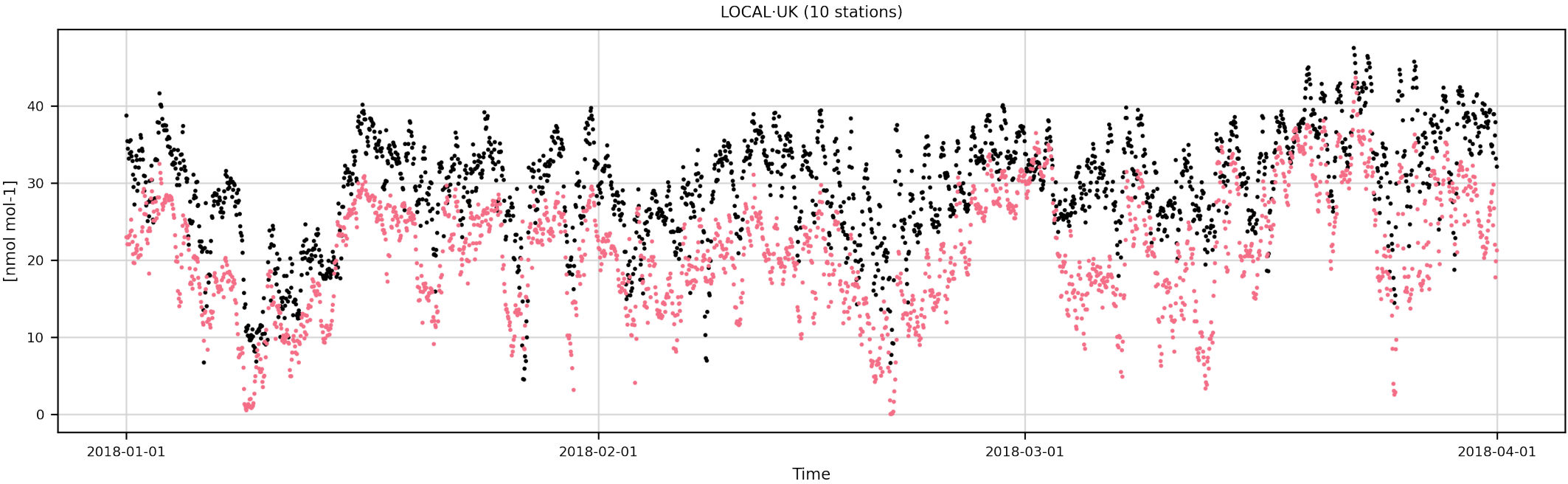


MONARCH
LOCAL·UK (10 stations)





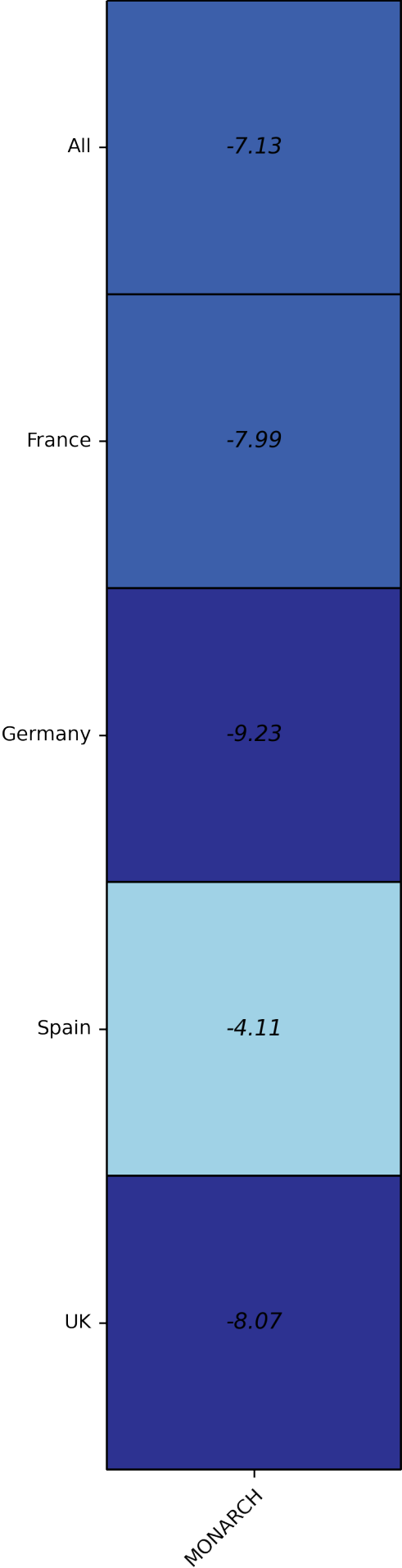
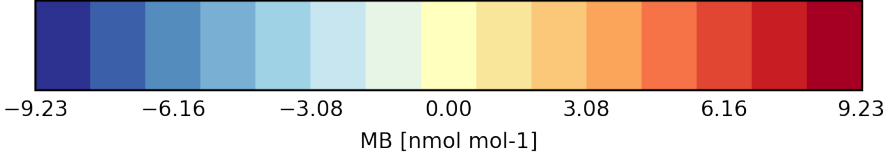




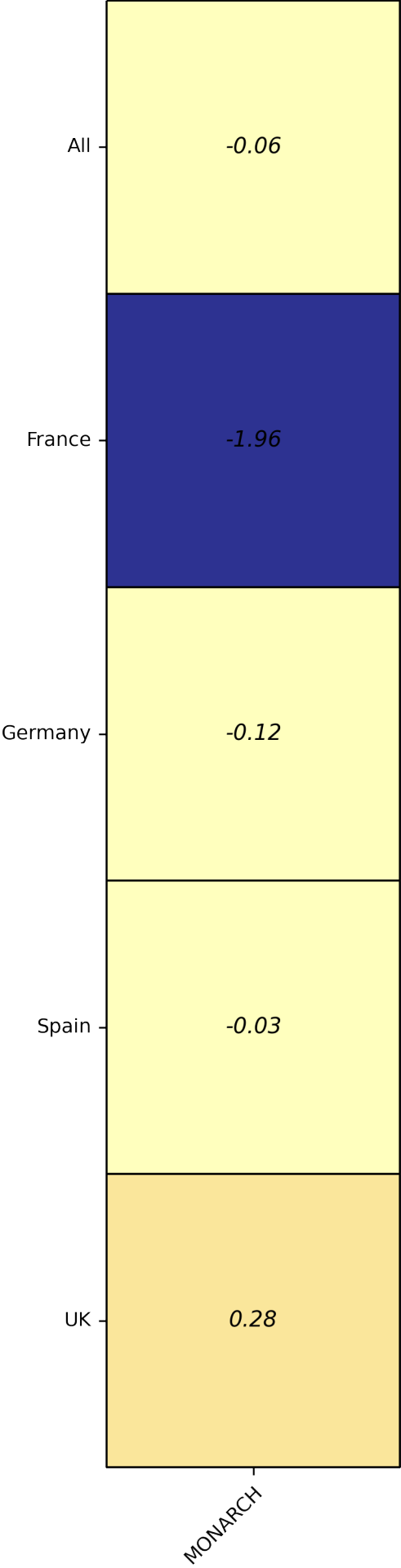
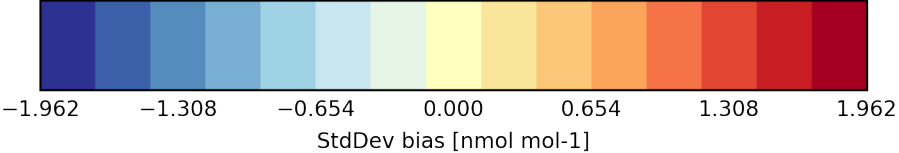
		Mean	StdDev	p5	Median	p95
All	observations	30.15	8.78	13.67	31.07	42.90
	MONARCH	25.25	8.84	9.70	25.48	38.21
France	observations	26.37	11.18	4.67	27.52	41.94
	MONARCH	18.37	9.22	1.79	19.54	31.94
Germany	observations	29.61	9.02	12.20	30.65	42.80
	MONARCH	19.55	8.86	4.86	19.65	35.43
Spain	observations	34.07	7.06	21.57	34.40	46.15
	MONARCH	31.00	7.66	19.42	30.79	43.19
UK	observations	29.19	8.52	13.80	30.34	41.01
	MONARCH	19.72	8.93	5.41	19.85	34.37

		Mean_bias	StdDev_bias	NMB	NME	RMSE	r
All	MONARCH	-7.13	-0.06	-23.58	31.50	10.54	0.66
France	MONARCH	-7.99	-1.96	-30.32	34.19	10.73	0.77
Germany	MONARCH	-9.23	-0.12	-33.59	36.01	11.80	0.72
Spain	MONARCH	-4.11	-0.03	-10.61	20.41	9.14	0.65
UK	MONARCH	-8.07	0.28	-29.58	32.45	10.63	0.70

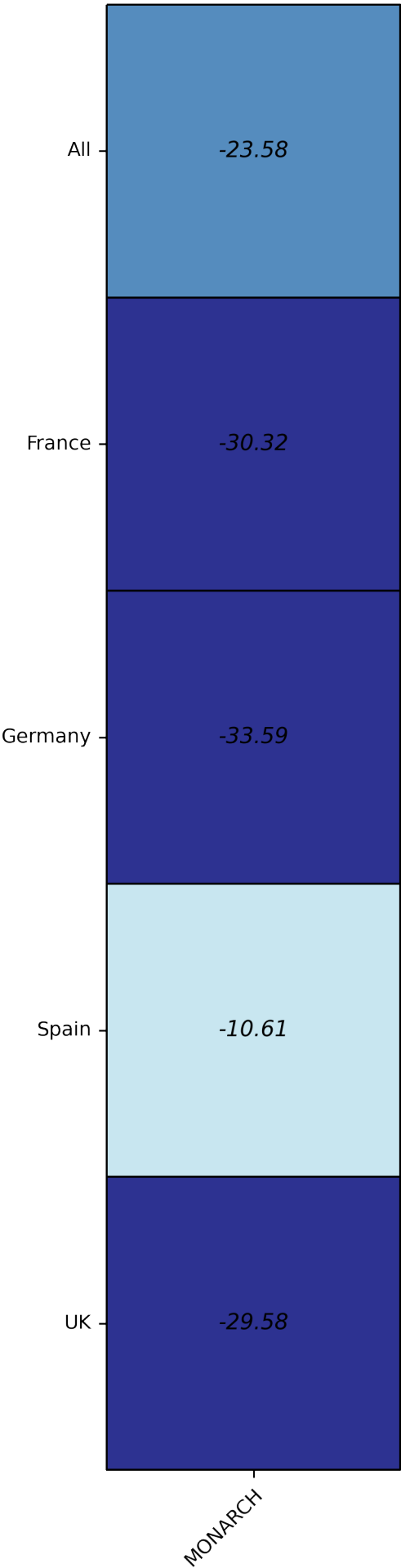
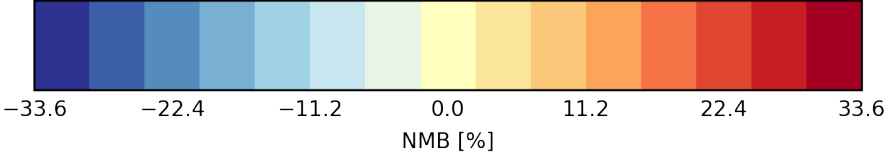
Heatmap MB (Summary)
EBAS|sconco3

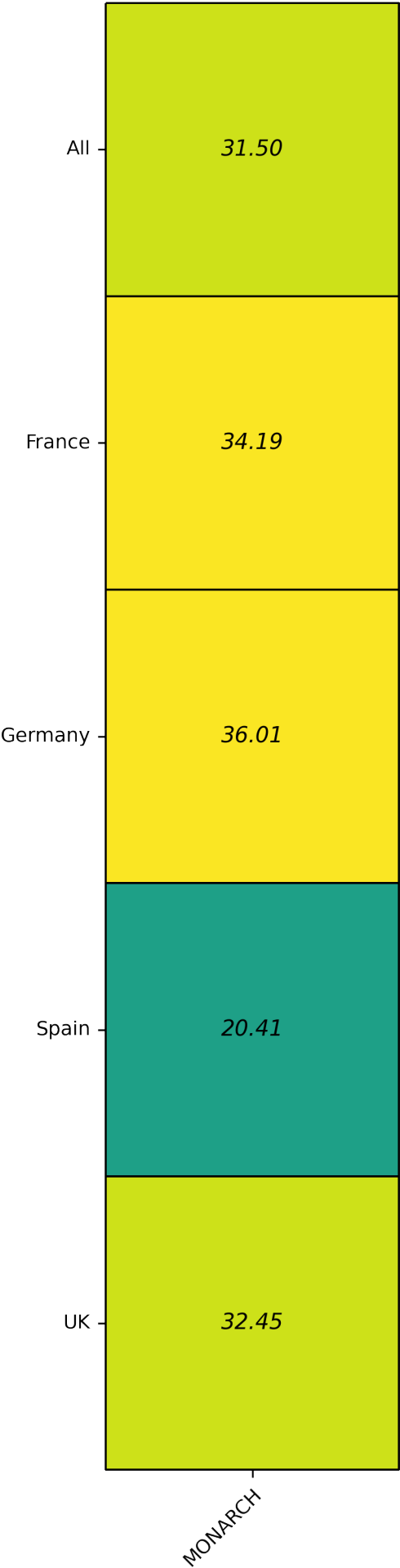
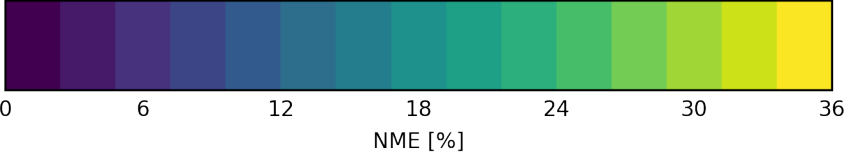


Heatmap StdDev bias (Summary)
EBAS|sconco3

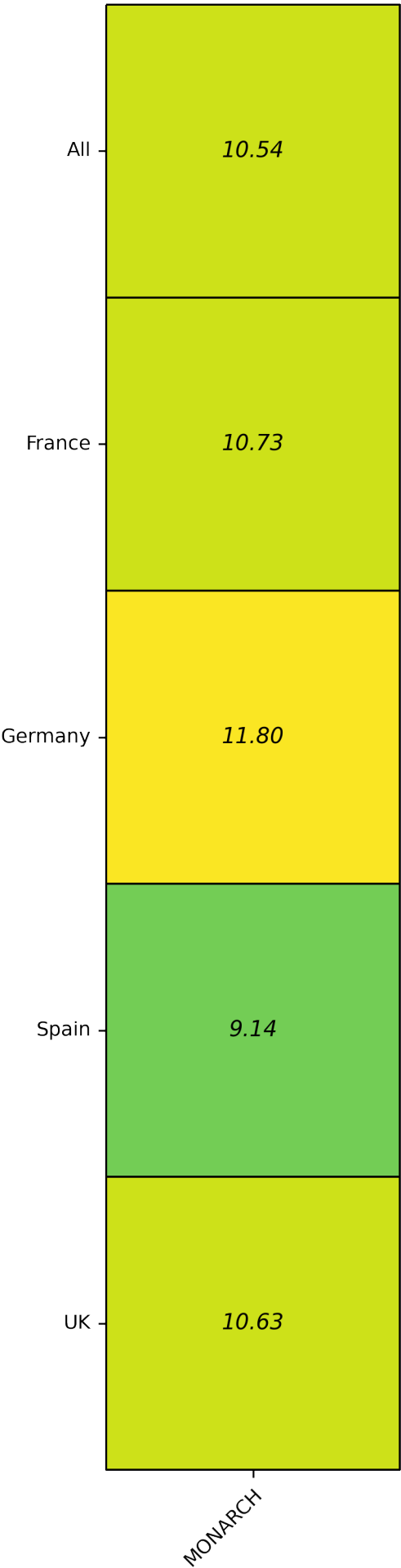
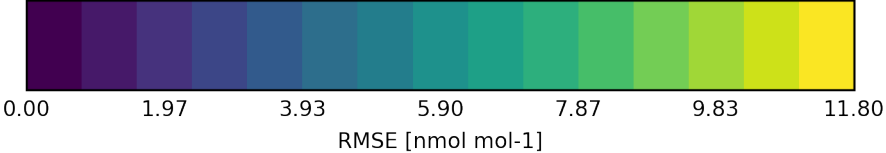


Heatmap NMB (Summary)
EBAS|sconco3

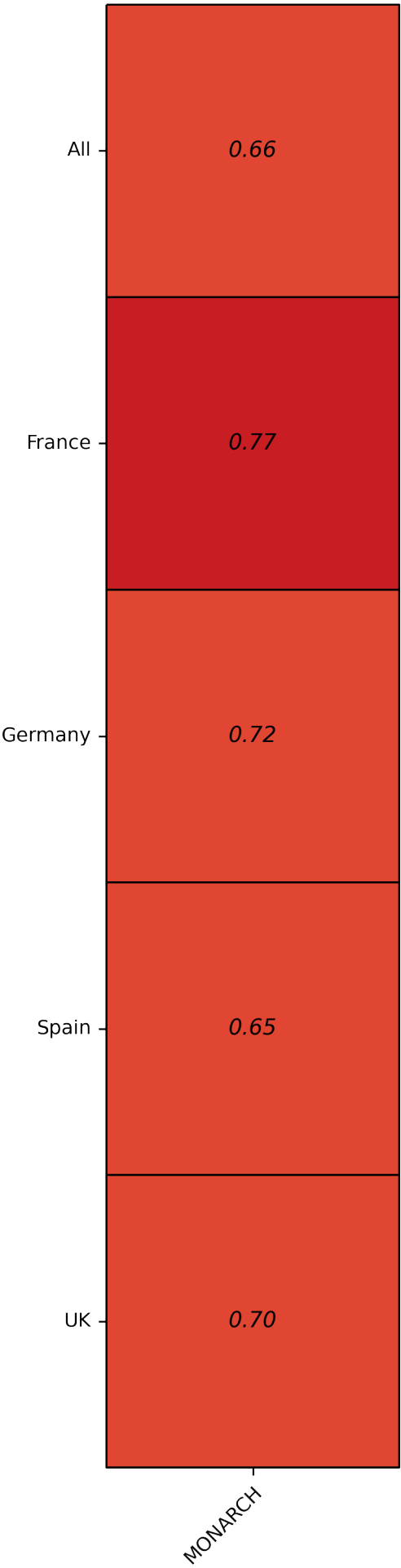
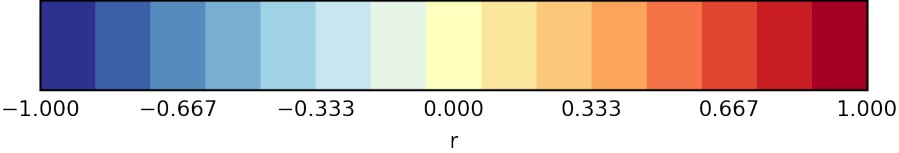




Heatmap RMSE (Summary)
EBAS|sconco3



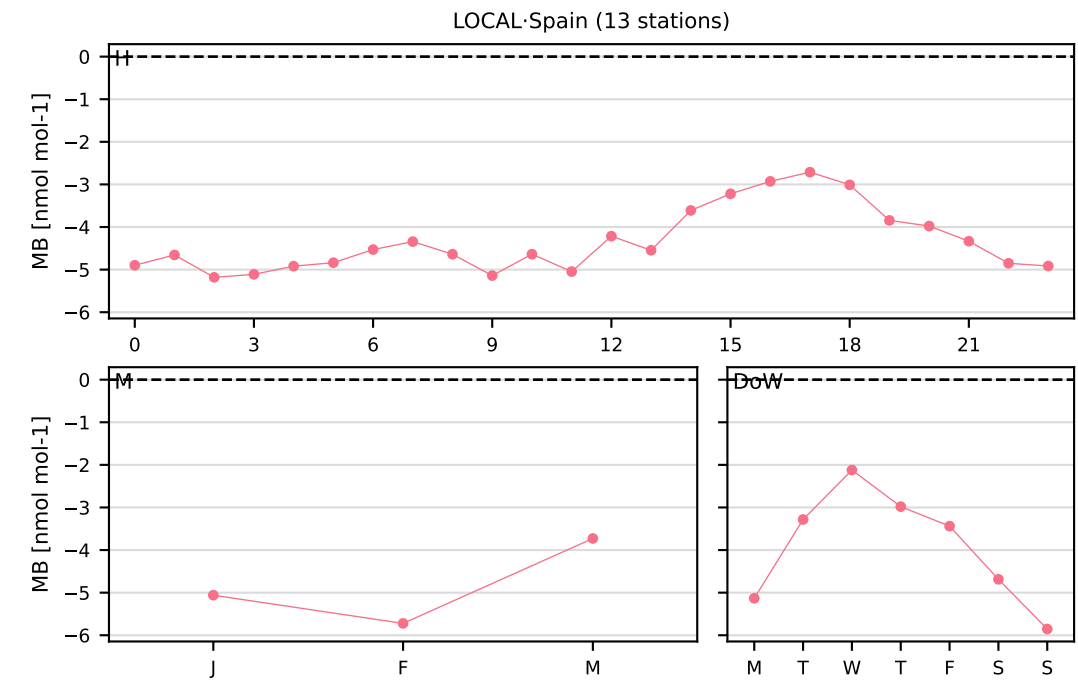
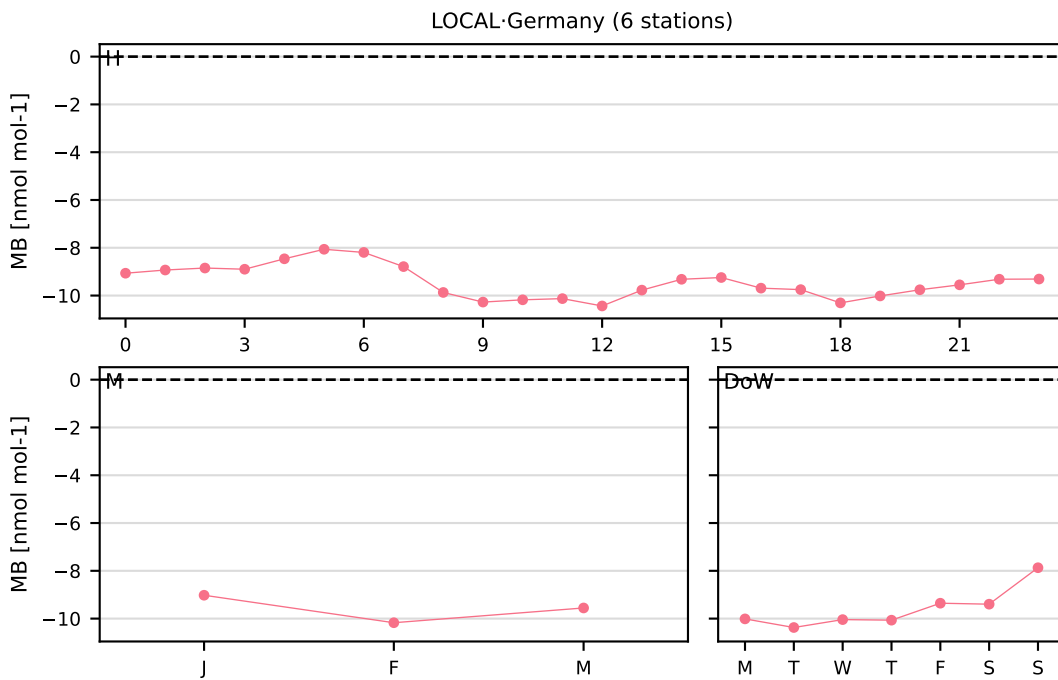
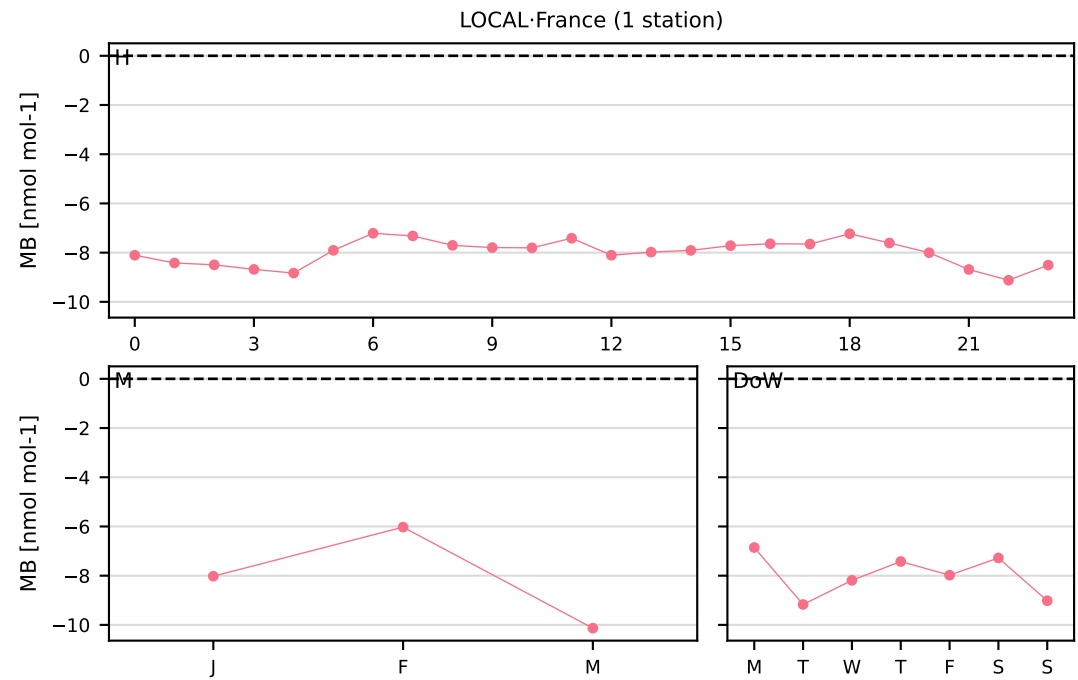
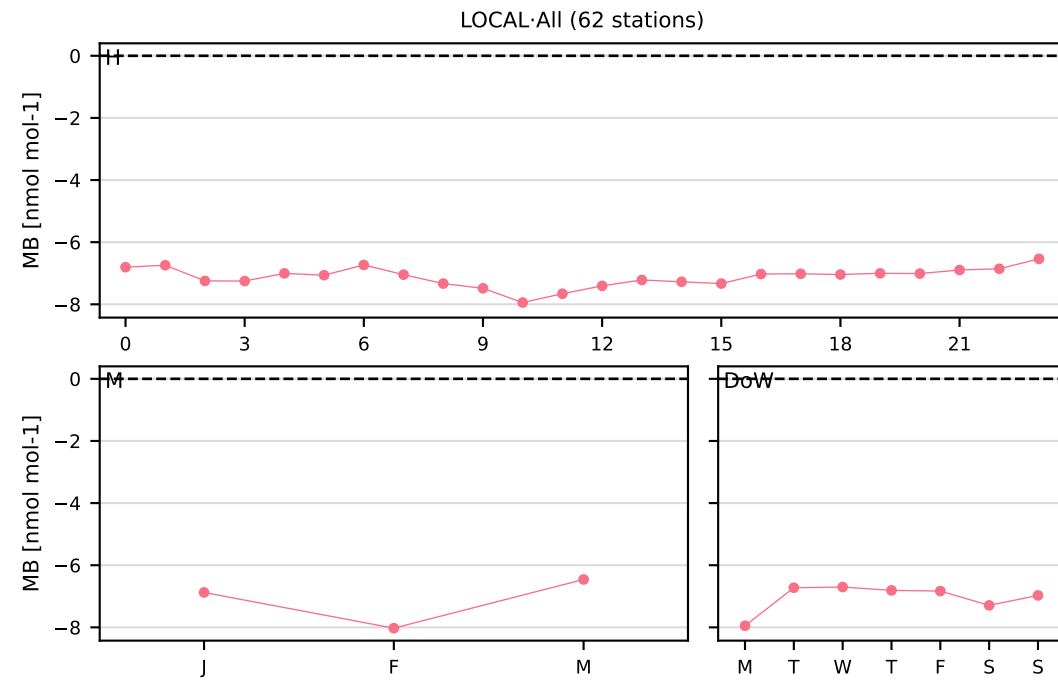
Heatmap r (Summary)
EBAS|sconco3



Periodic MB (Summary)

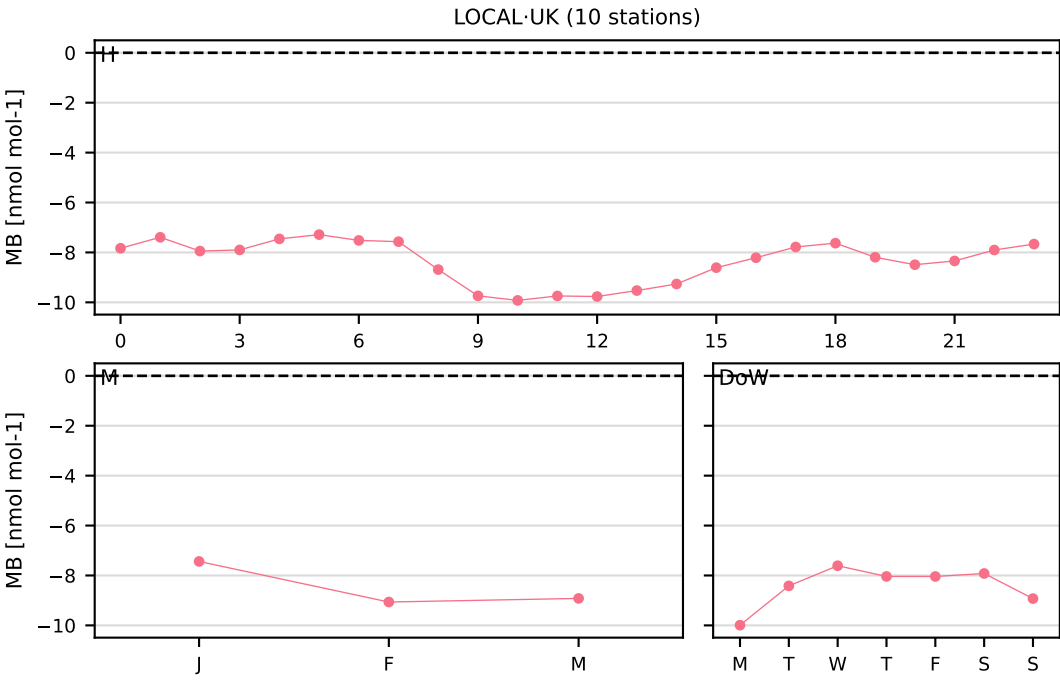
EBAS|sconco3

MONARCH



Periodic MB (Summary)
EBAS|sconco3

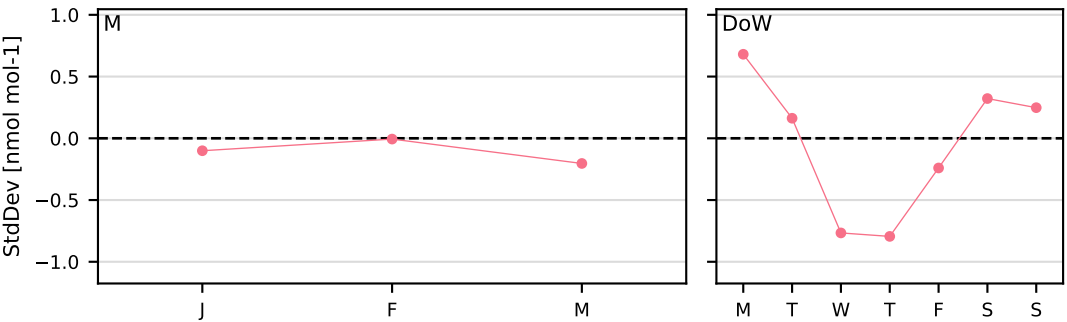
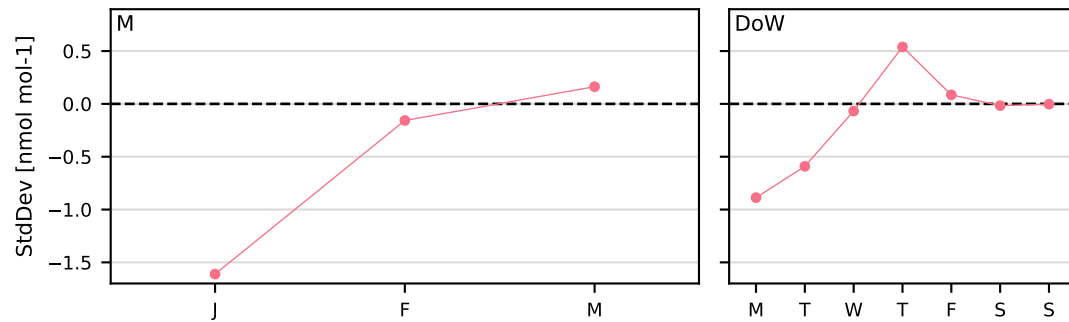
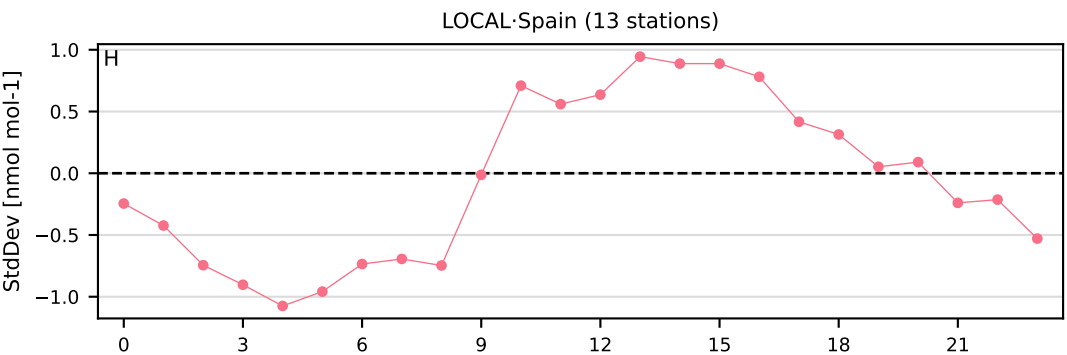
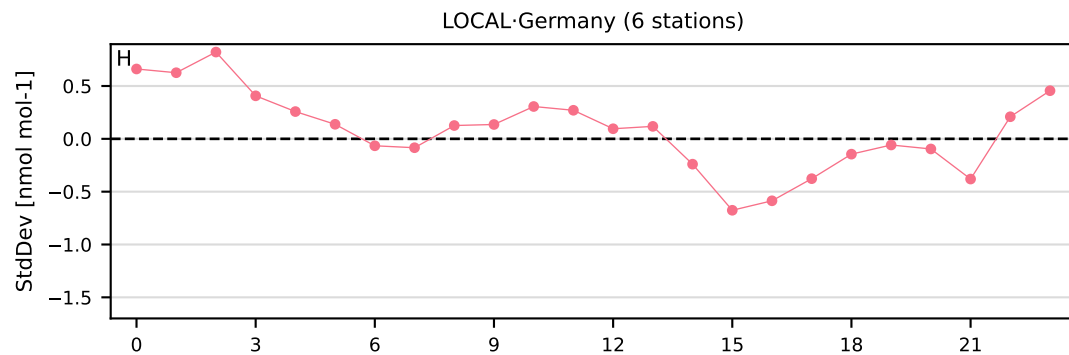
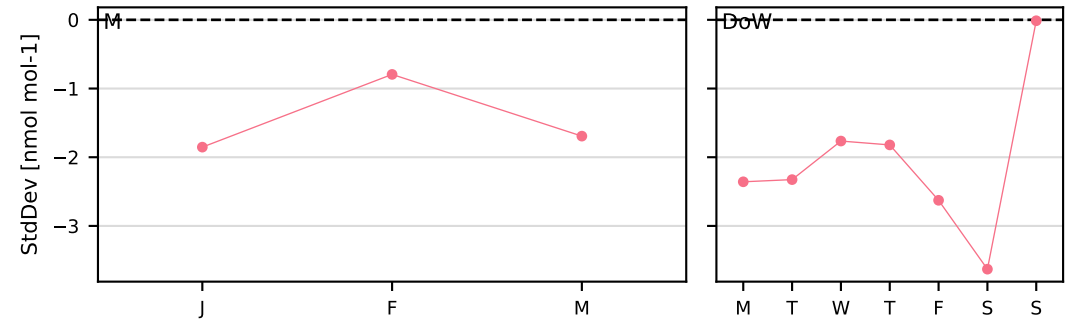
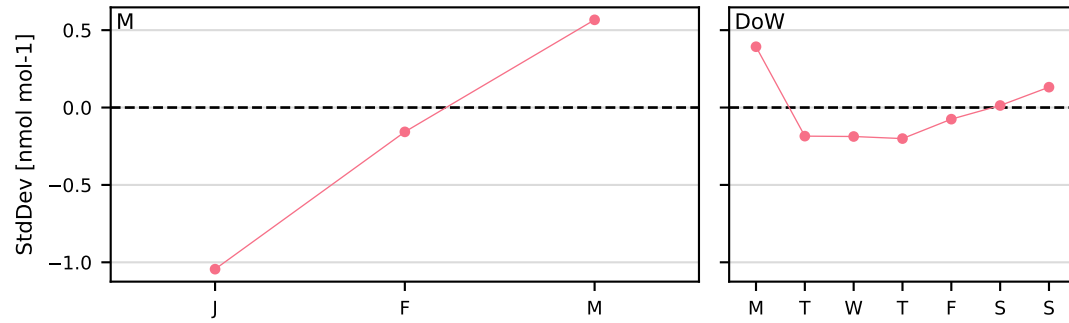
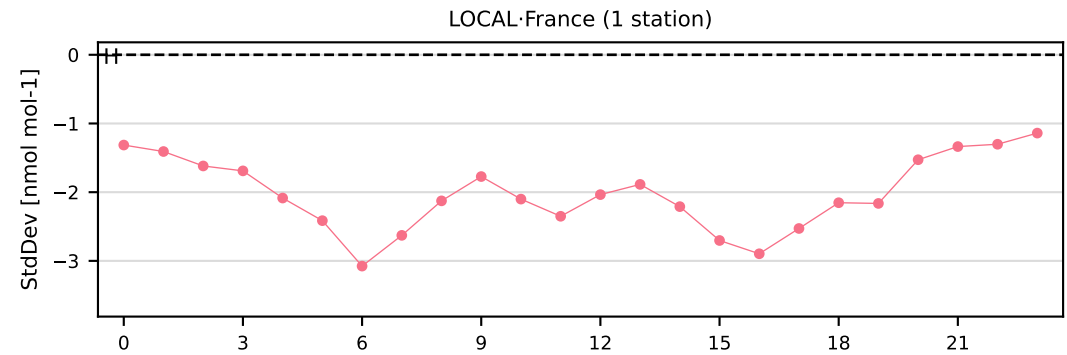
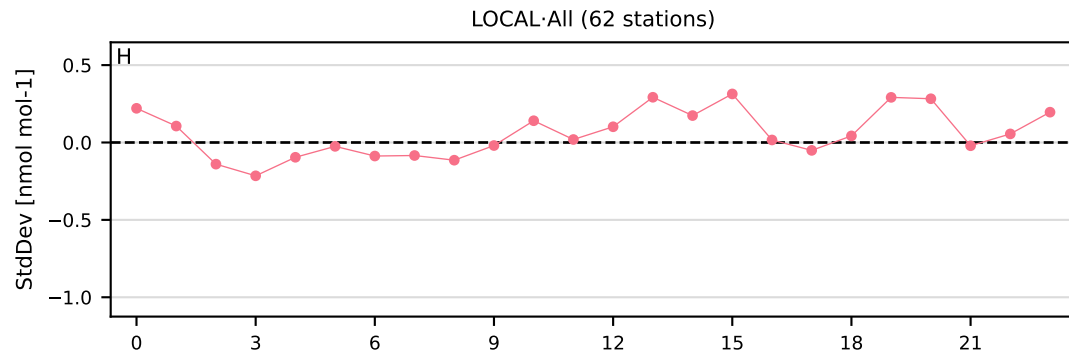
● MONARCH



Periodic StdDev bias (Summary)

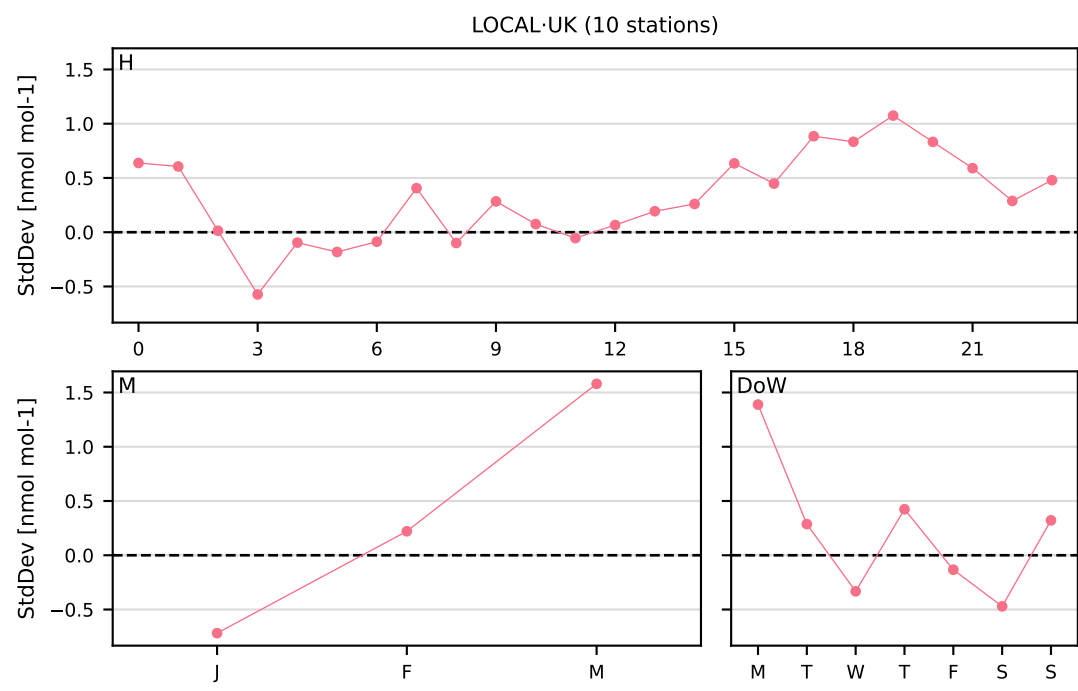
EBAS|sconco3

MONARCH



Periodic StdDev bias (Summary)
EBAS|sconco3

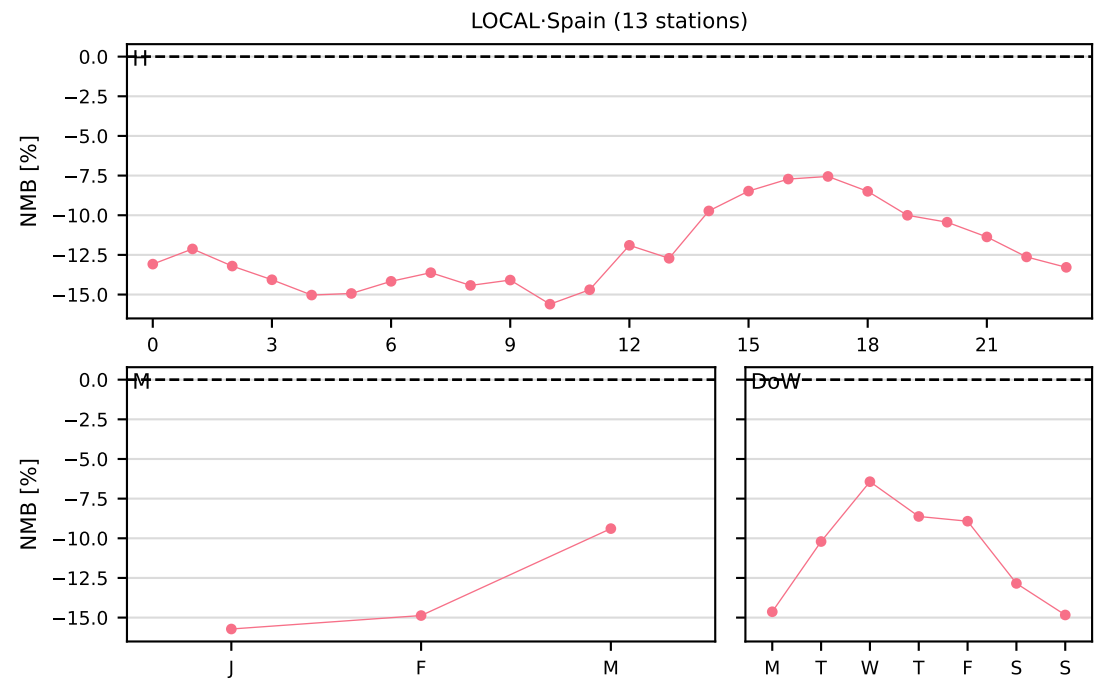
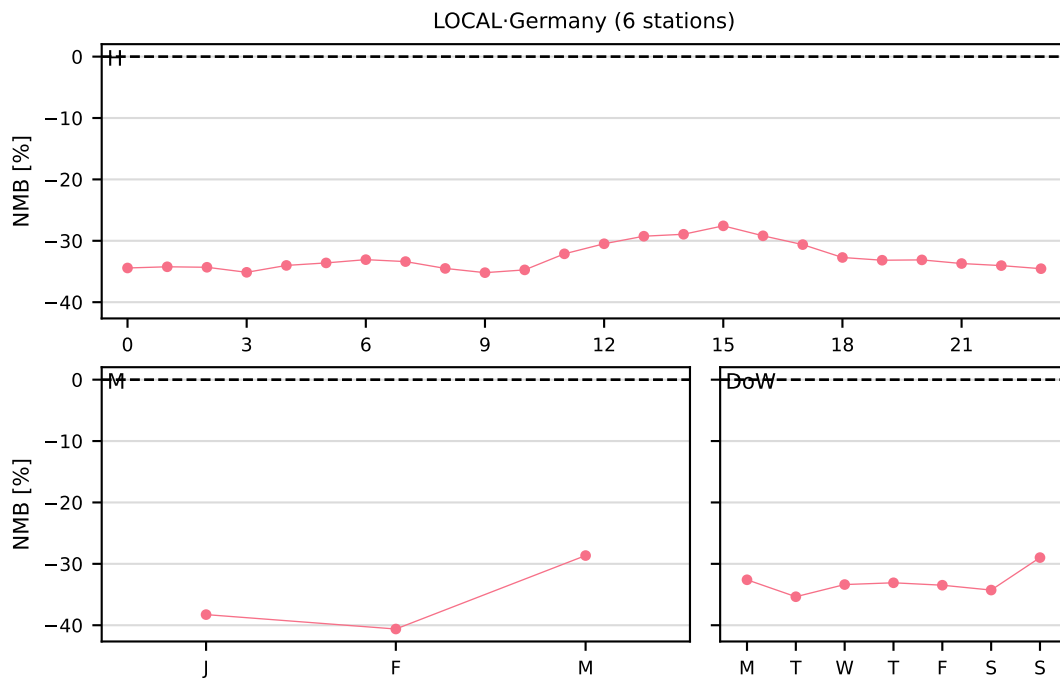
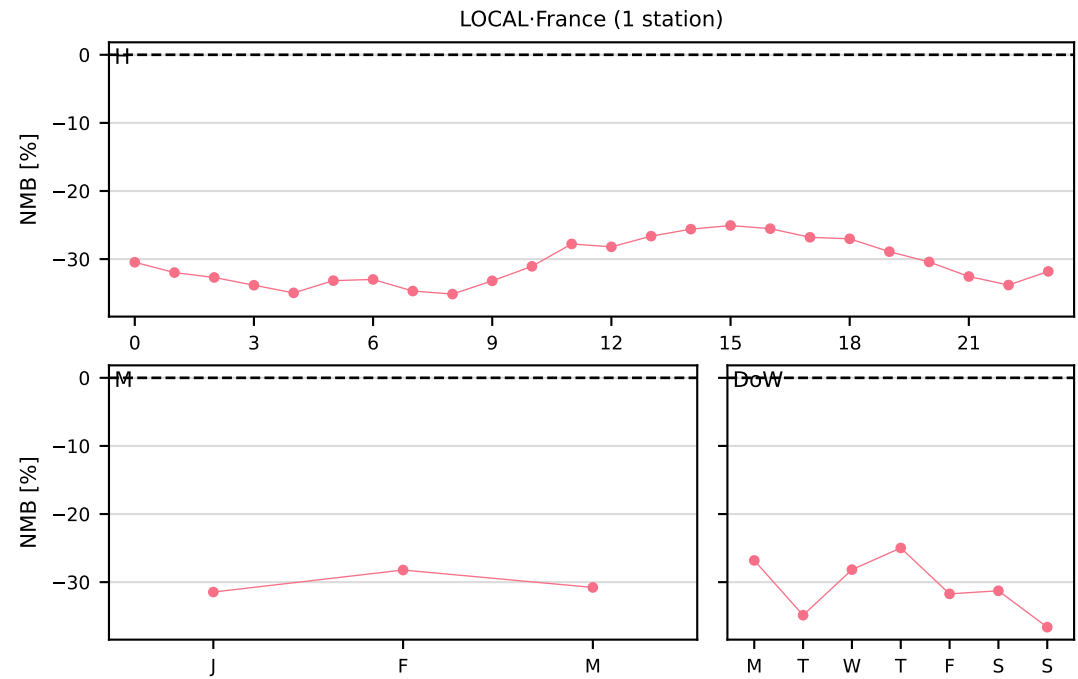
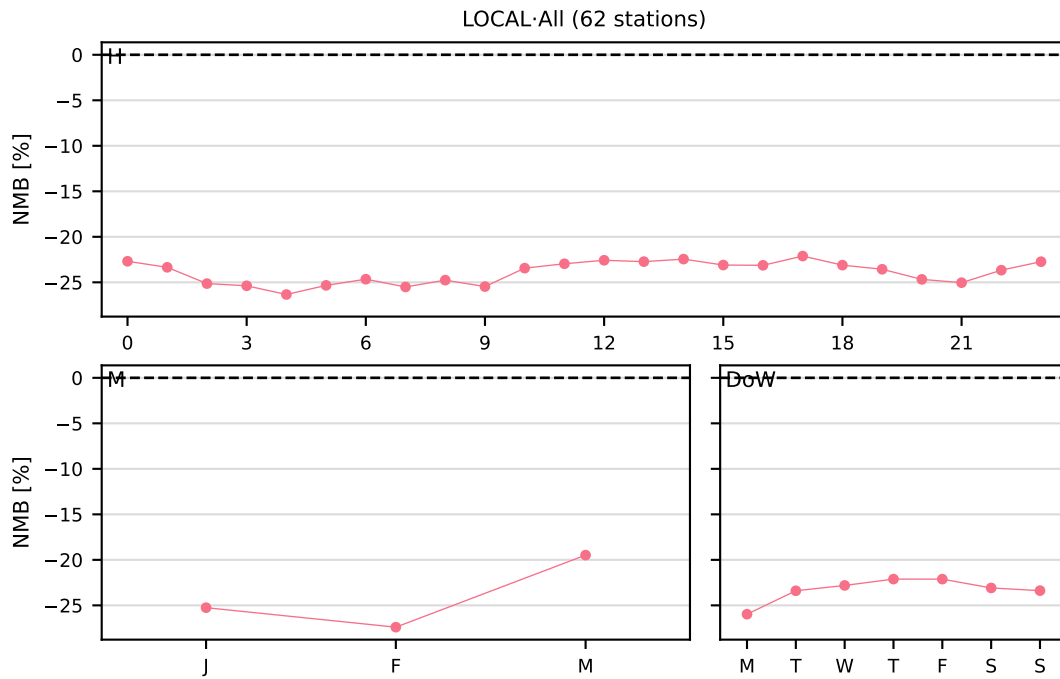
● MONARCH



Periodic NMB (Summary)

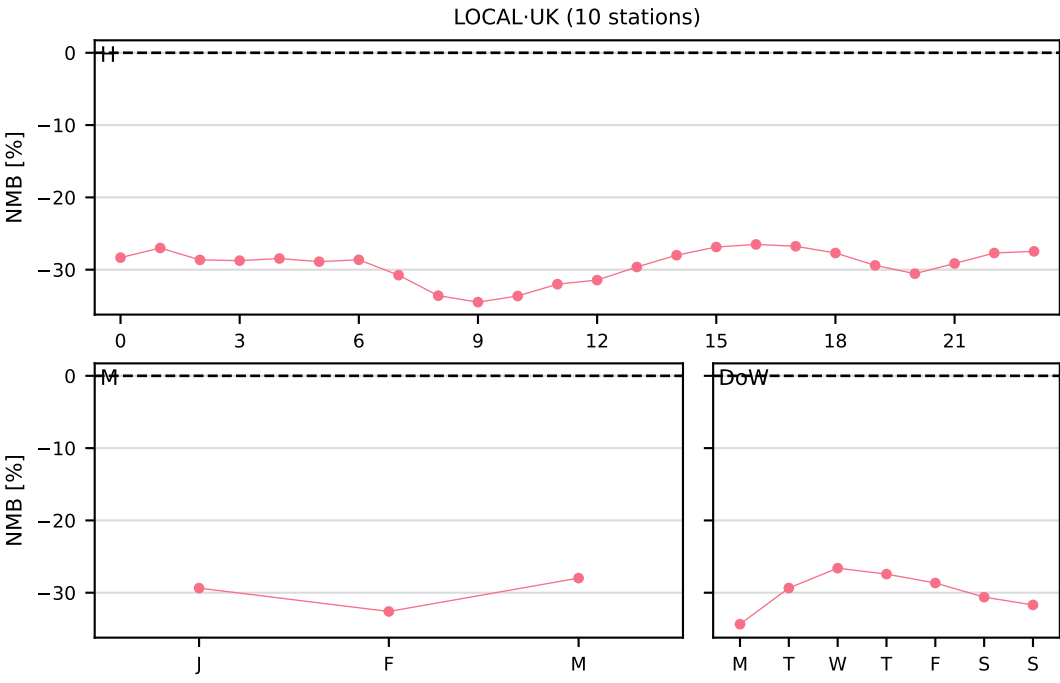
EBAS|sconco3

MONARCH



Periodic NMB (Summary)
EBAS|sconco3

● MONARCH

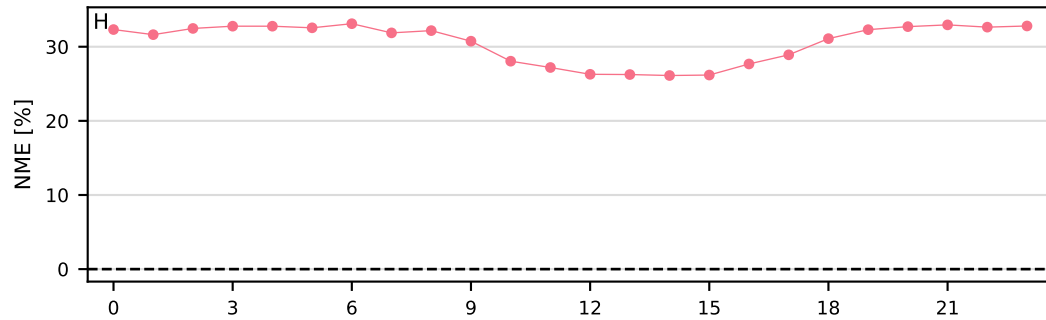


Periodic NME (Summary)

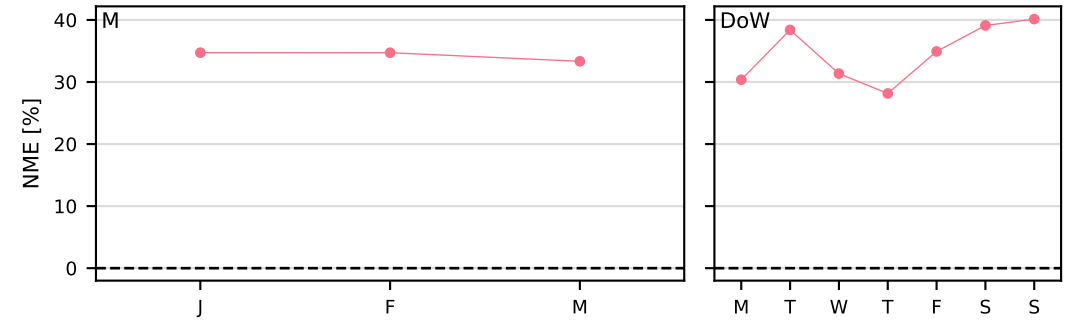
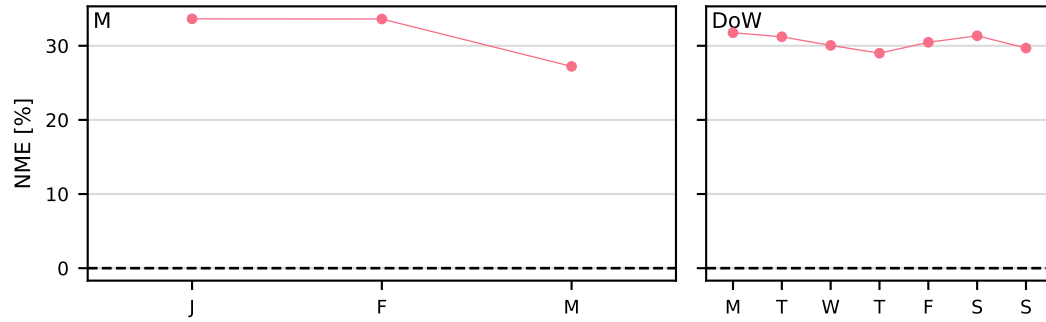
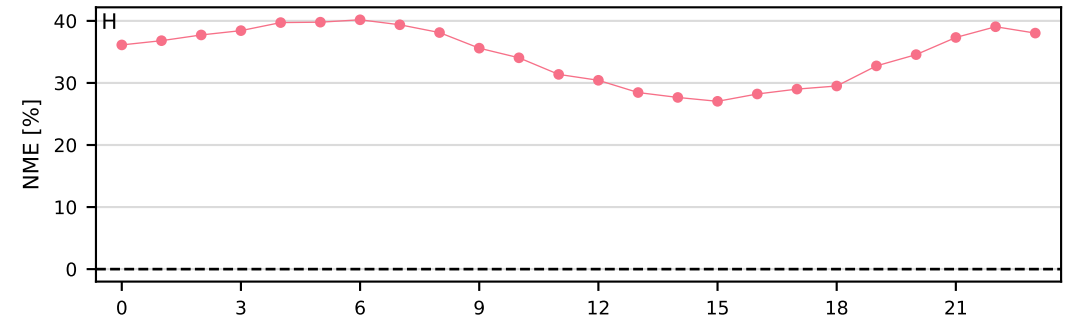
EBAS|sconco3

MONARCH

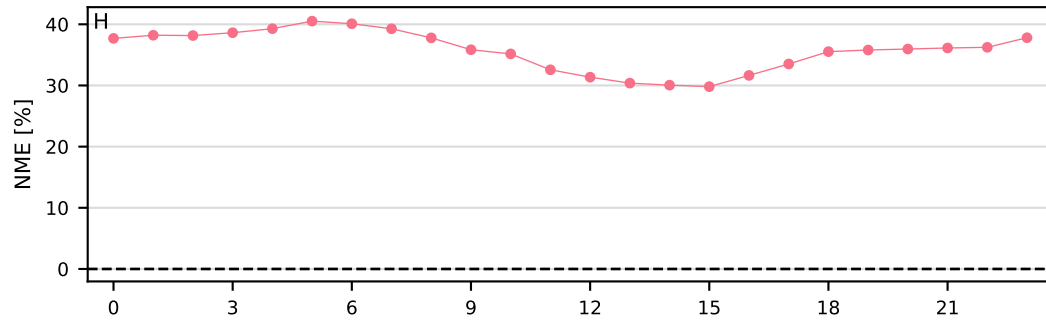
LOCAL·All (62 stations)



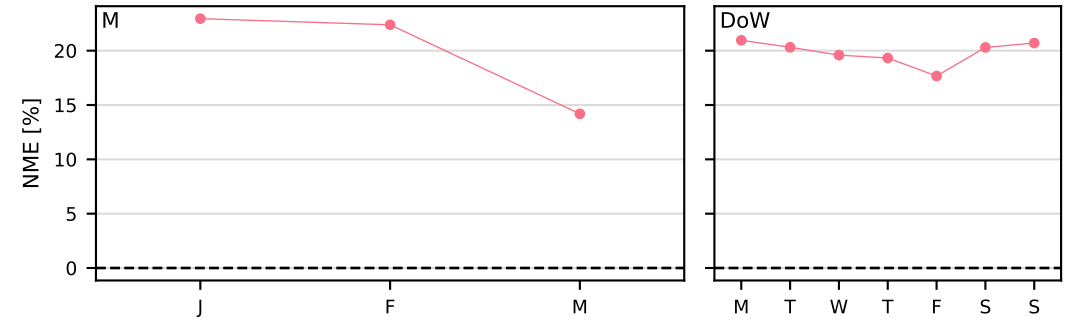
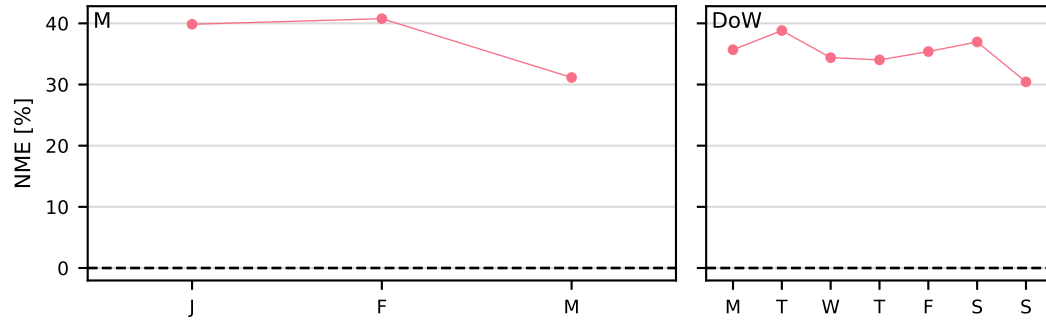
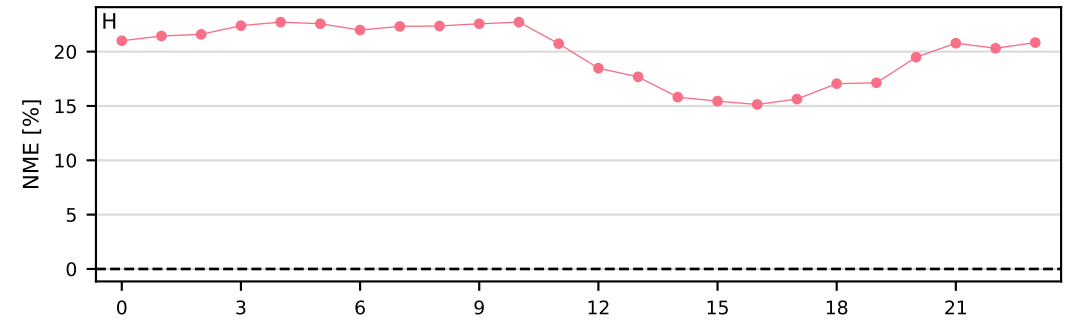
LOCAL·France (1 station)



LOCAL·Germany (6 stations)

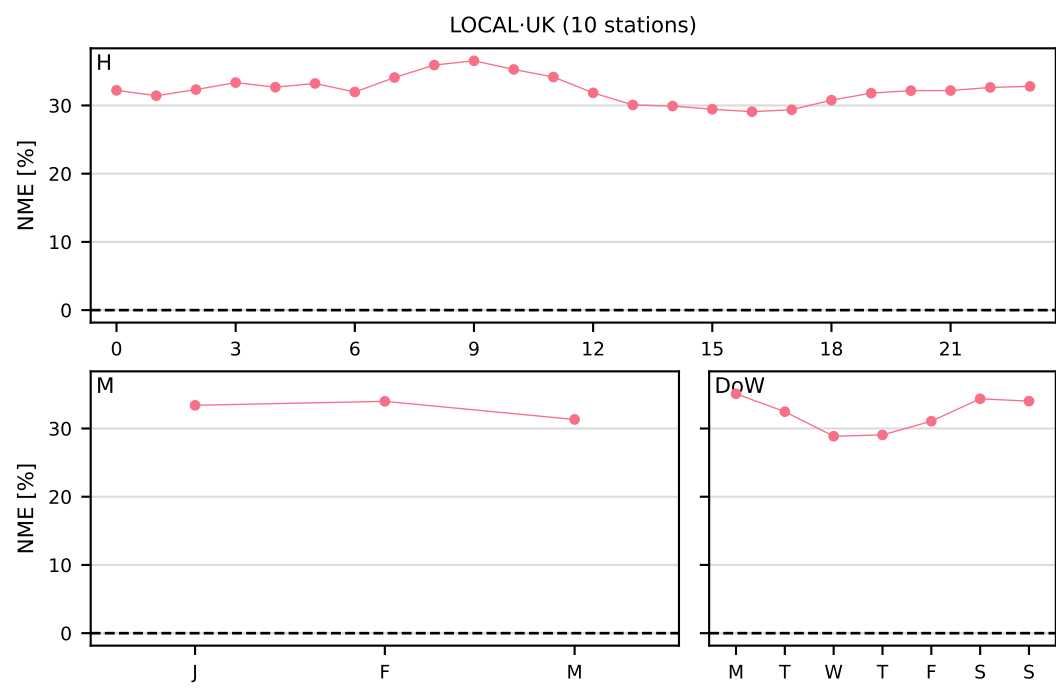


LOCAL·Spain (13 stations)



Periodic NME (Summary)
EBAS|sconco3

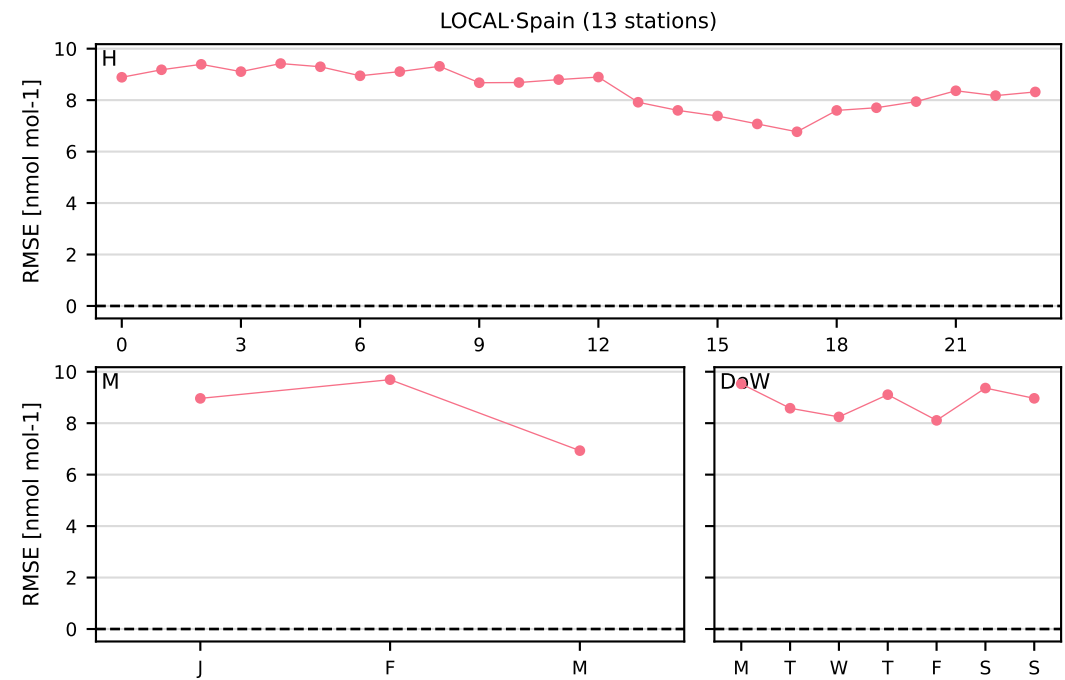
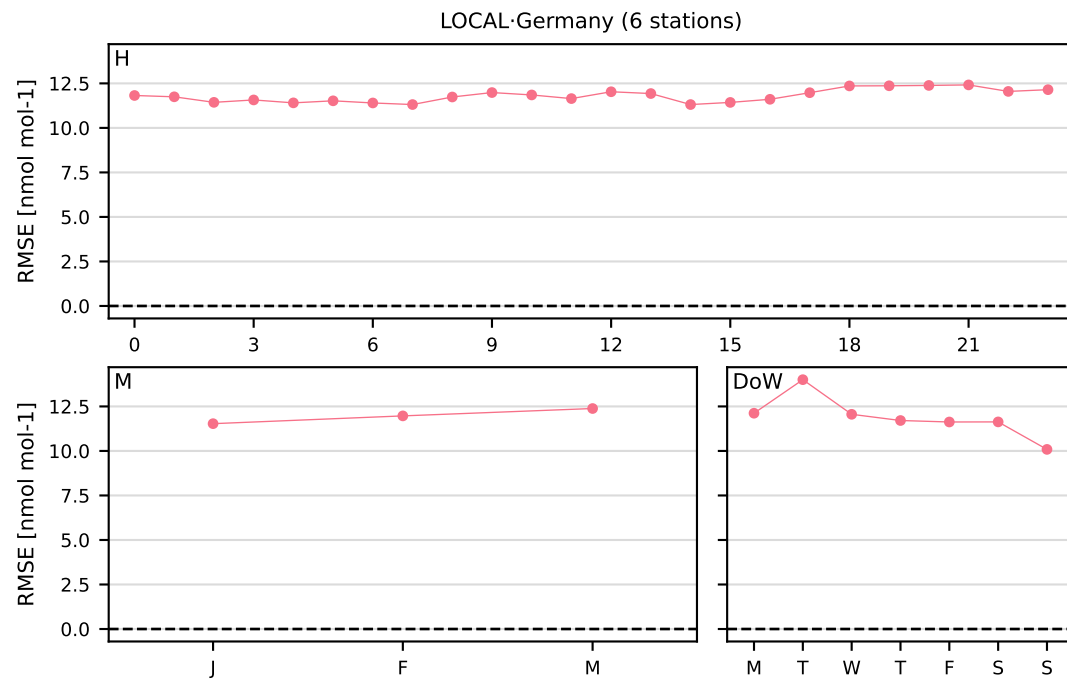
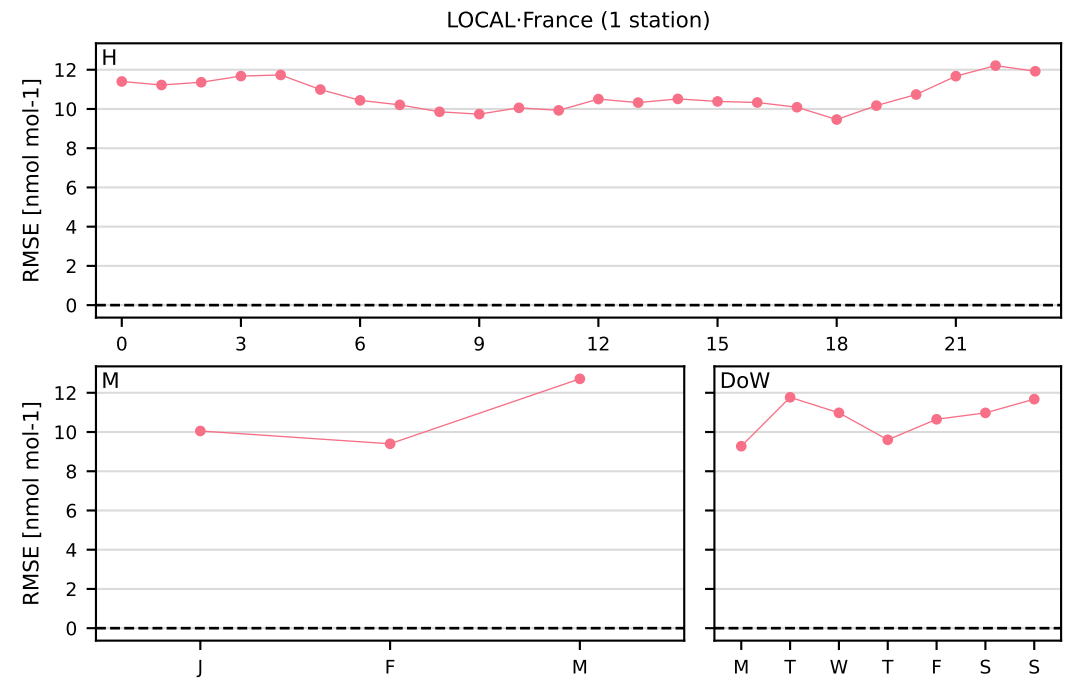
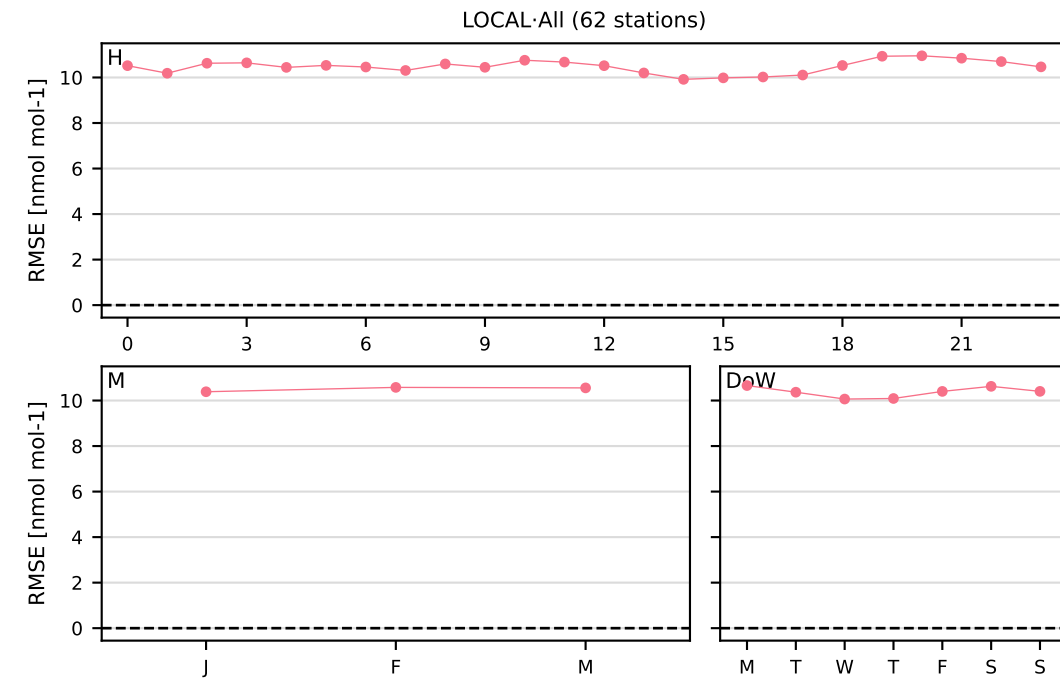
● MONARCH



Periodic RMSE (Summary)

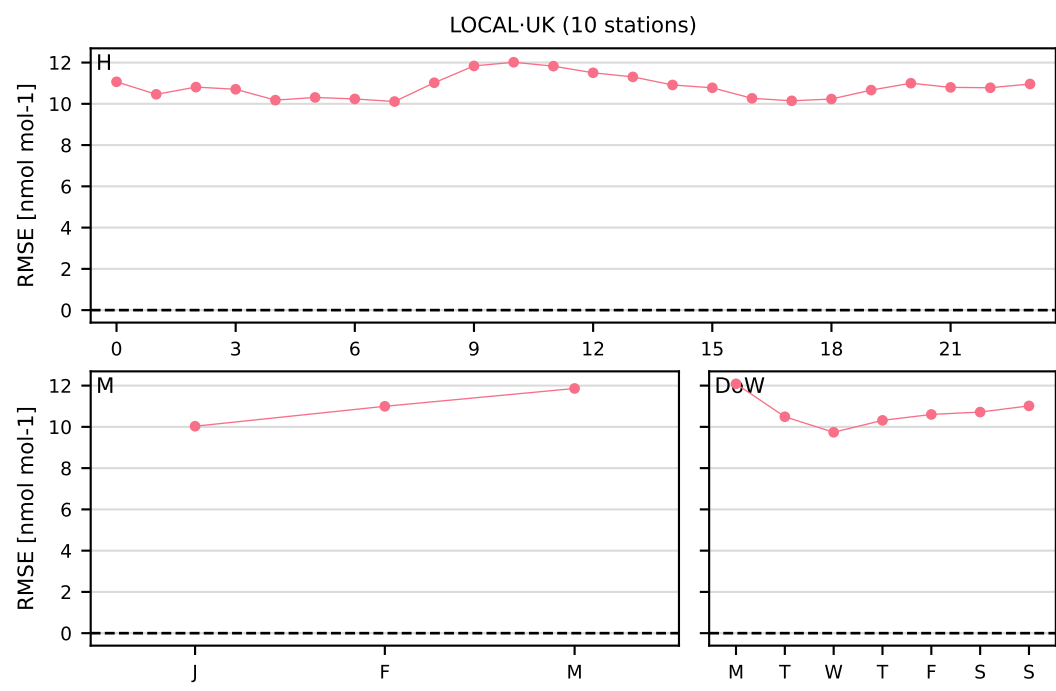
EBAS|sconco3

MONARCH



Periodic RMSE (Summary)
EBAS|sconco3

● MONARCH

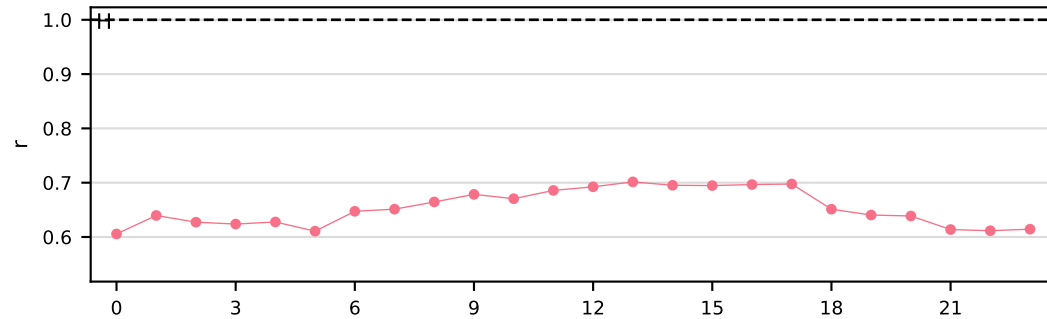


Periodic r (Summary)

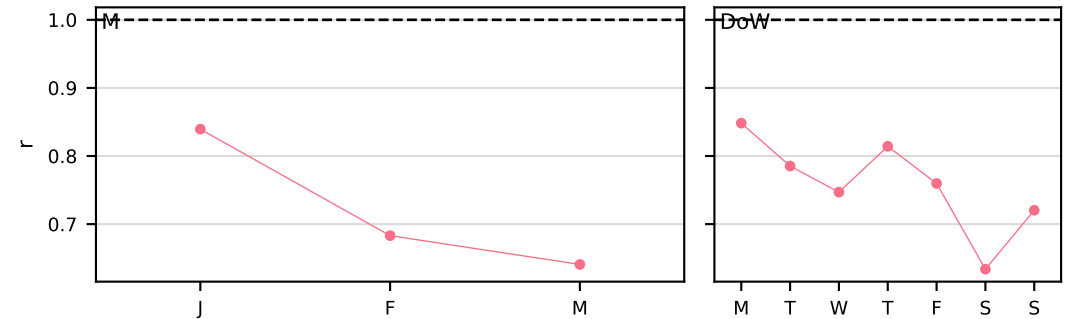
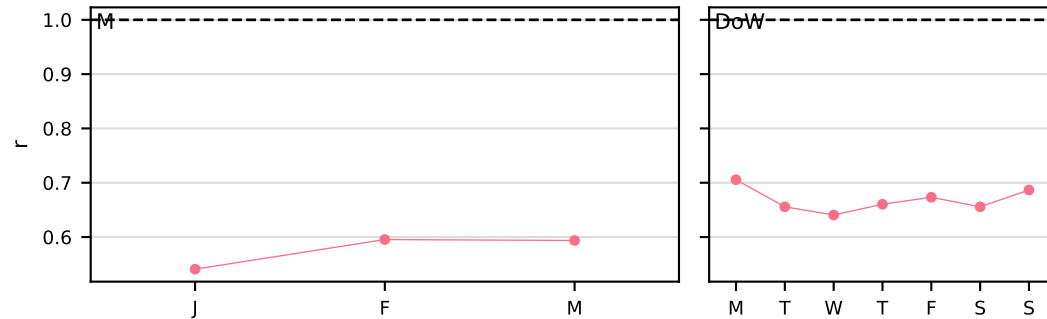
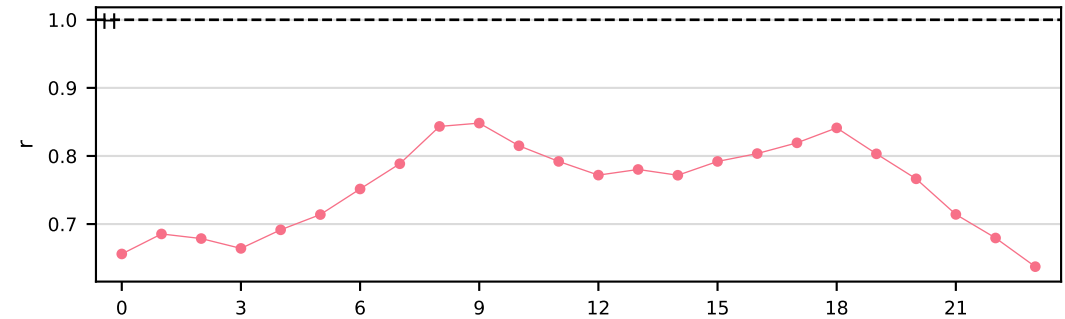
EBAS|sconco3

MONARCH

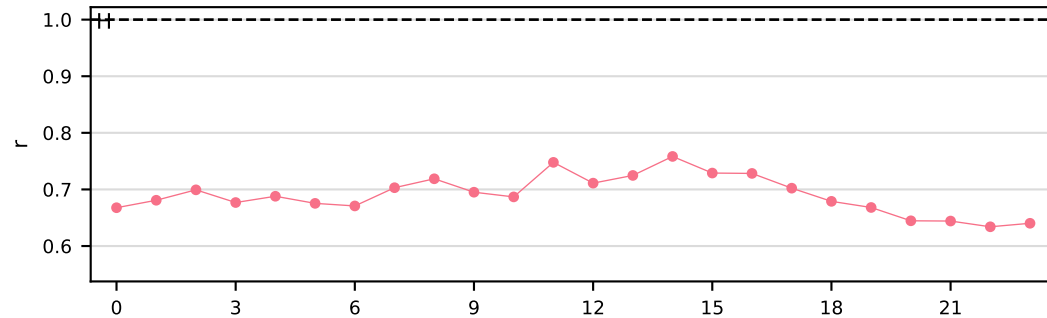
LOCAL·All (62 stations)



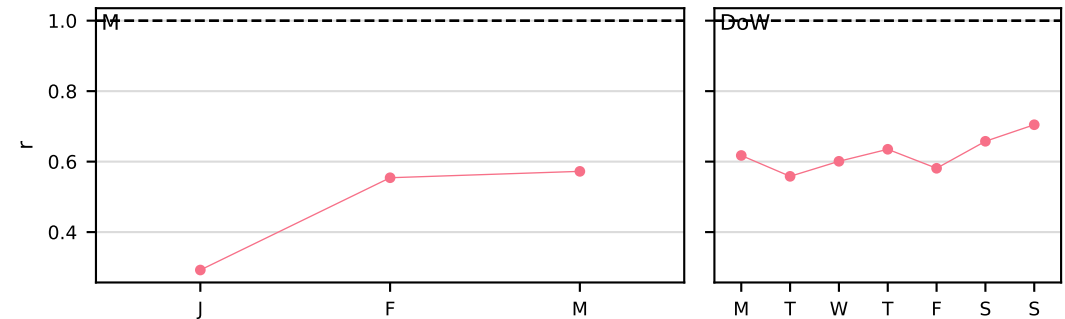
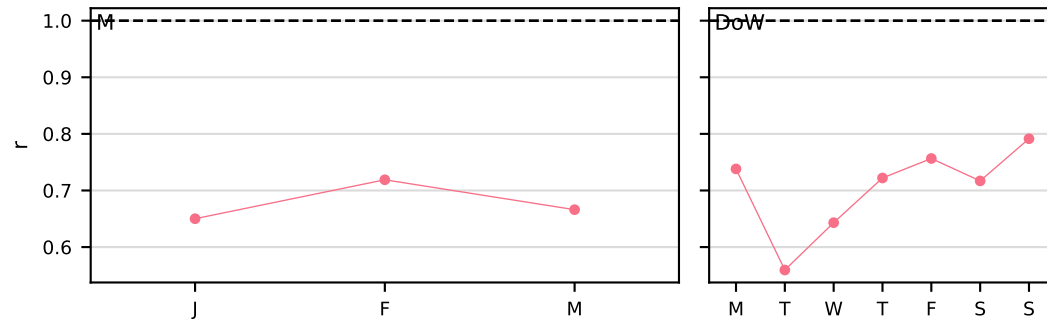
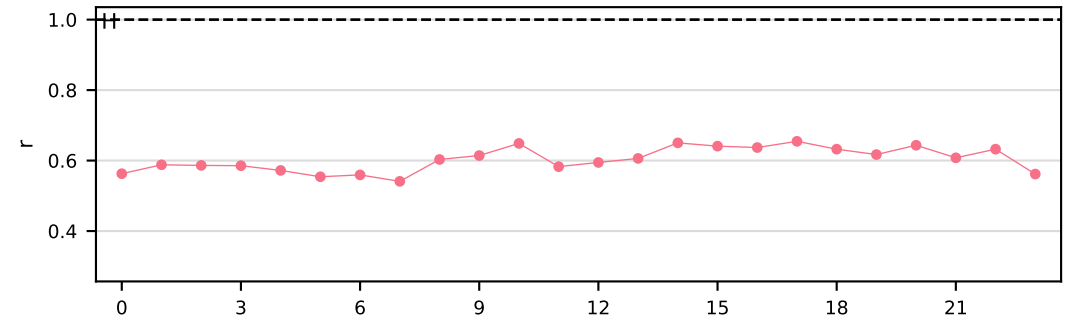
LOCAL·France (1 station)



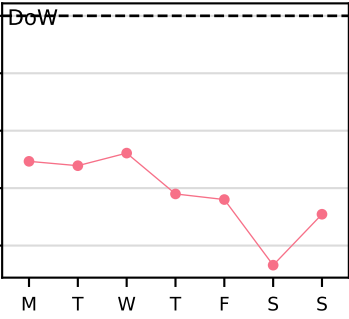
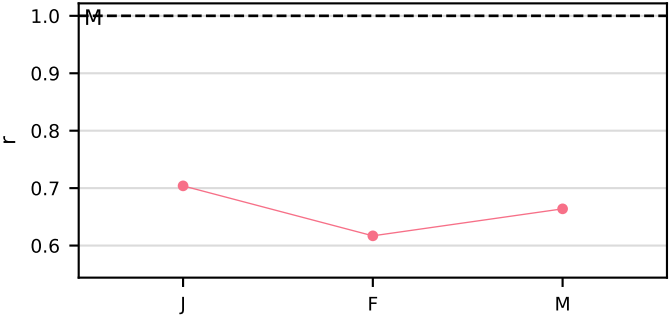
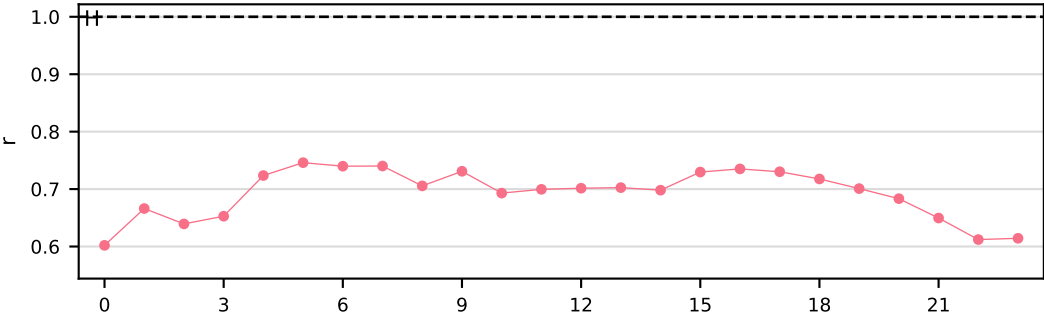
LOCAL·Germany (6 stations)



LOCAL·Spain (13 stations)



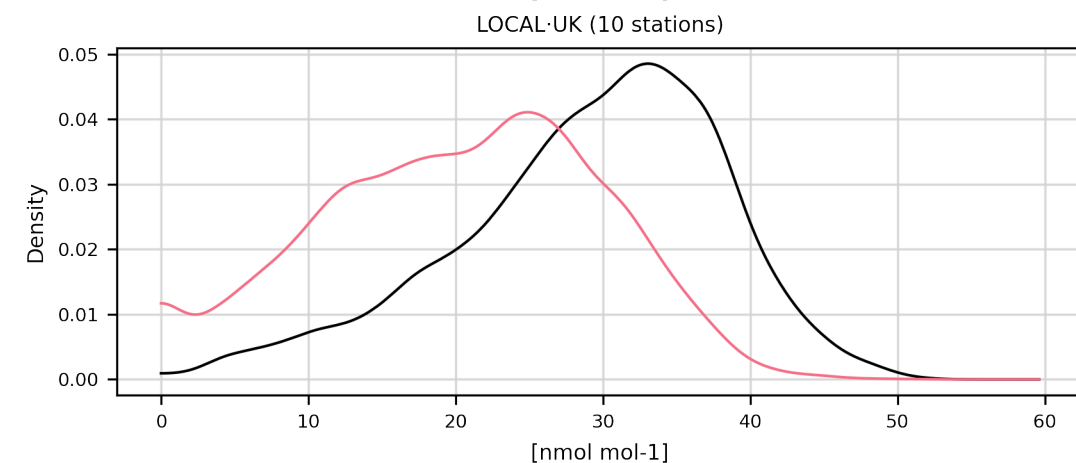
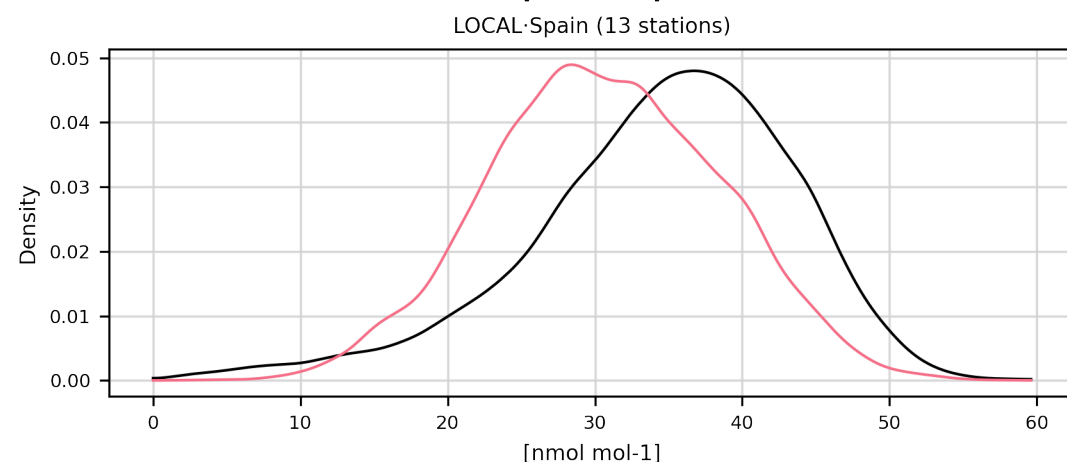
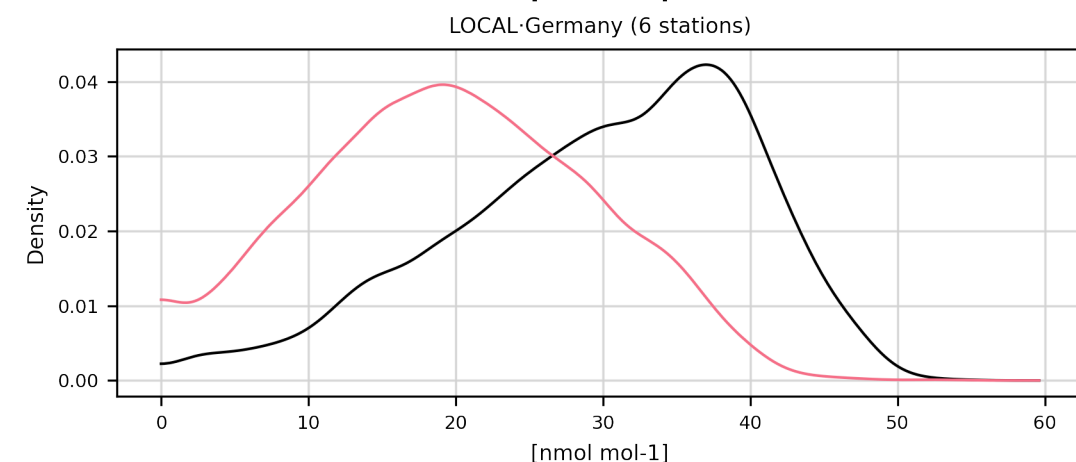
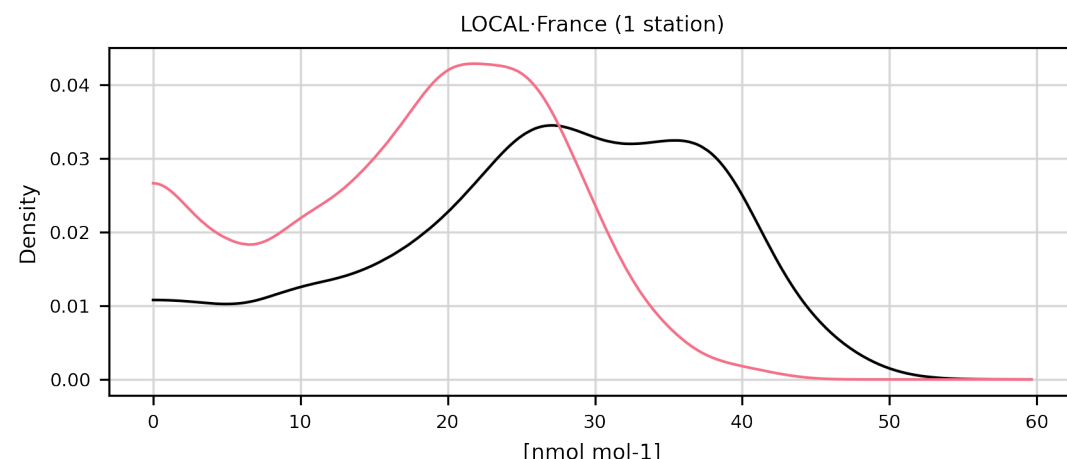
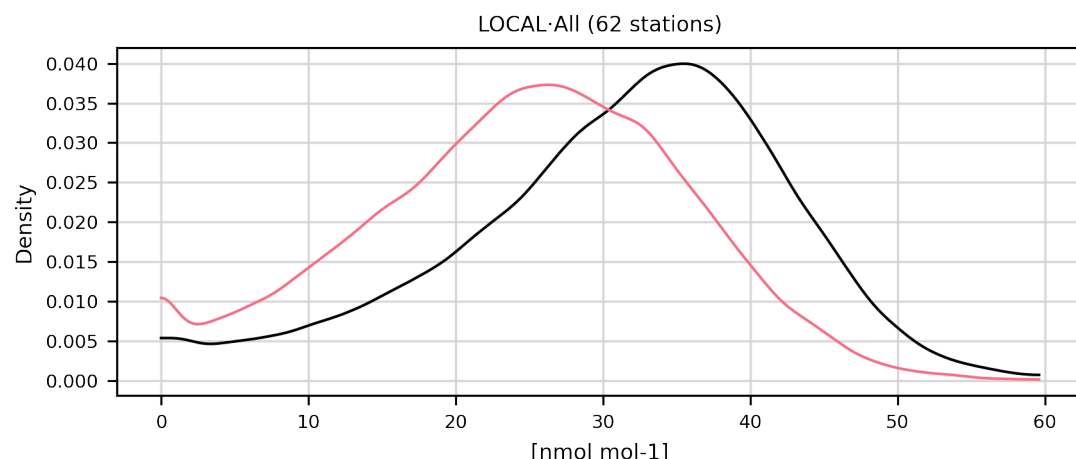
LOCAL·UK (10 stations)



Distribution (Summary)

EBAS|sconco3

● observations ● MONARCH

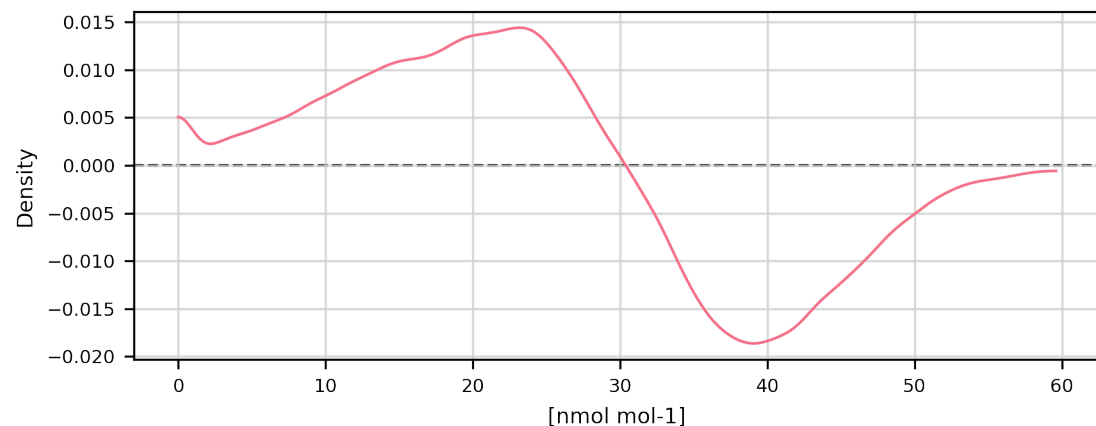


Distribution bias (Summary)

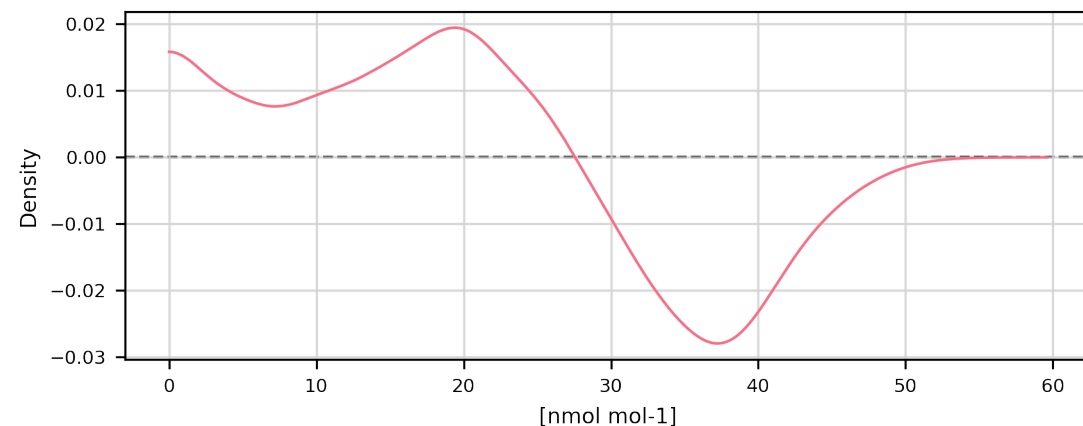
EBAS|sconco3

● MONARCH

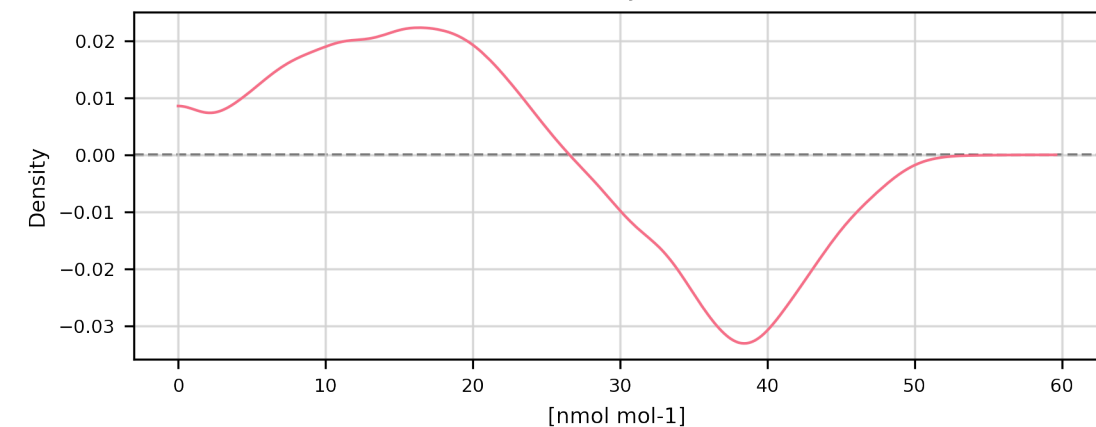
LOCAL·All (62 stations)



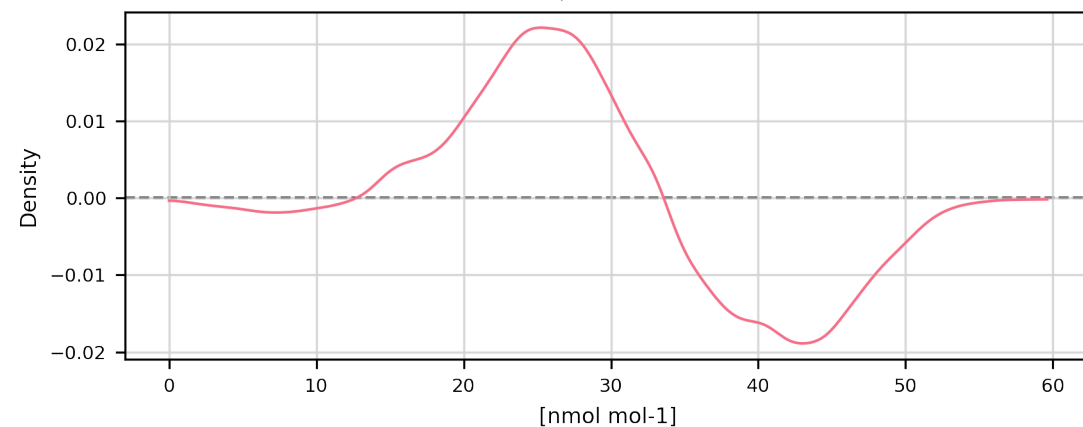
LOCAL·France (1 station)



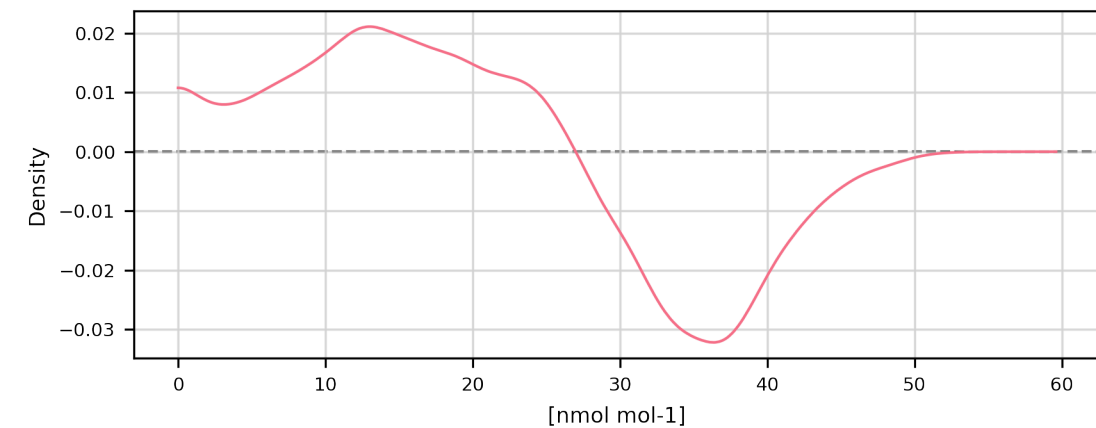
LOCAL·Germany (6 stations)

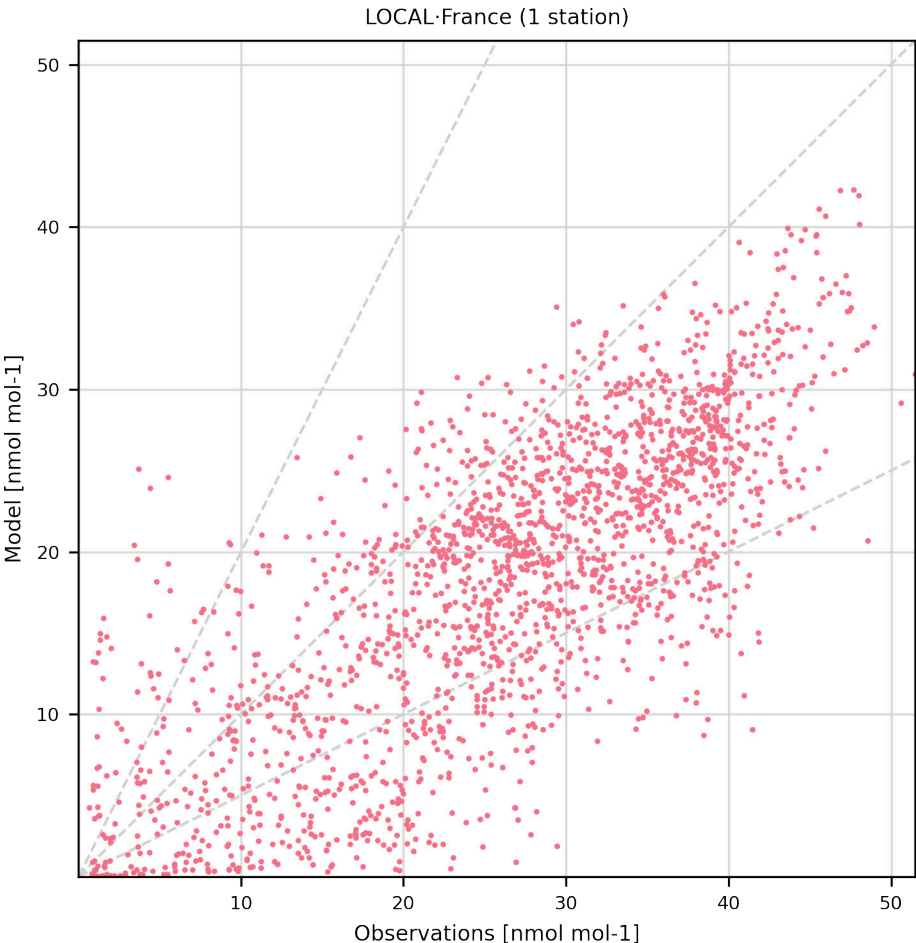
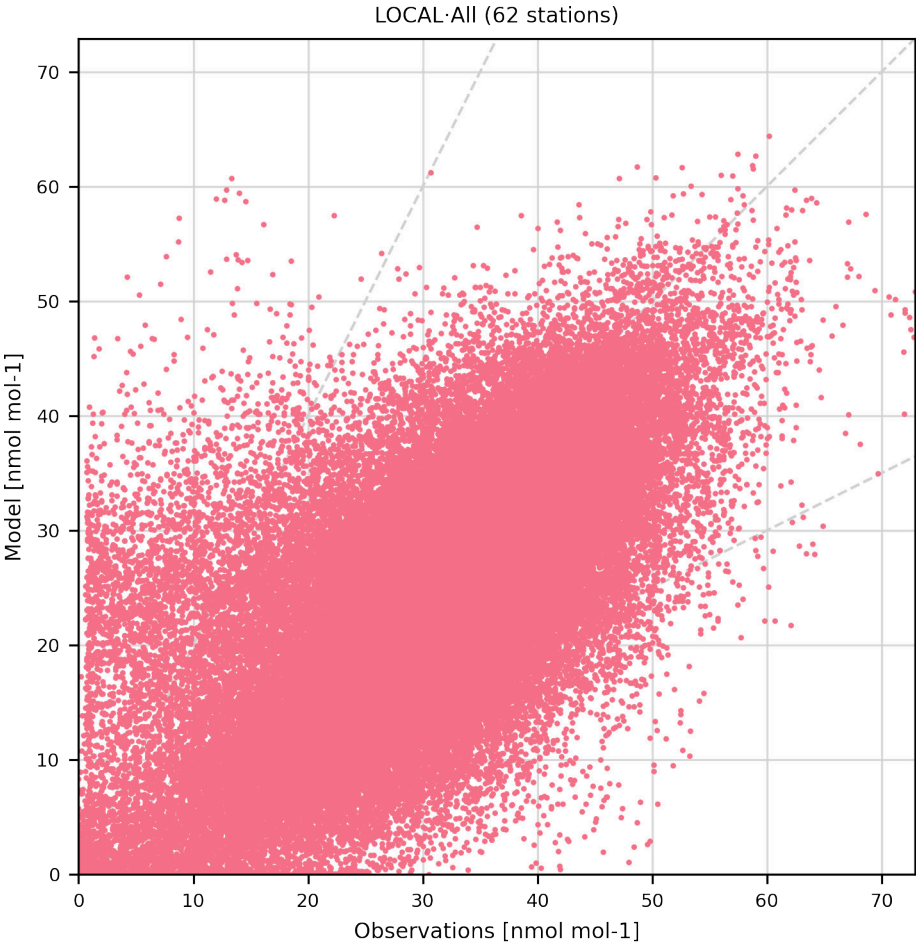


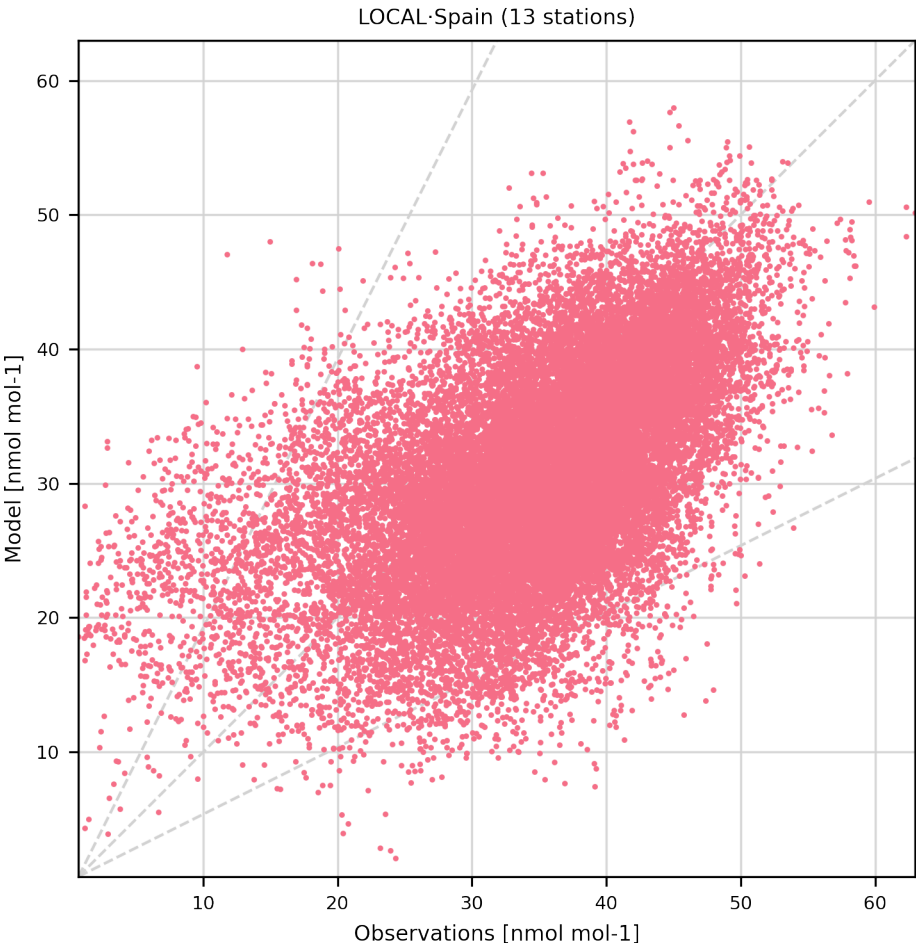
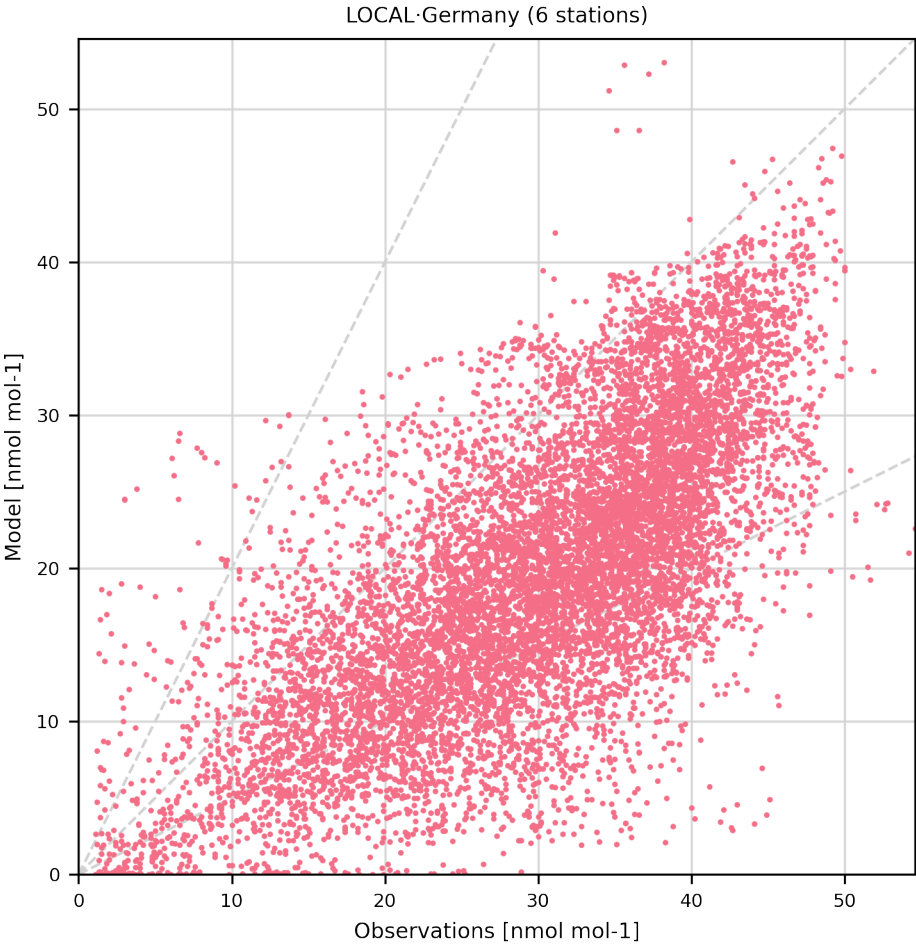
LOCAL·Spain (13 stations)

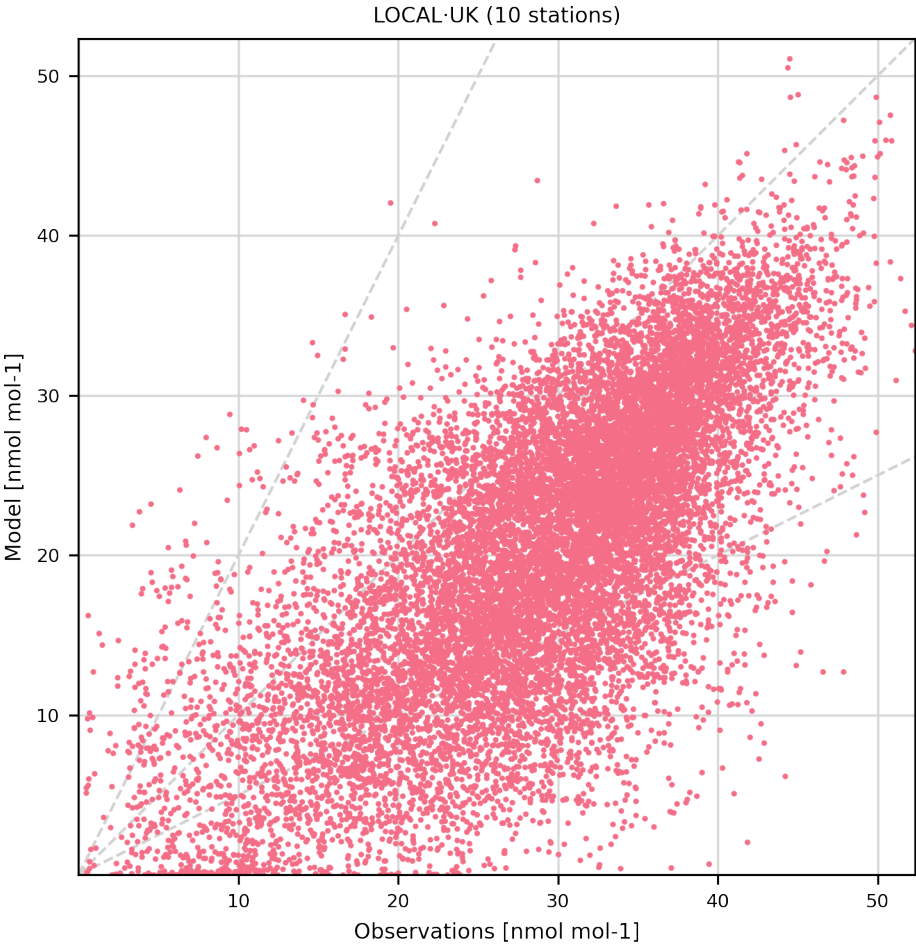


LOCAL·UK (10 stations)

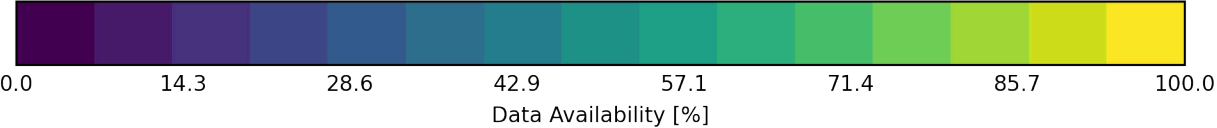




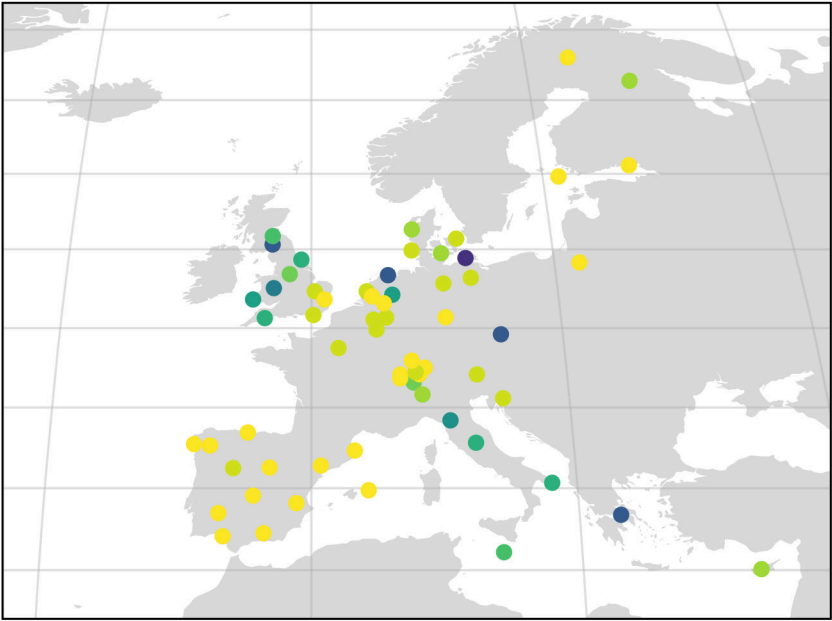




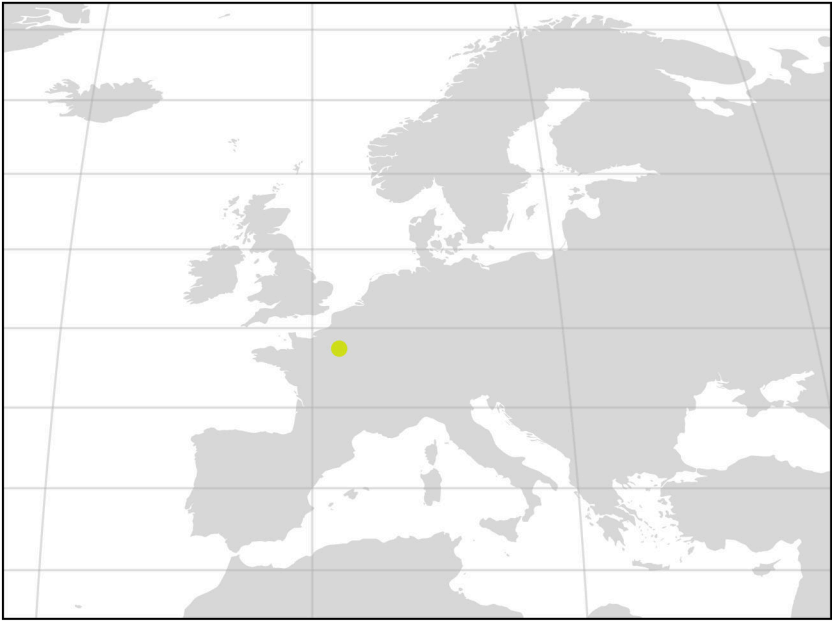
Map Data Availability (Summary)
EBAS|sconcn02



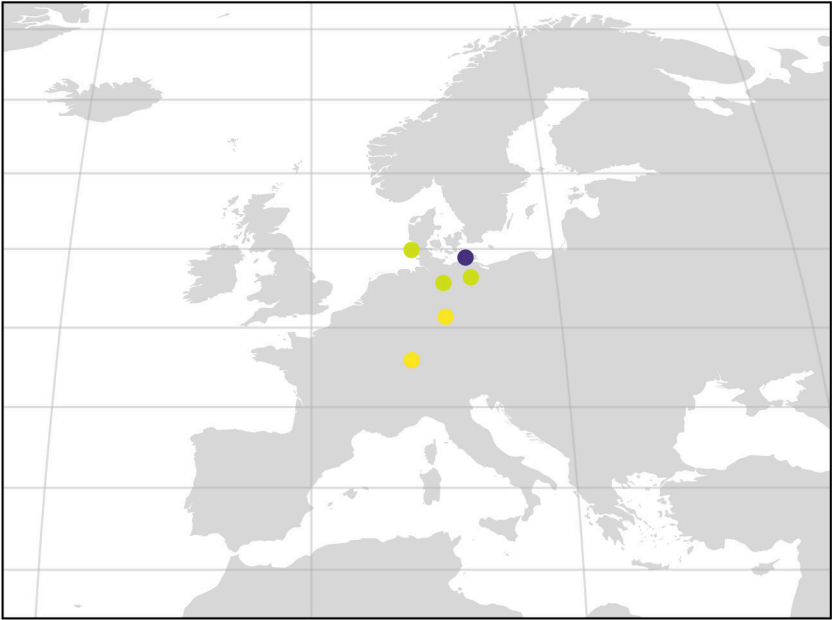
observations
LOCAL·All (62 stations)



observations
LOCAL·France (1 station)



observations
LOCAL·Germany (6 stations)

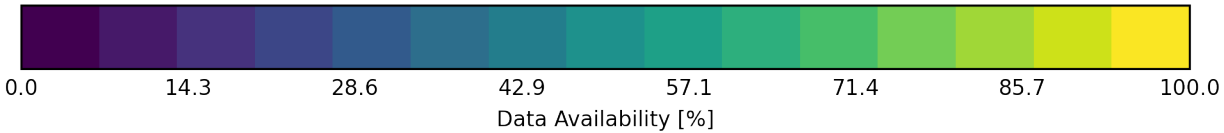


observations
LOCAL·Spain (13 stations)

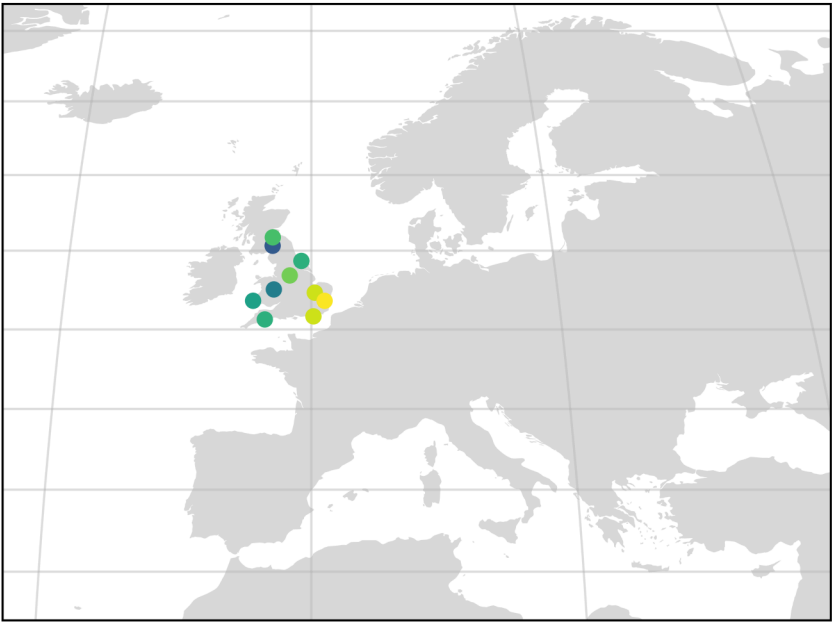


Map Data Availability (Summary)

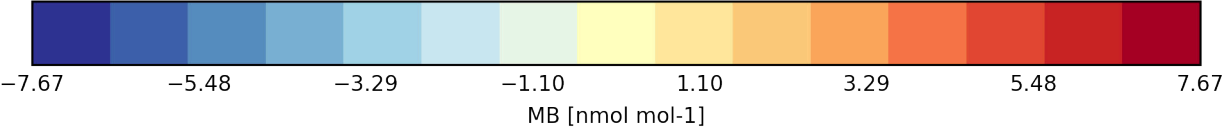
EBAS|sconcn02



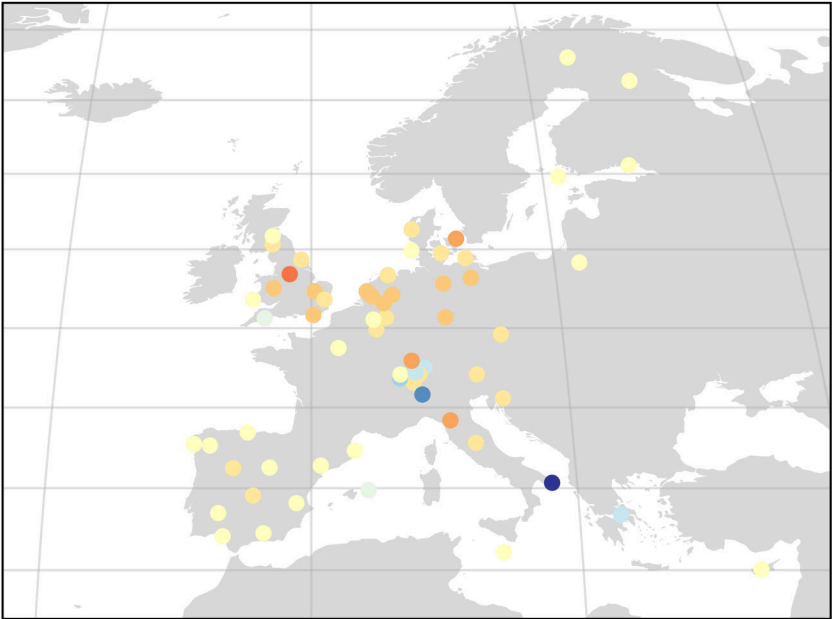
observations
LOCAL-UK (10 stations)



Map MB (Summary)
EBAS|sconcn02



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



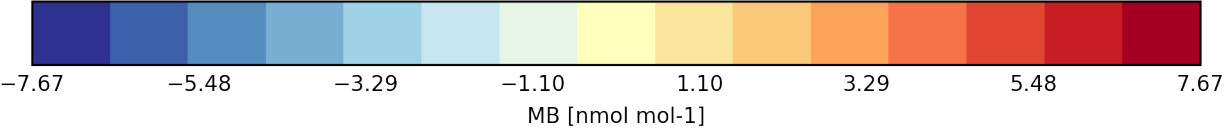
MONARCH
LOCAL·Germany (6 stations)



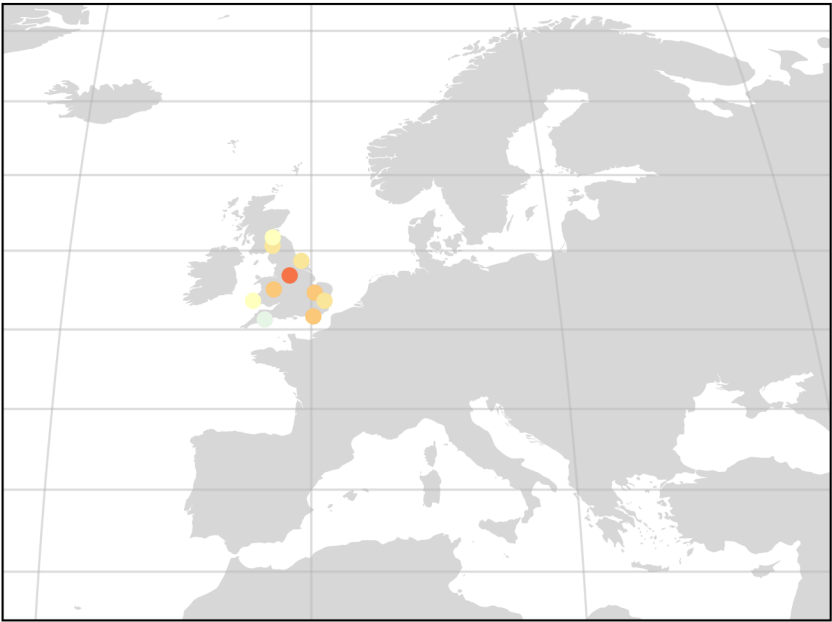
MONARCH
LOCAL·Spain (13 stations)



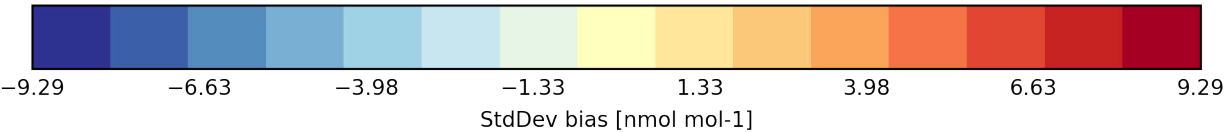
Map MB (Summary)
EBAS|sconcn02



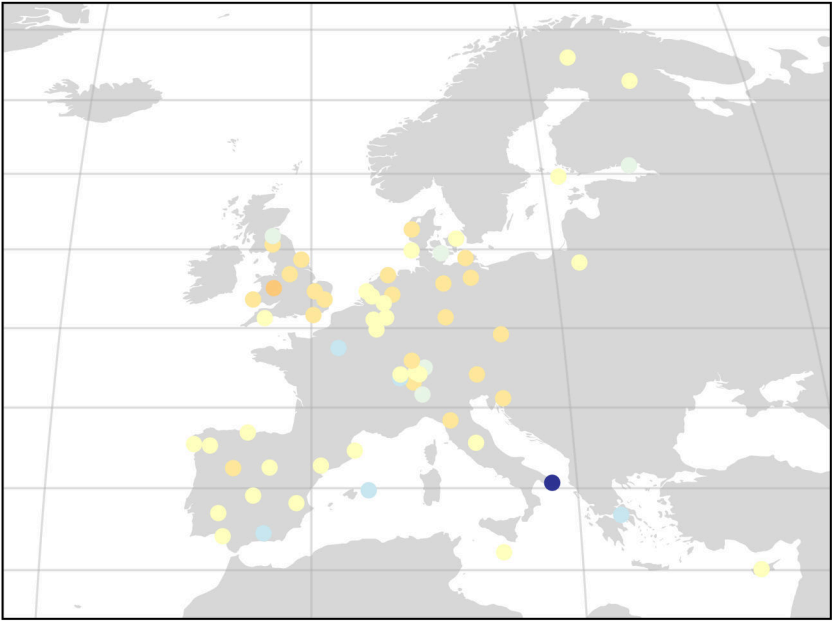
MONARCH
LOCAL·UK (10 stations)



Map StdDev bias (Summary)
EBAS|sconcn02



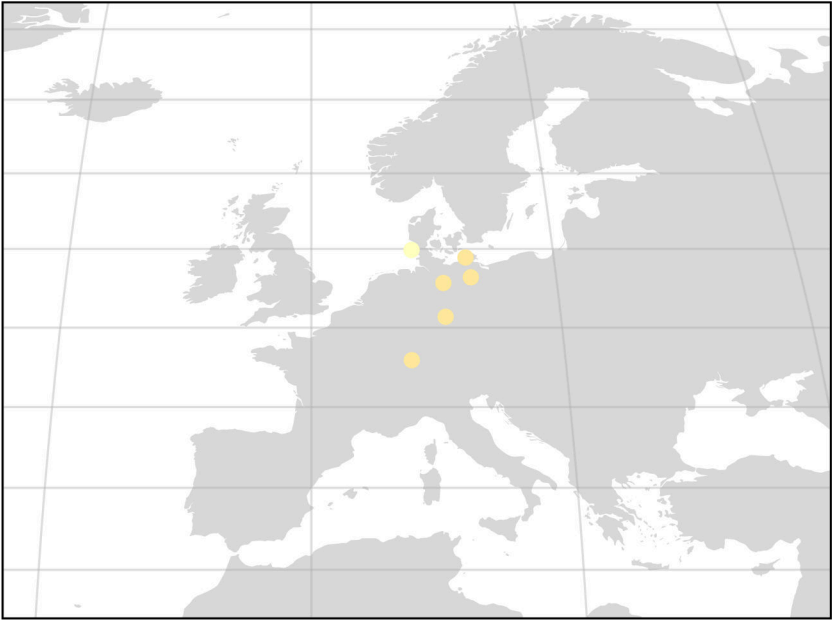
MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



MONARCH
LOCAL·Germany (6 stations)

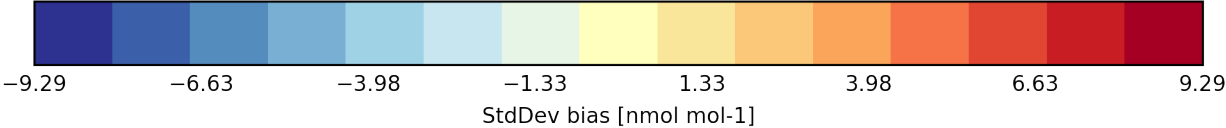


MONARCH
LOCAL·Spain (13 stations)

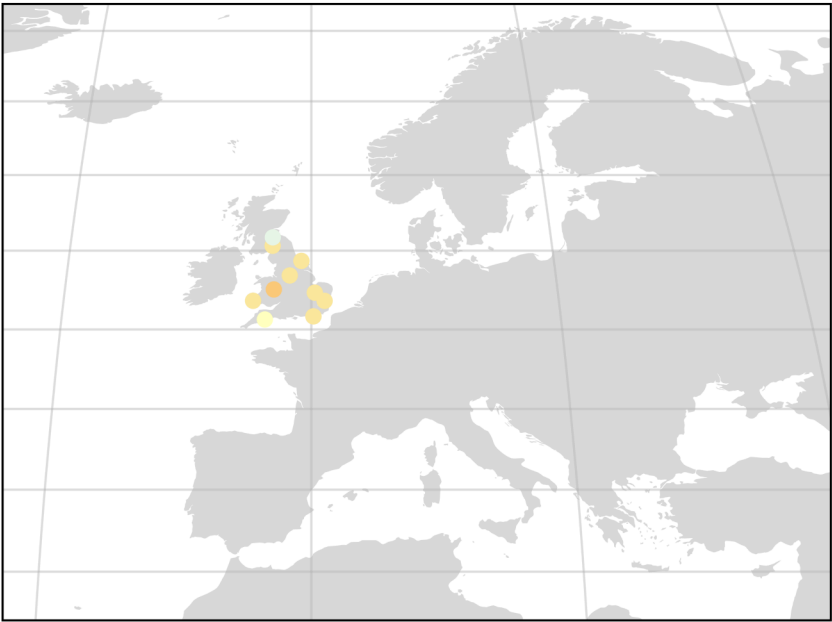


Map StdDev bias (Summary)

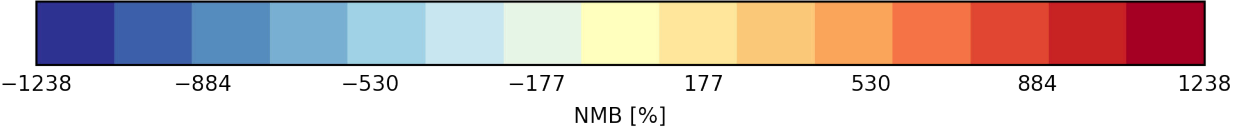
EBAS|sconcn02



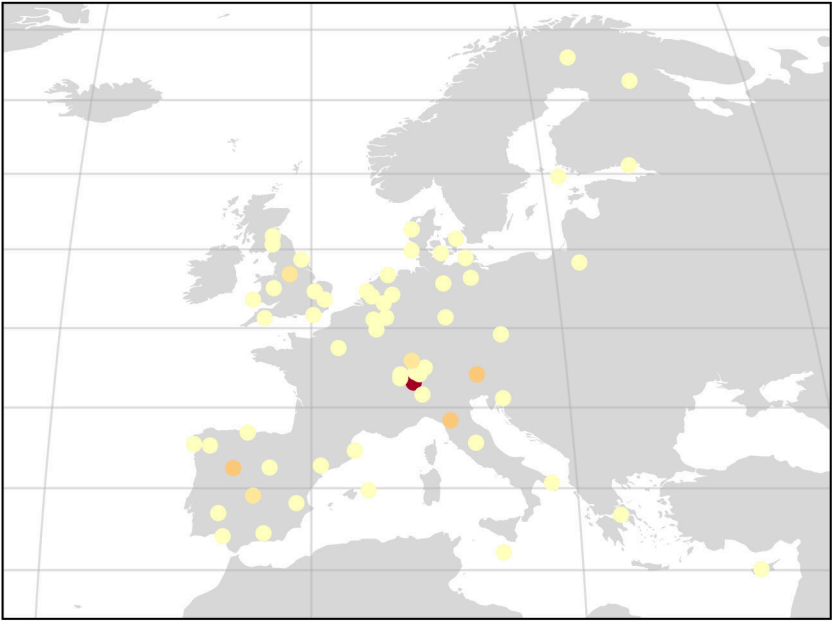
MONARCH
LOCAL·UK (10 stations)



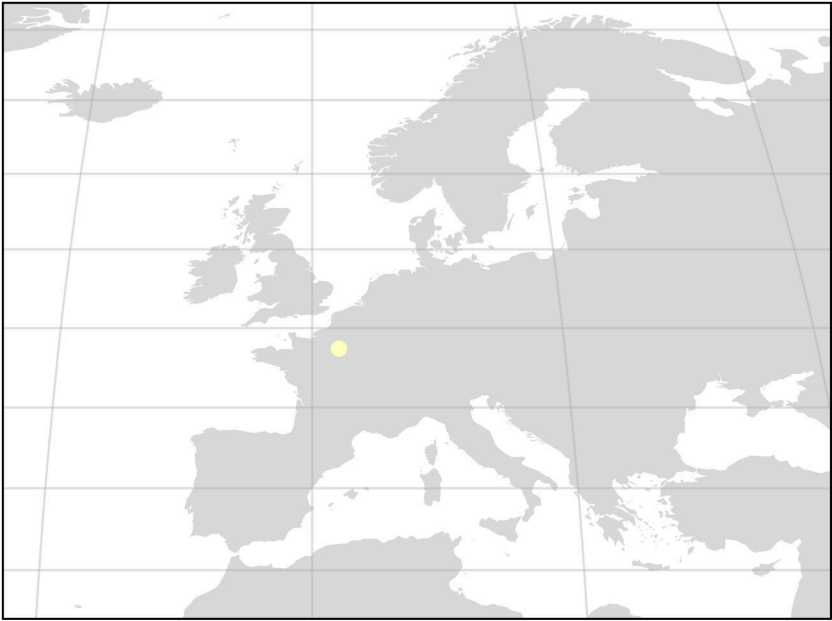
Map NMB (Summary)
EBAS|sconcn02



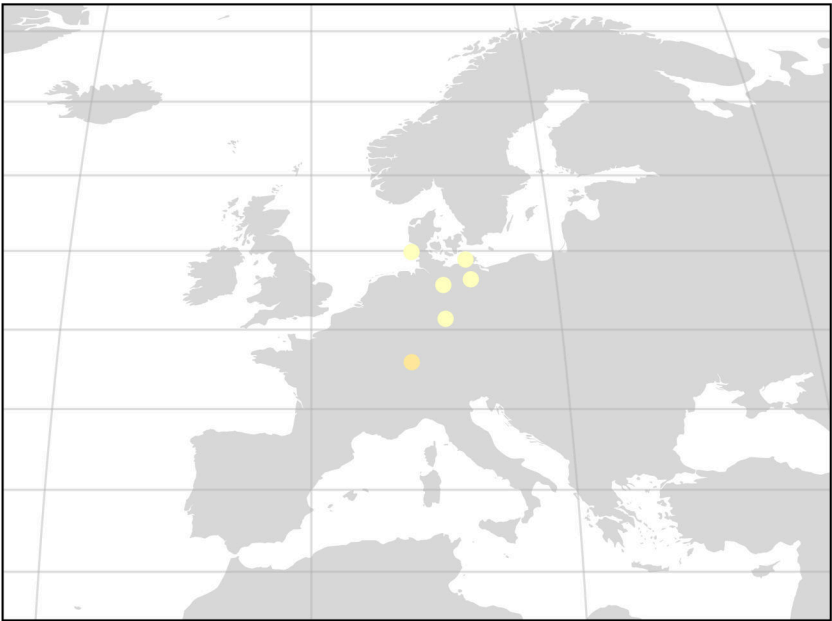
MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



MONARCH
LOCAL·Germany (6 stations)

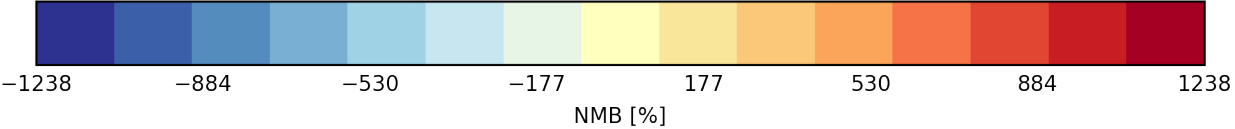


MONARCH
LOCAL·Spain (13 stations)

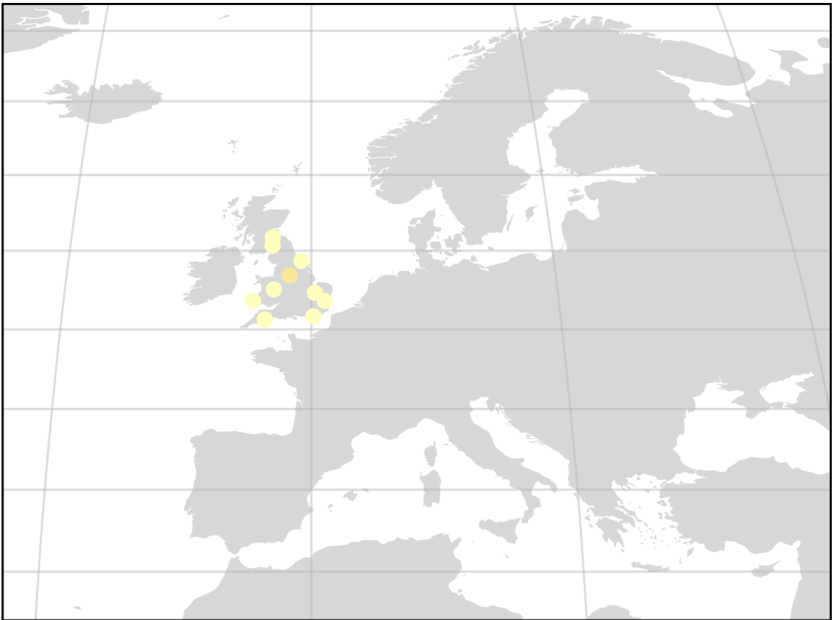


Map NMB (Summary)

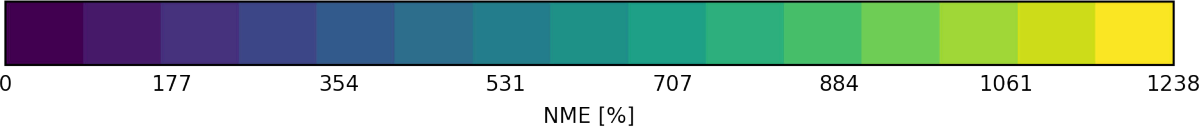
EBAS|sconcn02



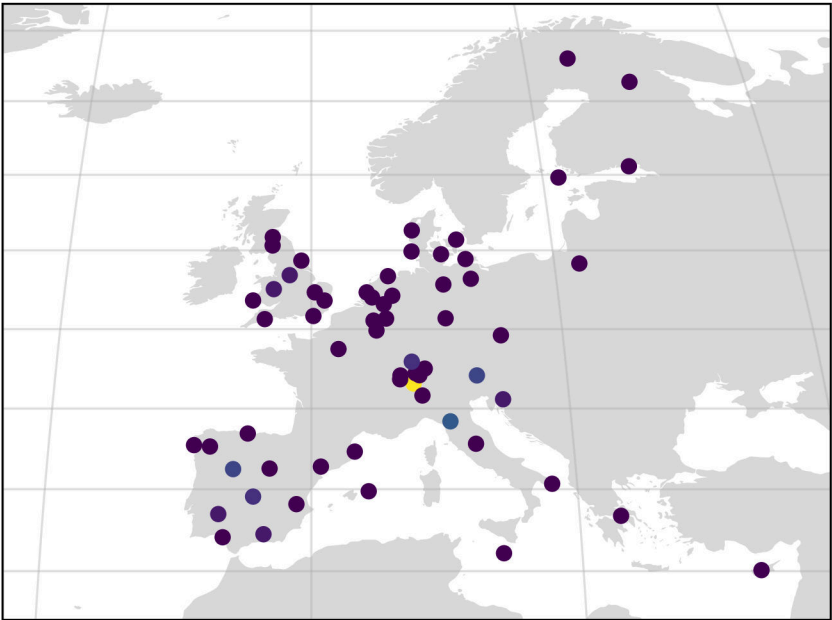
MONARCH
LOCAL·UK (10 stations)



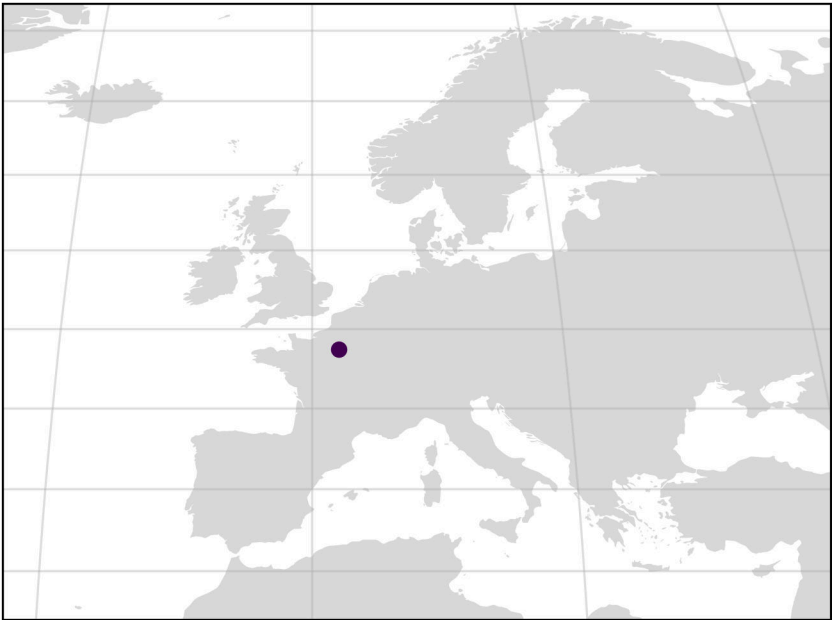
Map NME (Summary)
EBAS|sconcn02



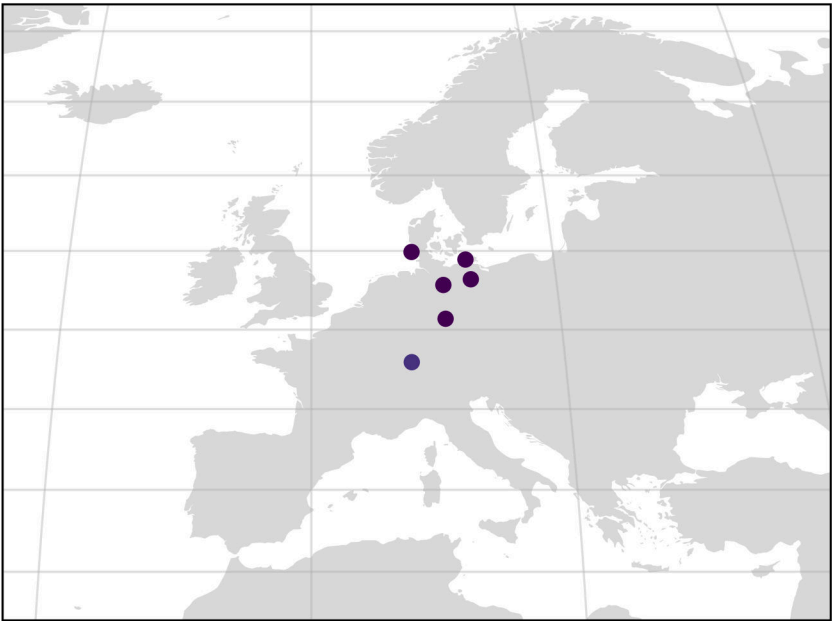
MONARCH
LOCAL·All (62 stations)



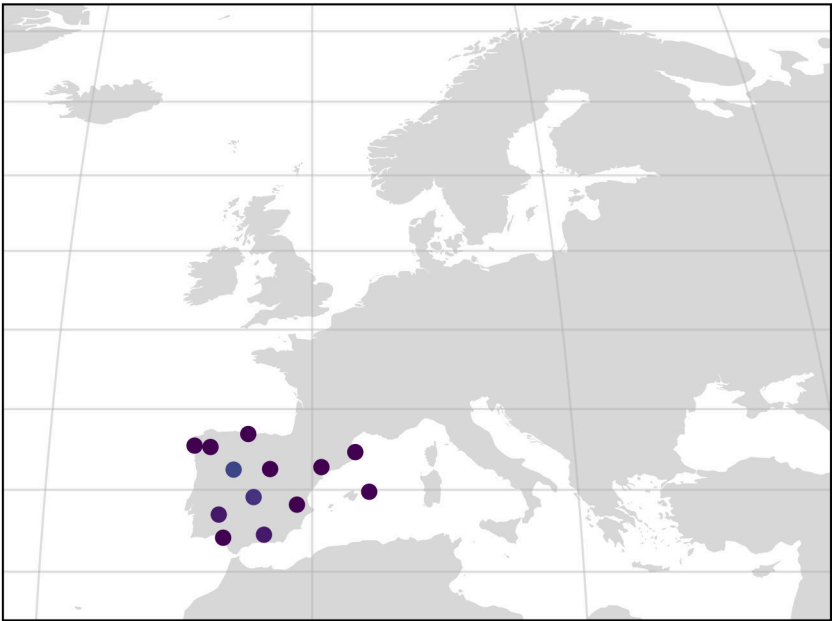
MONARCH
LOCAL·France (1 station)



MONARCH
LOCAL·Germany (6 stations)

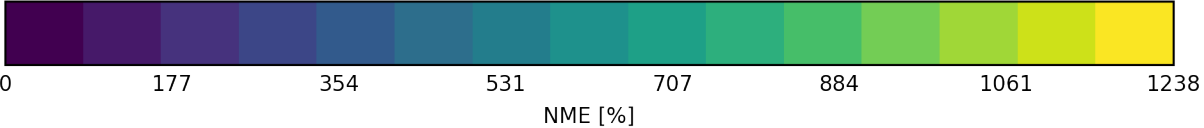


MONARCH
LOCAL·Spain (13 stations)

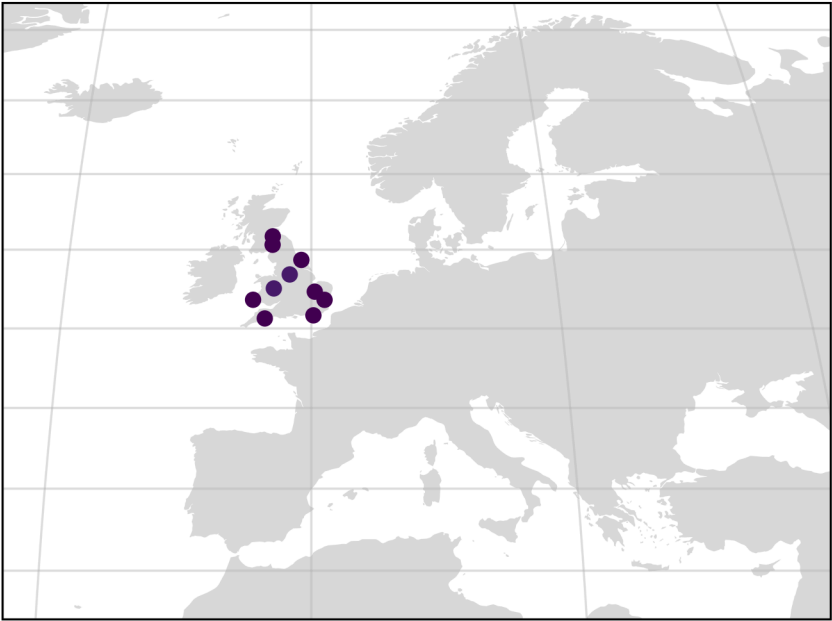


Map NME (Summary)

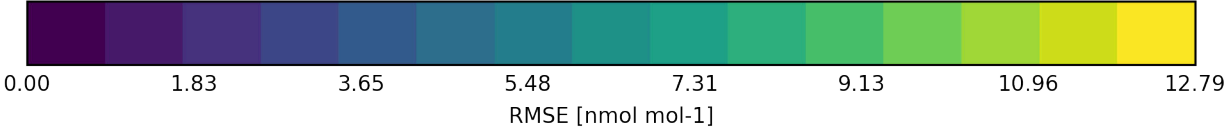
EBAS|sconcn02



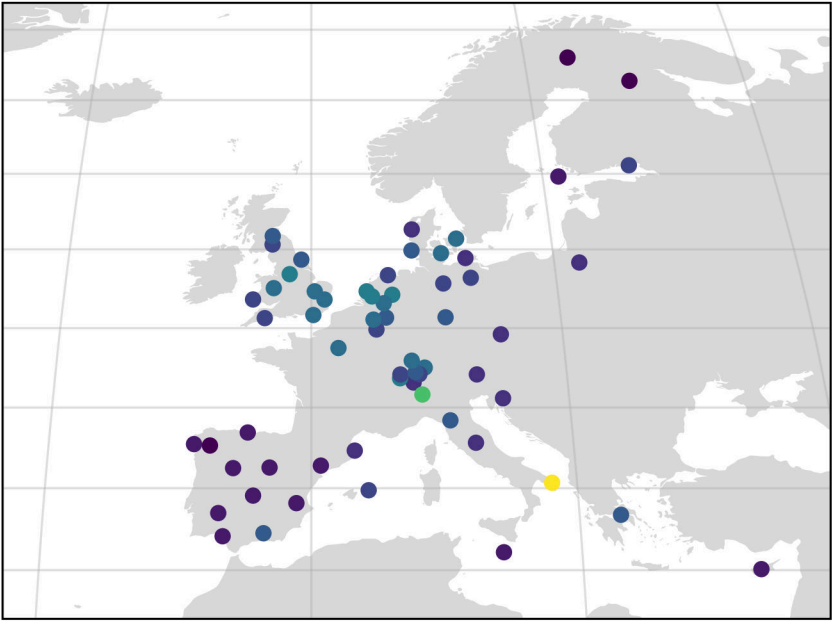
MONARCH
LOCAL·UK (10 stations)



Map RMSE (Summary)
EBAS|sconcn02



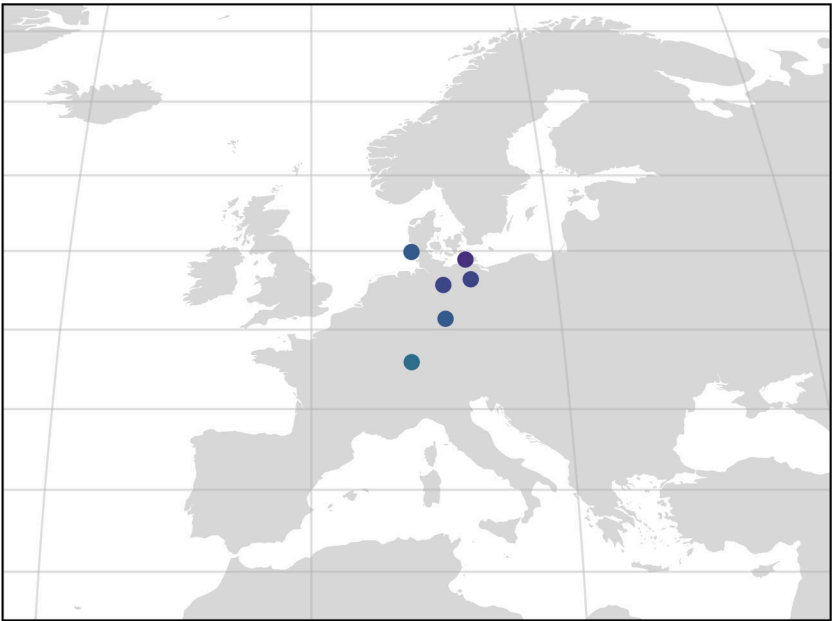
MONARCH
LOCAL·All (62 stations)



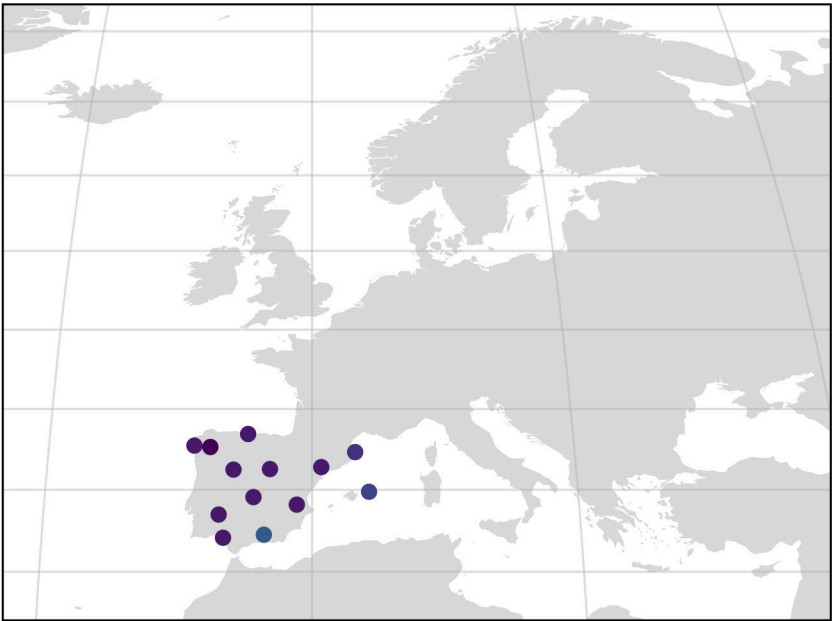
MONARCH
LOCAL·France (1 station)



MONARCH
LOCAL·Germany (6 stations)

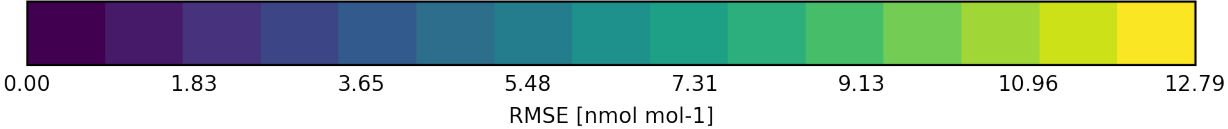


MONARCH
LOCAL·Spain (13 stations)

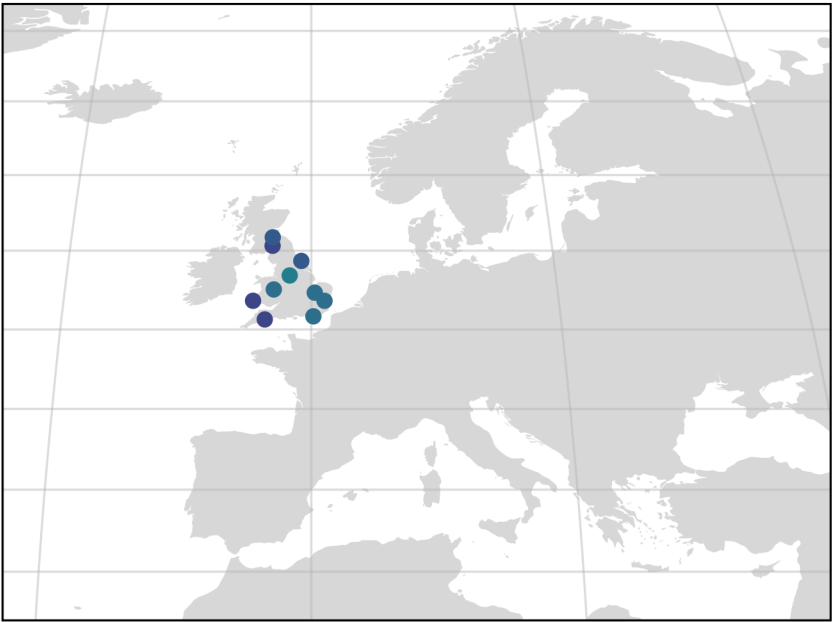


Map RMSE (Summary)

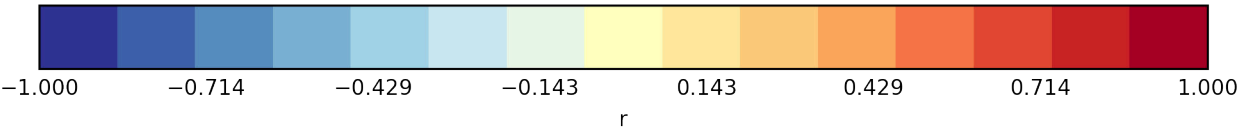
EBAS|sconcnO2



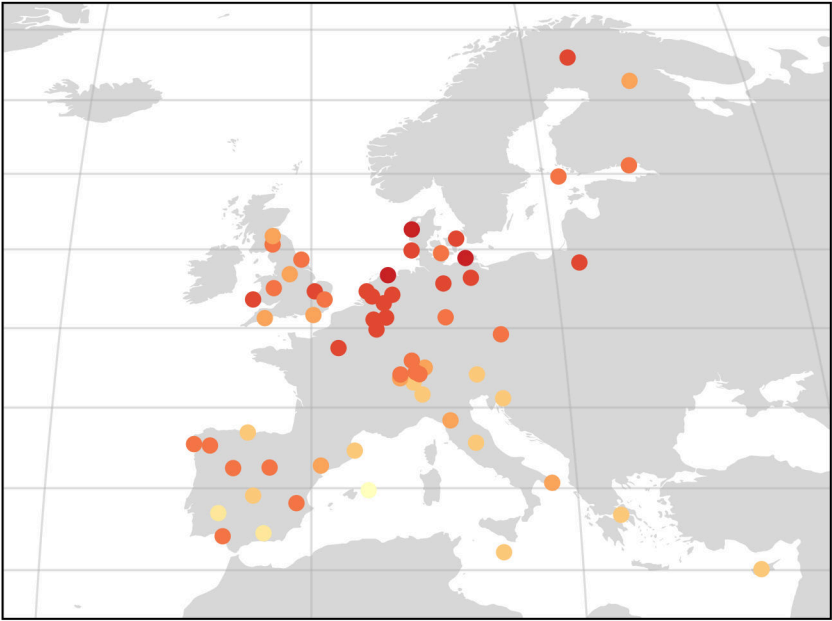
MONARCH
LOCAL·UK (10 stations)



Map r (Summary)
EBAS|sconcn02



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



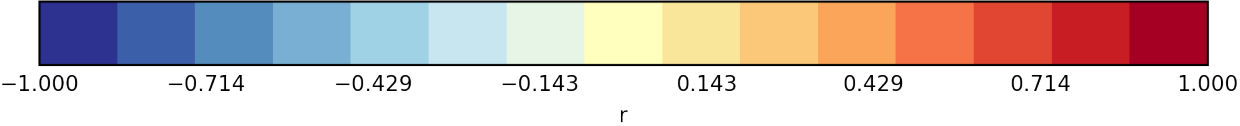
MONARCH
LOCAL·Germany (6 stations)



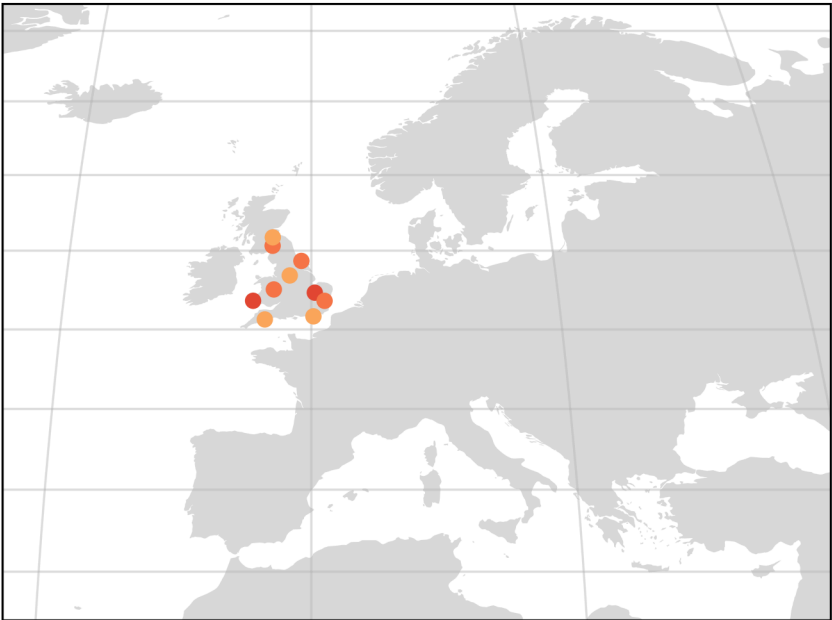
MONARCH
LOCAL·Spain (13 stations)

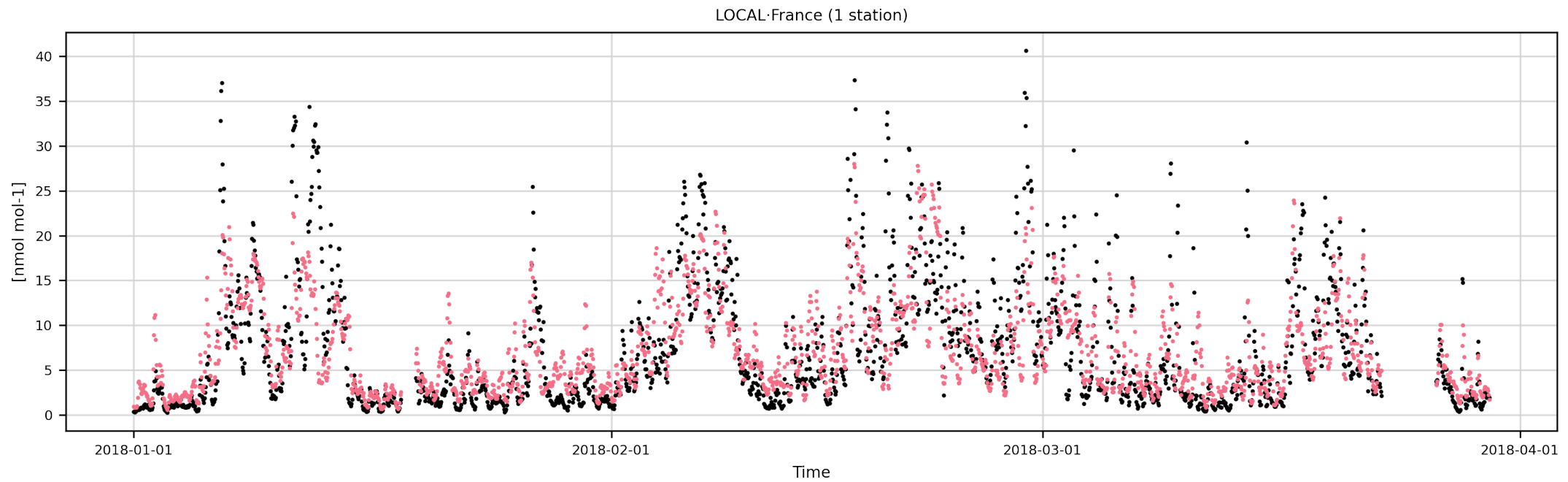
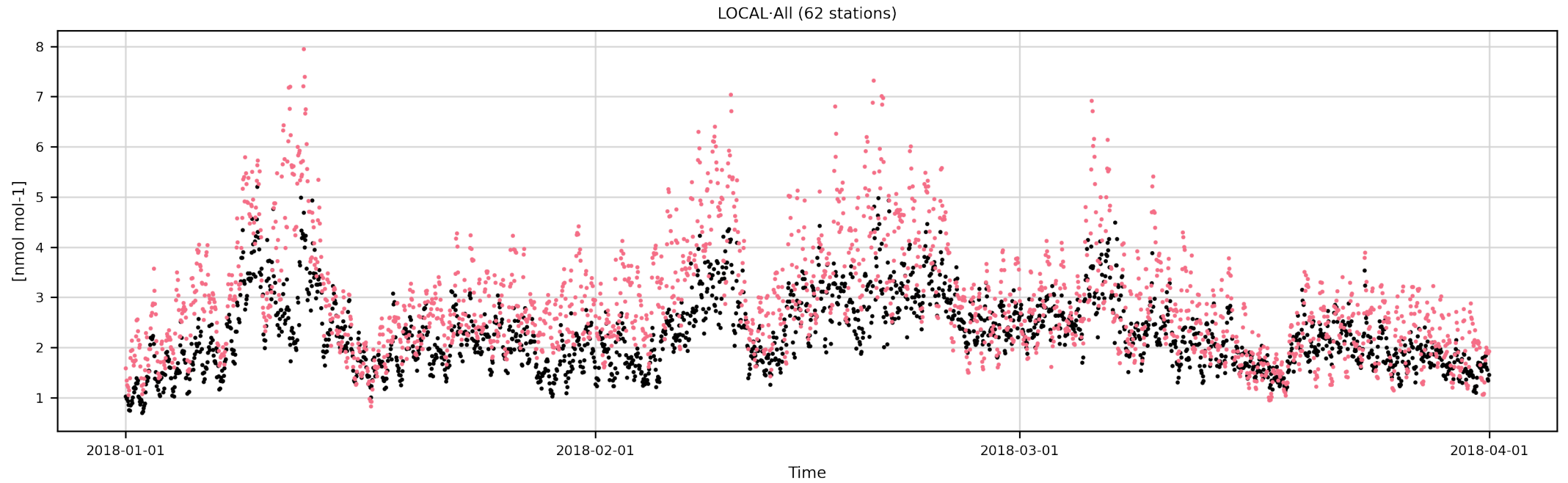


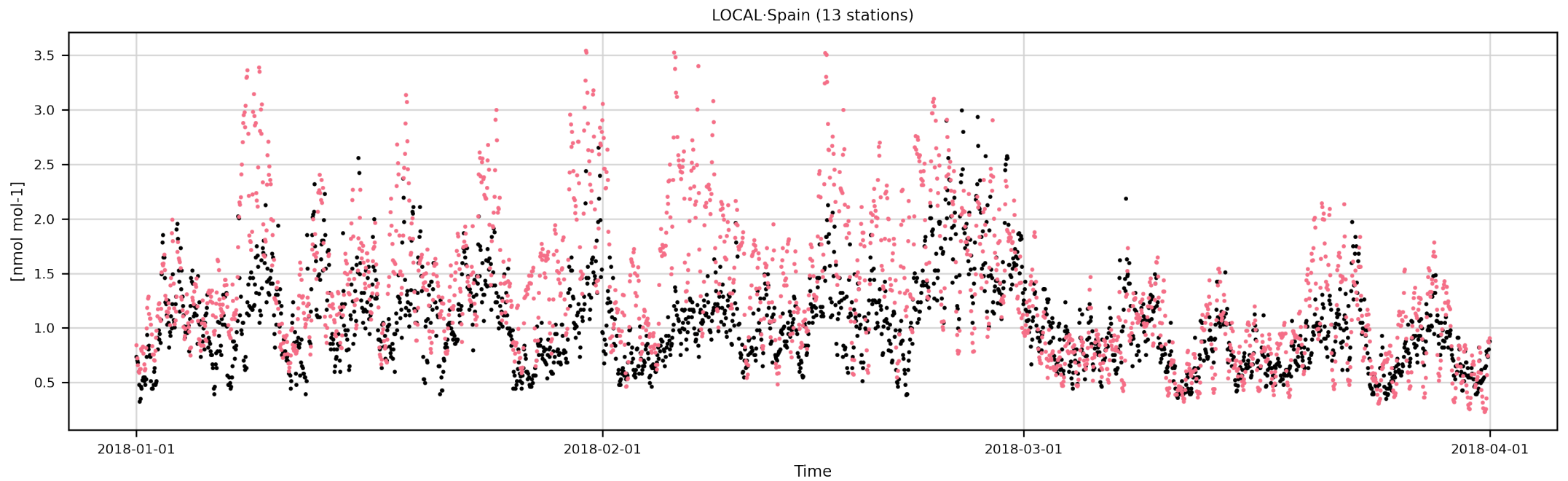
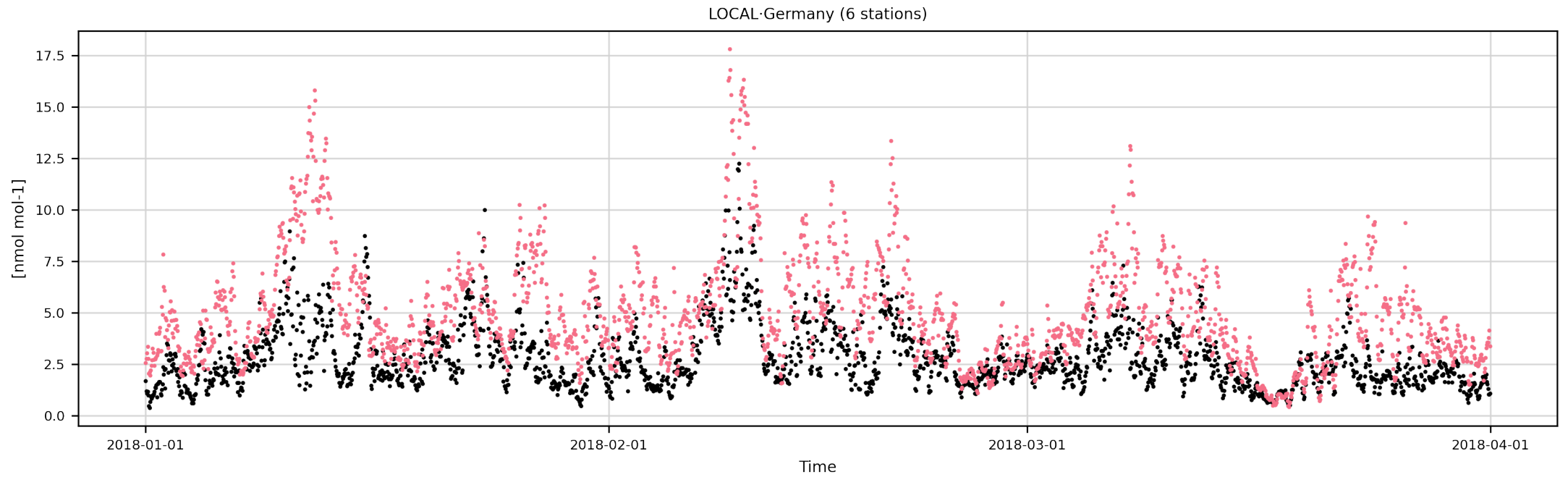
Map r (Summary)
EBAS|sconcn02

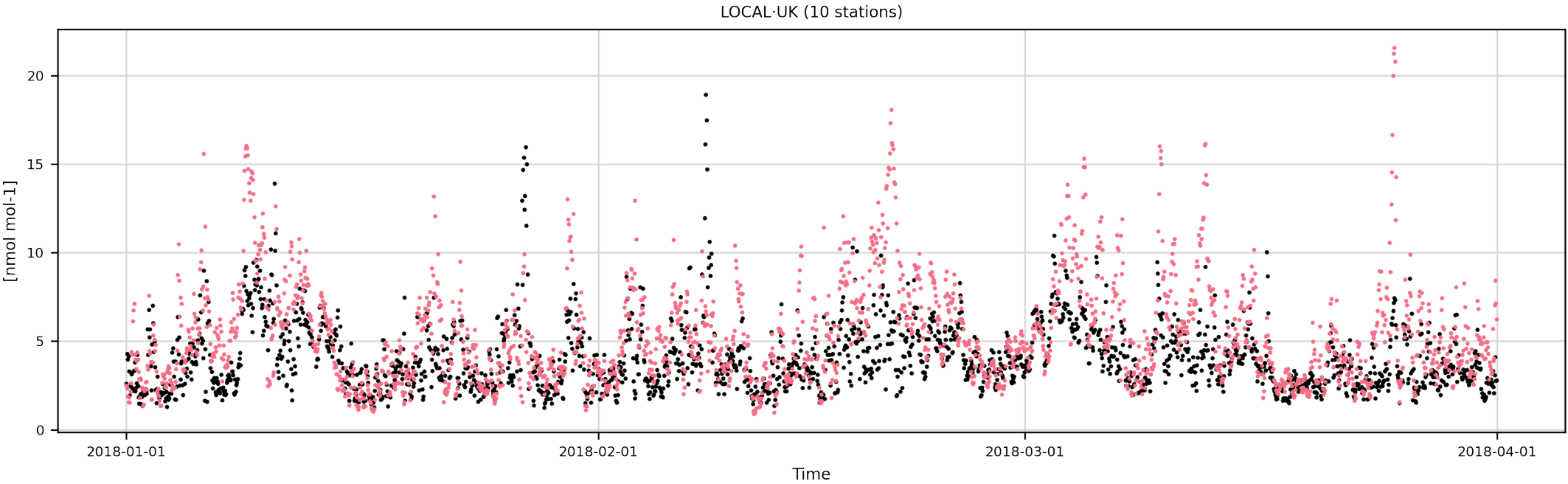


MONARCH
LOCAL·UK (10 stations)





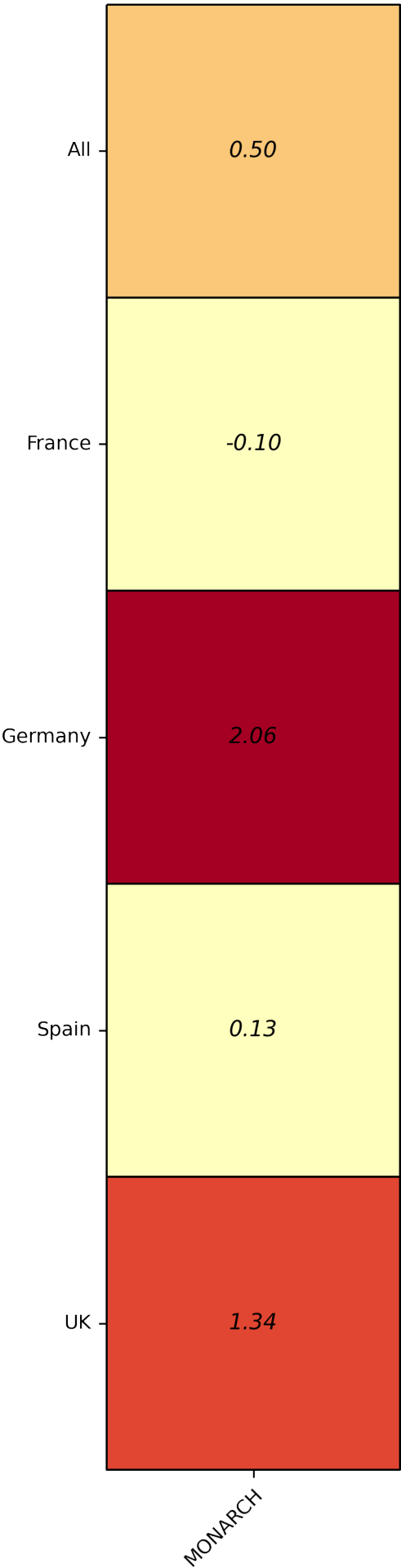
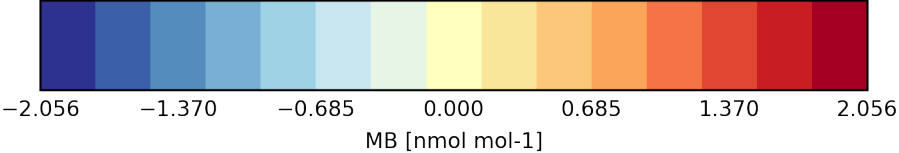




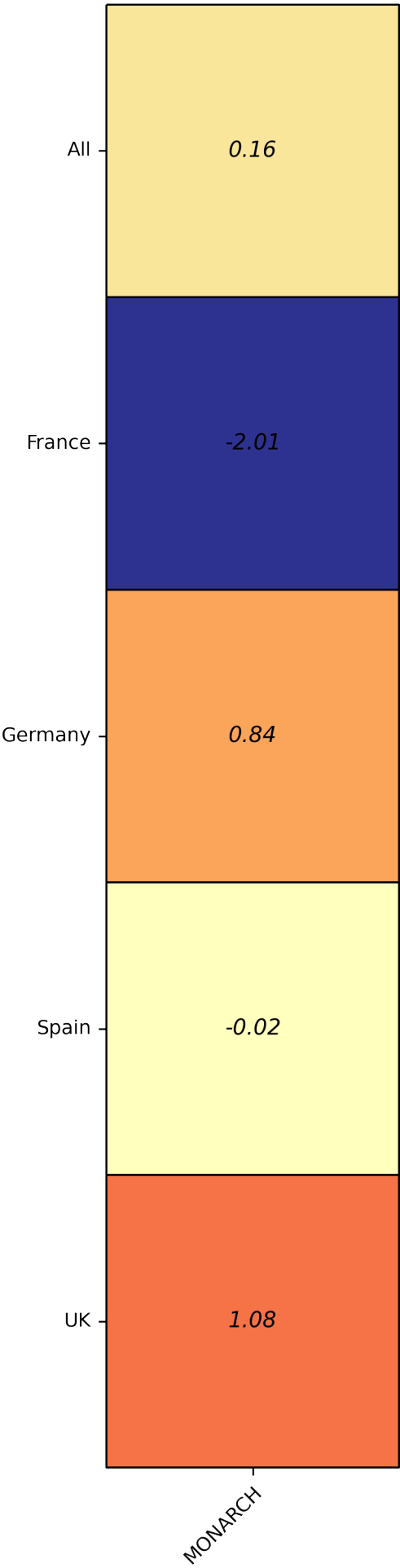
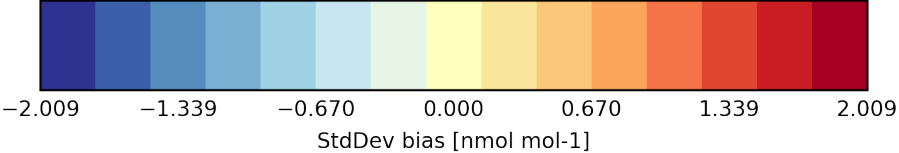
		Mean	StdDev	p5	Median	p95
All	observations	3.16	2.78	0.72	2.39	8.35
	MONARCH	3.71	2.86	0.87	3.00	9.38
France	observations	7.31	7.06	0.76	4.60	22.01
	MONARCH	7.21	5.06	1.67	5.63	17.38
Germany	observations	3.05	2.28	0.55	2.48	7.63
	MONARCH	5.13	3.25	1.33	4.20	11.79
Spain	observations	1.28	1.04	0.33	1.03	3.07
	MONARCH	1.37	1.01	0.38	1.11	3.33
UK	observations	4.39	3.18	1.34	3.26	10.54
	MONARCH	5.92	4.20	1.22	4.71	14.74

		Mean_bias	StdDev_bias	NMB	NME	RMSE	r
All	MONARCH	0.50	0.16	15.13	57.89	3.03	0.48
France	MONARCH	-0.10	-2.01	-1.33	43.36	4.84	0.73
Germany	MONARCH	2.06	0.84	62.05	71.81	3.35	0.61
Spain	MONARCH	0.13	-0.02	9.48	59.86	1.36	0.42
UK	MONARCH	1.34	1.08	30.46	58.78	4.13	0.48

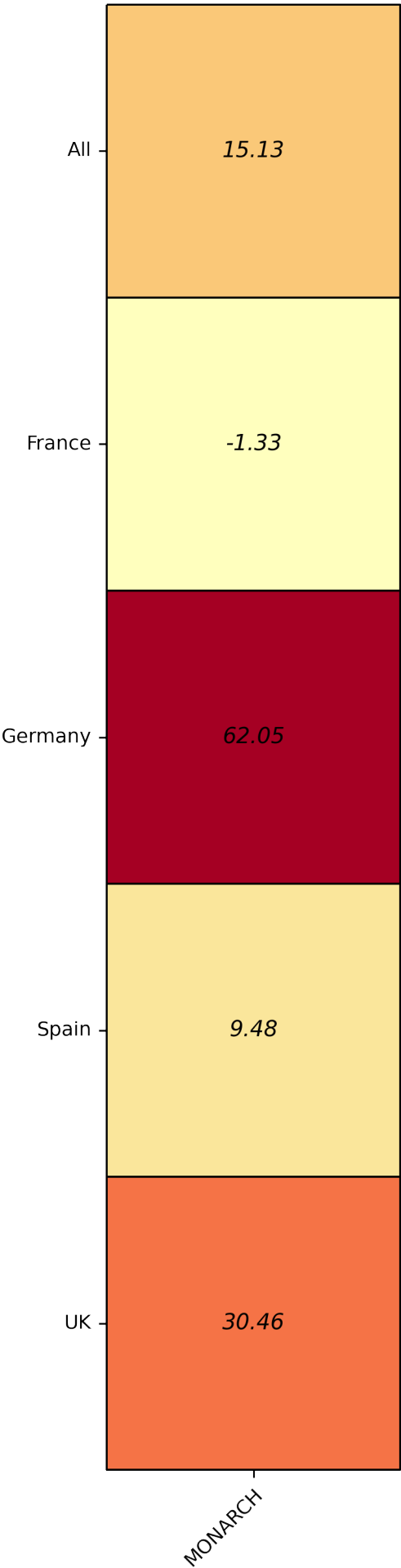
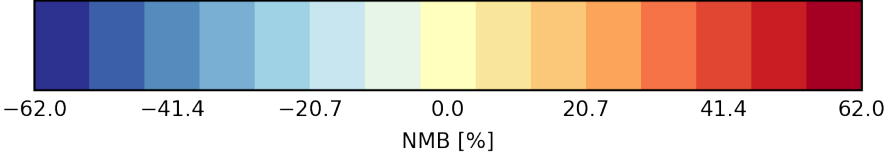
Heatmap MB (Summary)
EBAS|sconcnO2



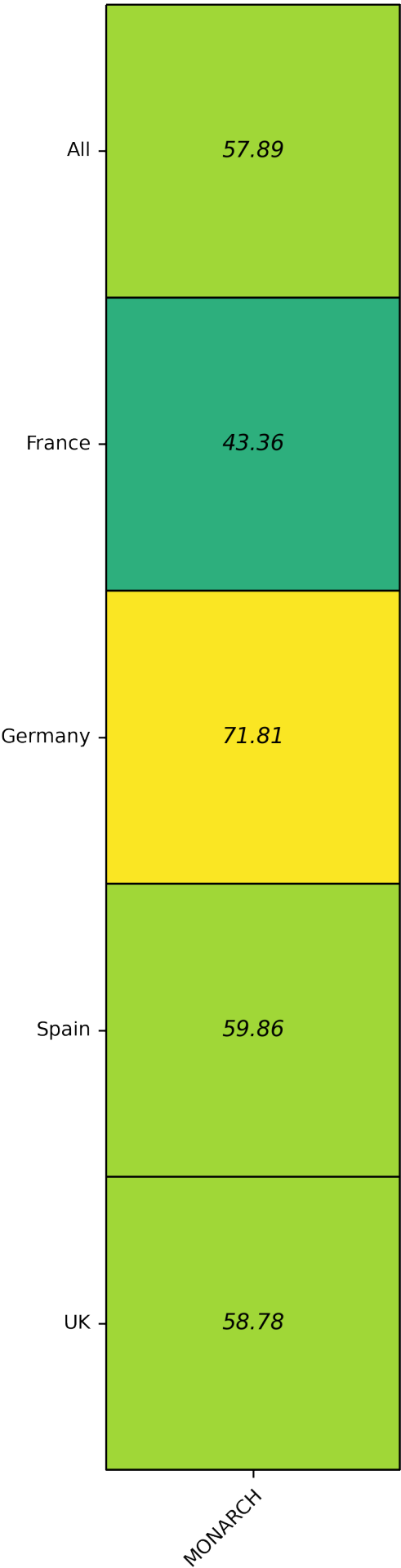
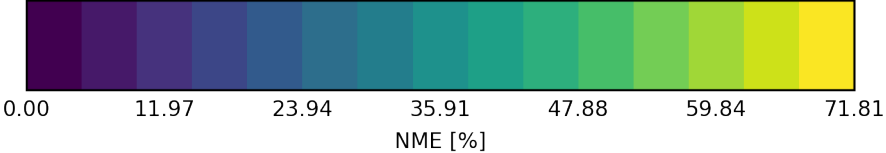
Heatmap StdDev bias (Summary)
EBAS|sconcnO2



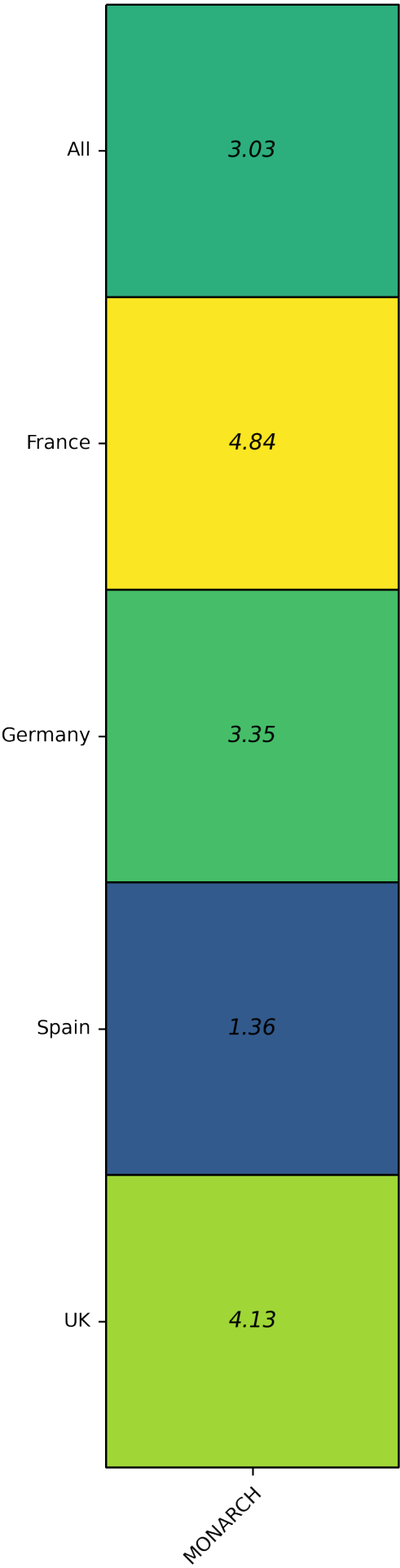
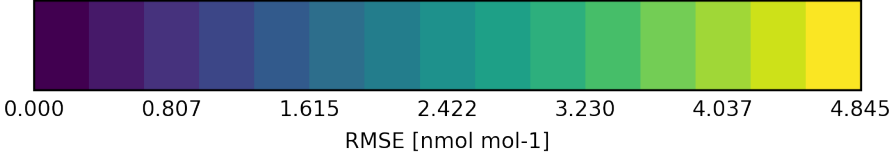
Heatmap NMB (Summary)
EBAS|sconcn02



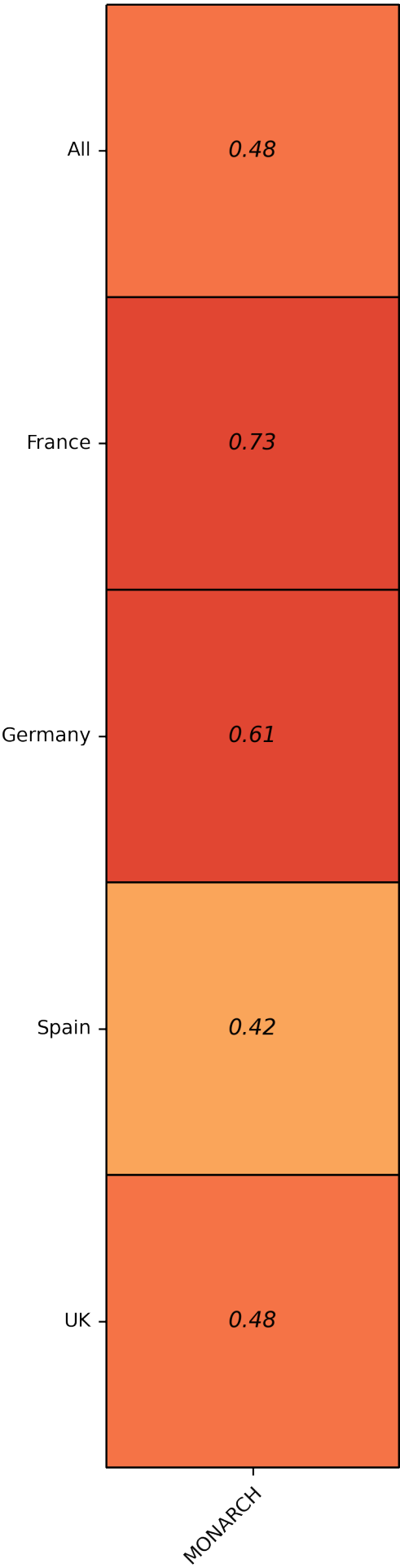
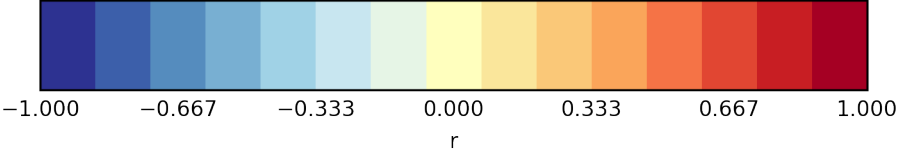
Heatmap NME (Summary)
EBAS|sconcn02



Heatmap RMSE (Summary)
EBAS|sconcnO2



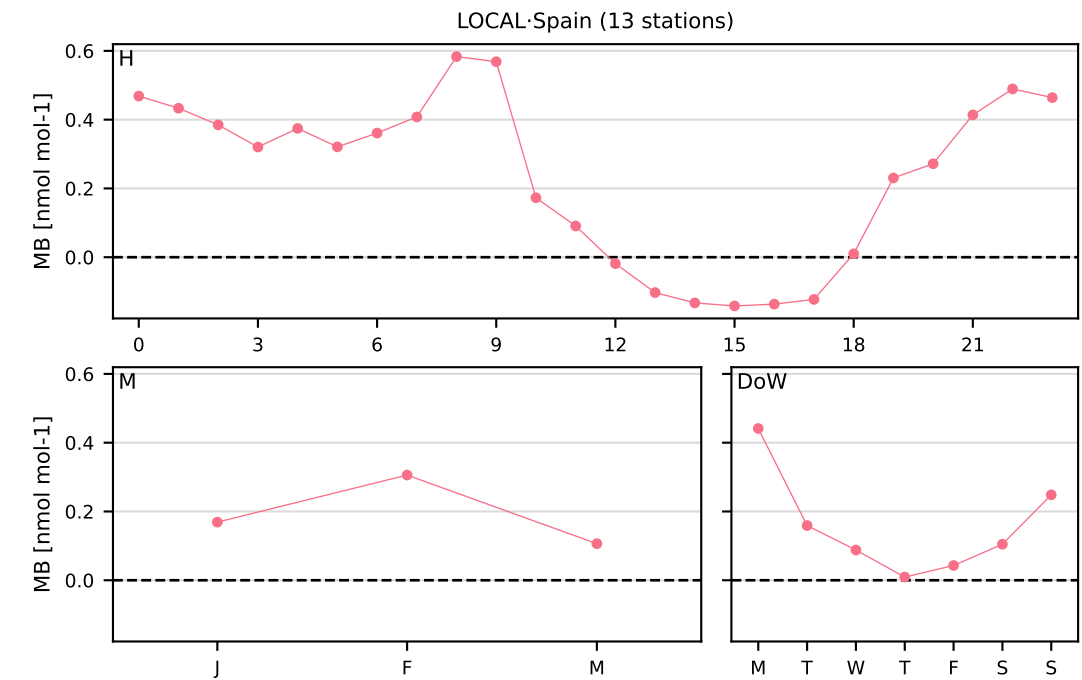
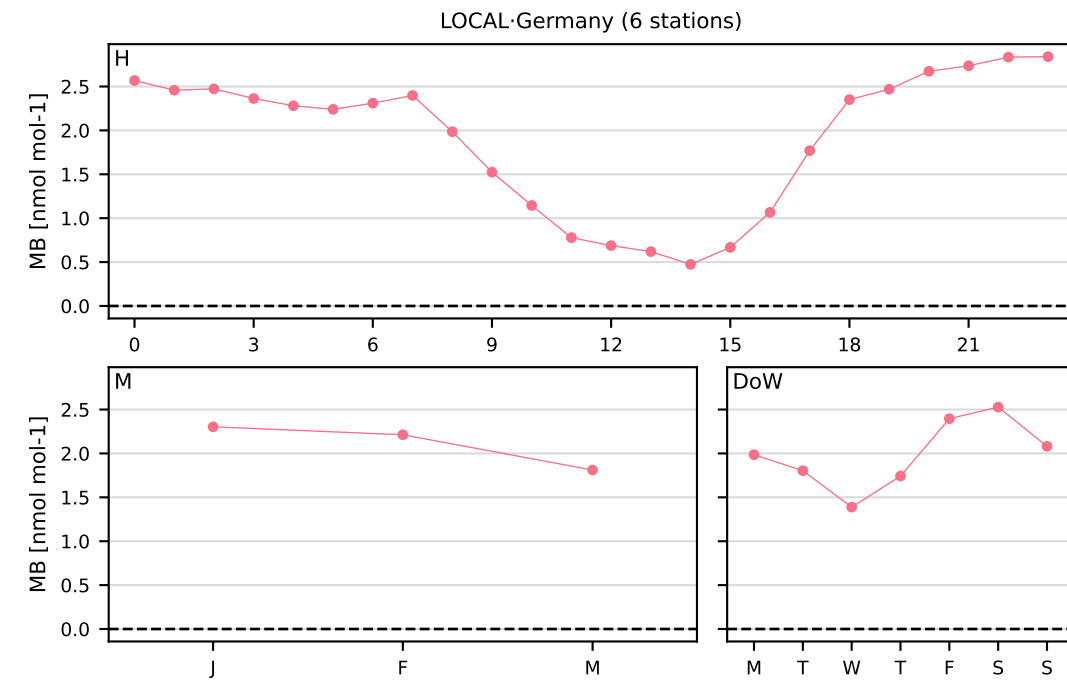
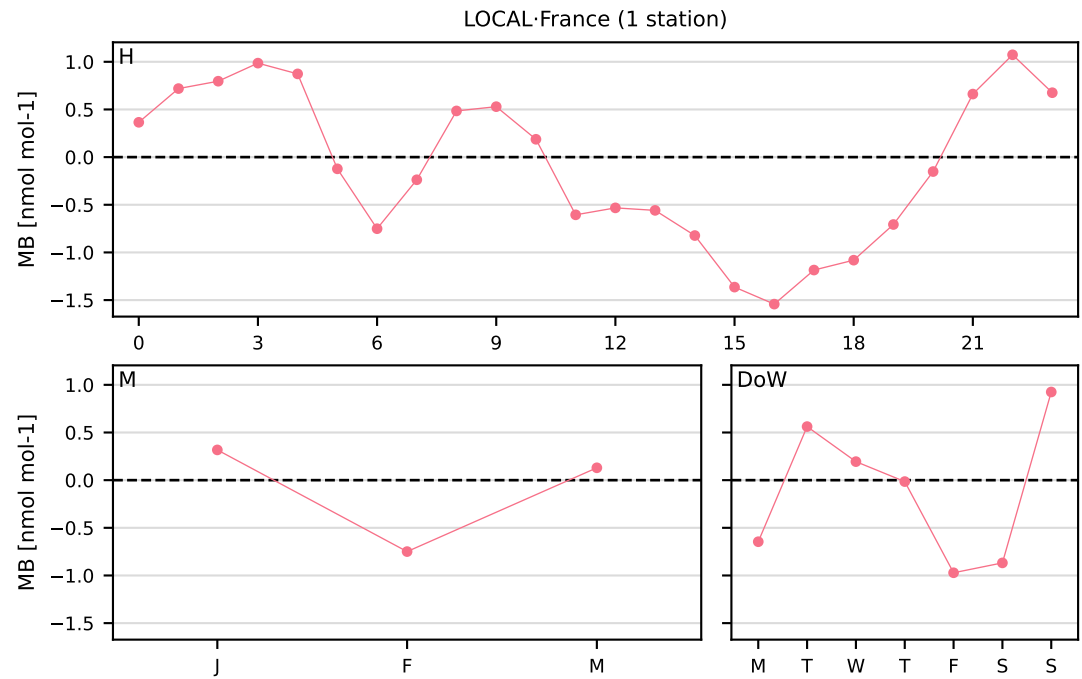
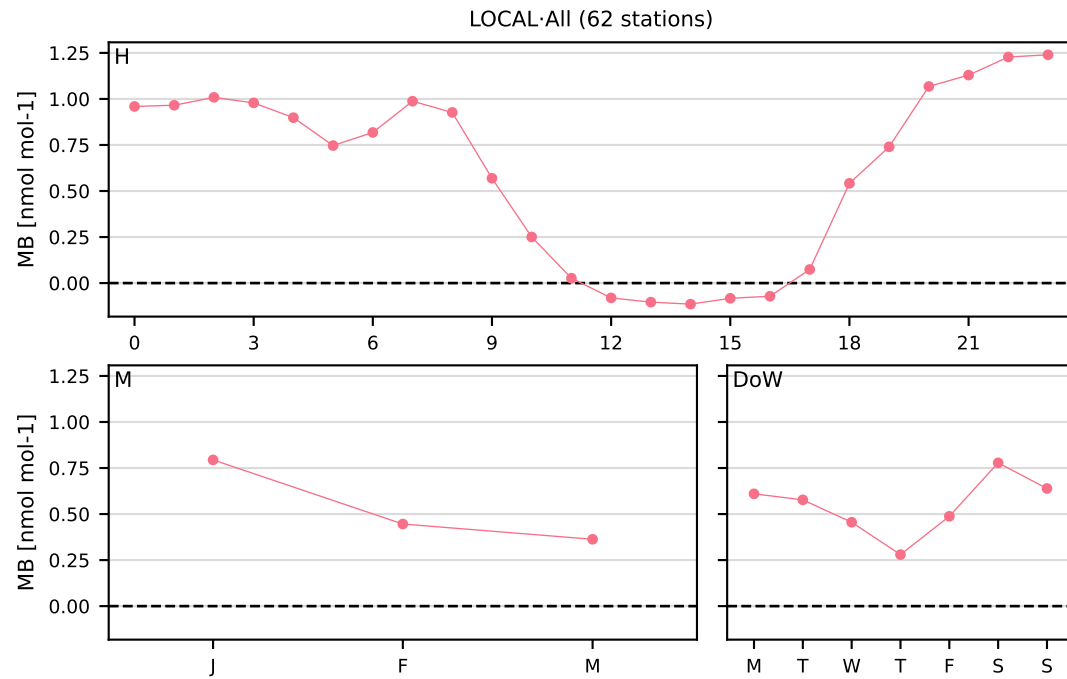
Heatmap r (Summary)
EBAS|sconcn02



Periodic MB (Summary)

EBAS|sconcn02

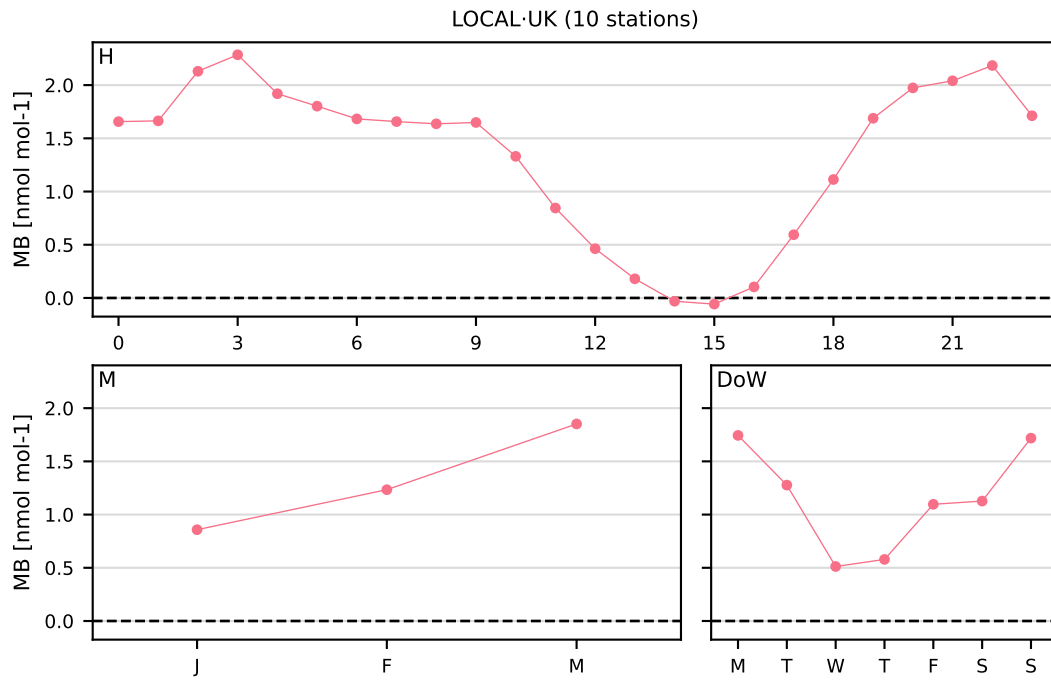
MONARCH



Periodic MB (Summary)

EBAS|sconcn02

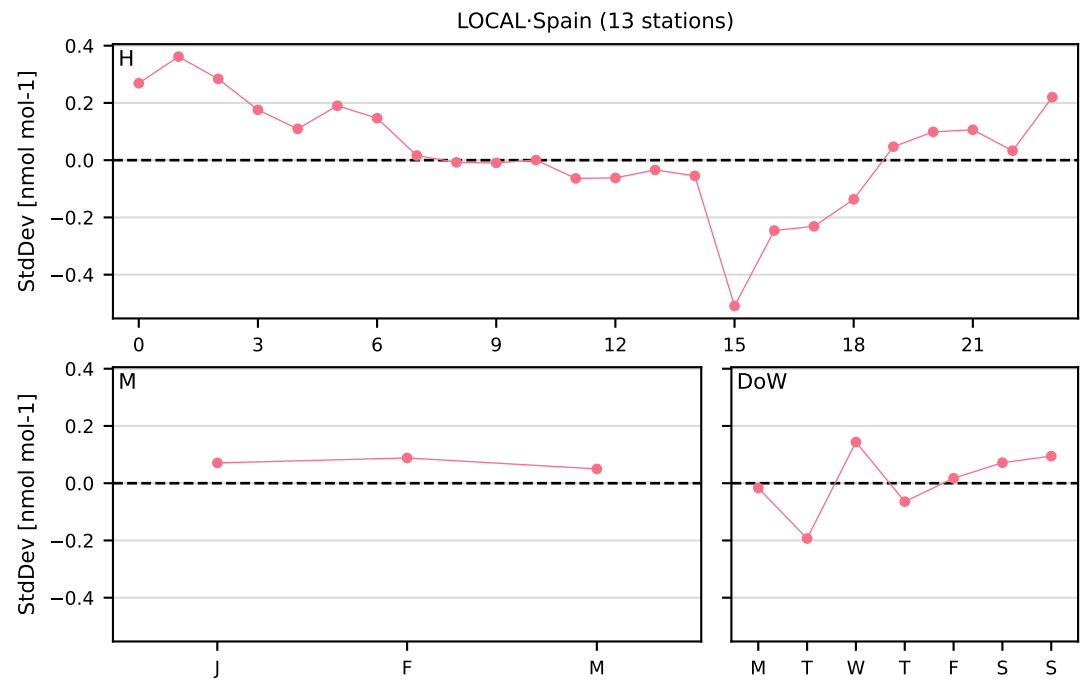
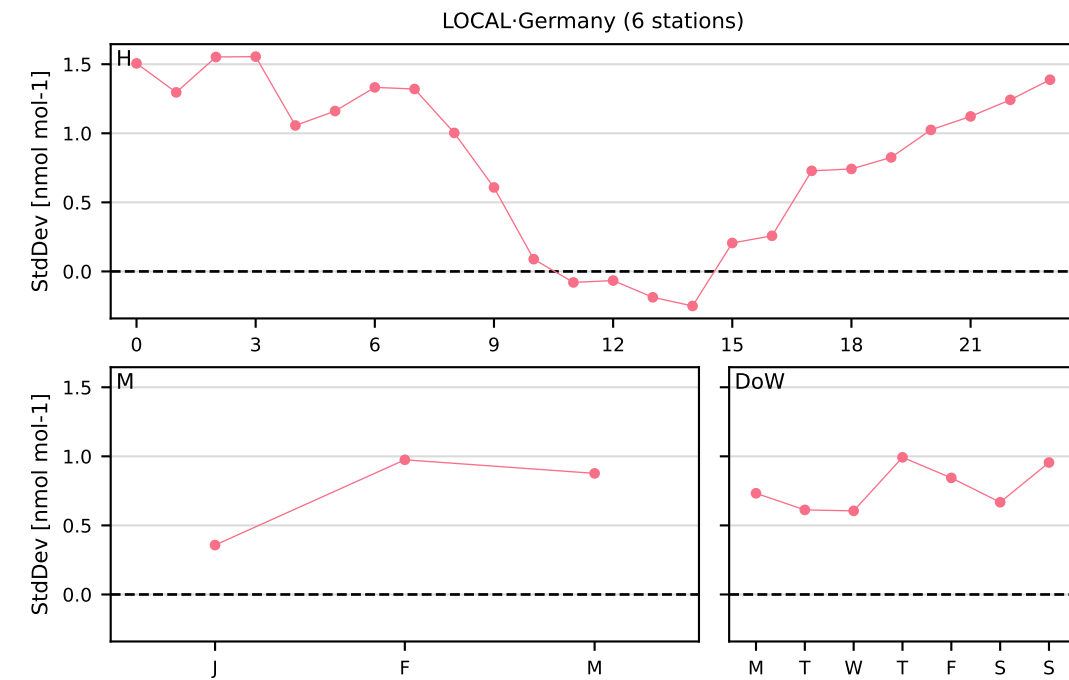
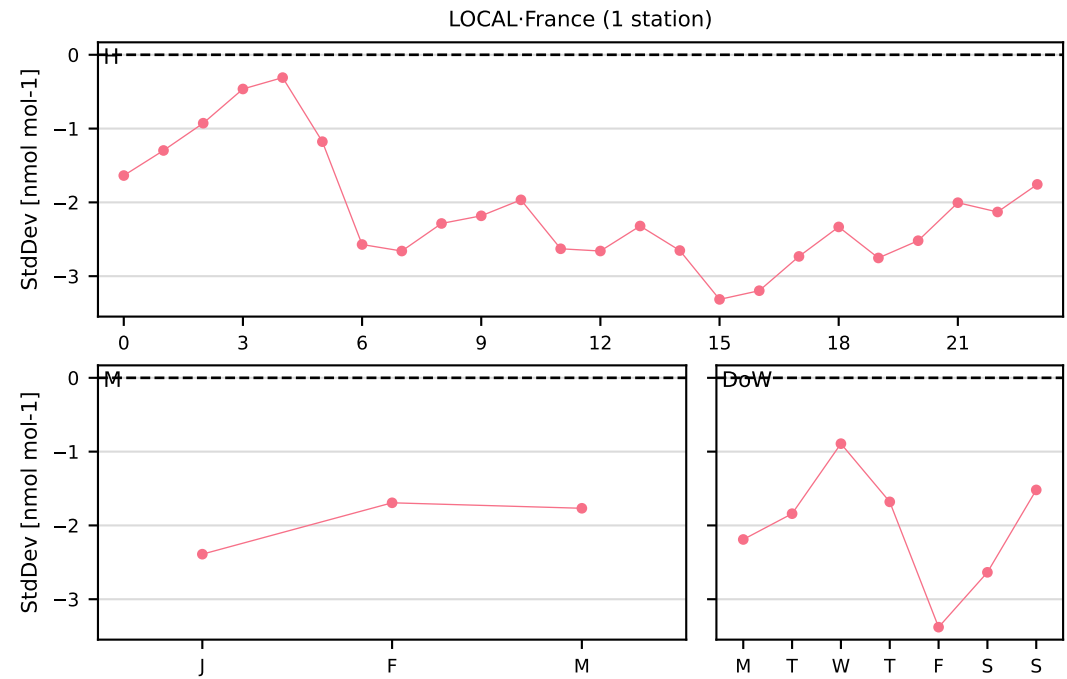
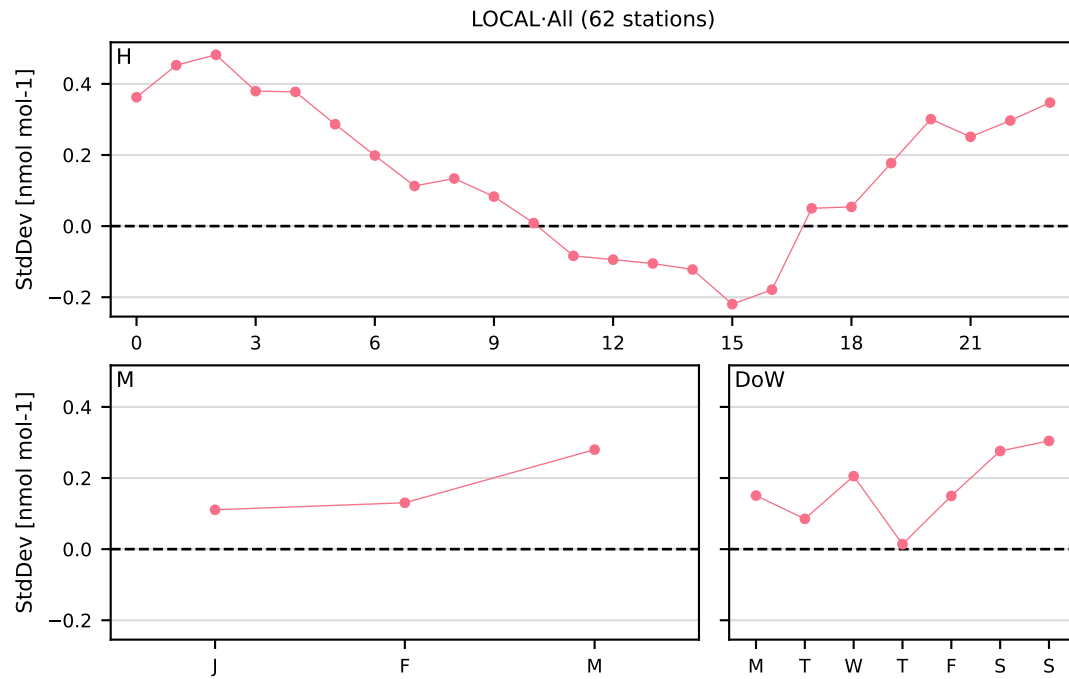
● MONARCH



Periodic StdDev bias (Summary)

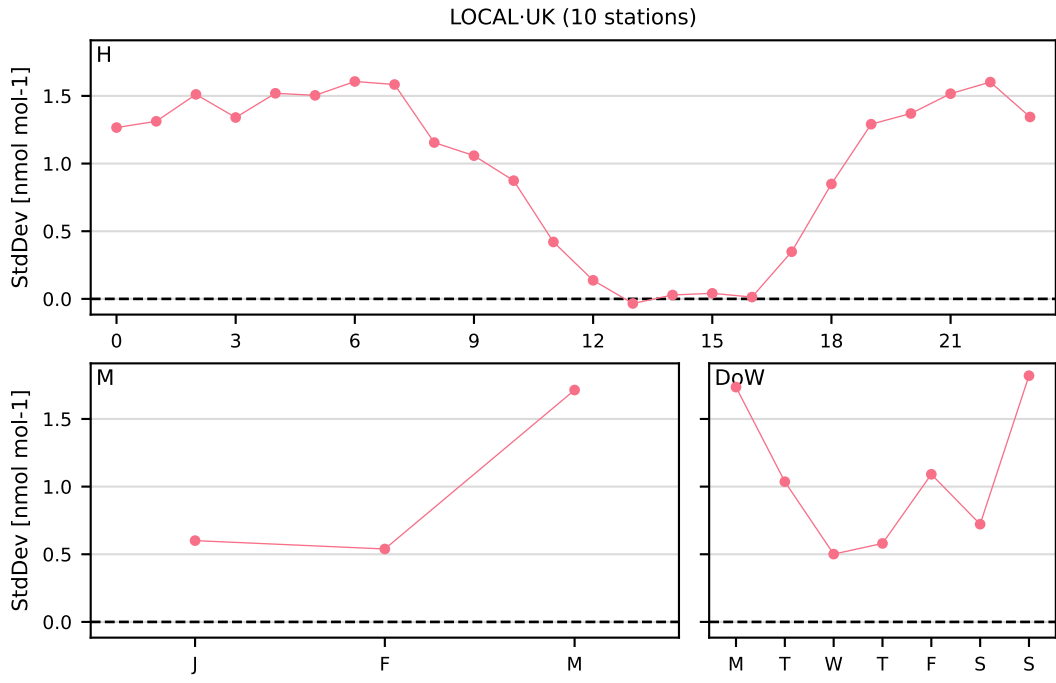
EBAS|sconcn02

MONARCH



Periodic StdDev bias (Summary)
EBAS|sconcn02

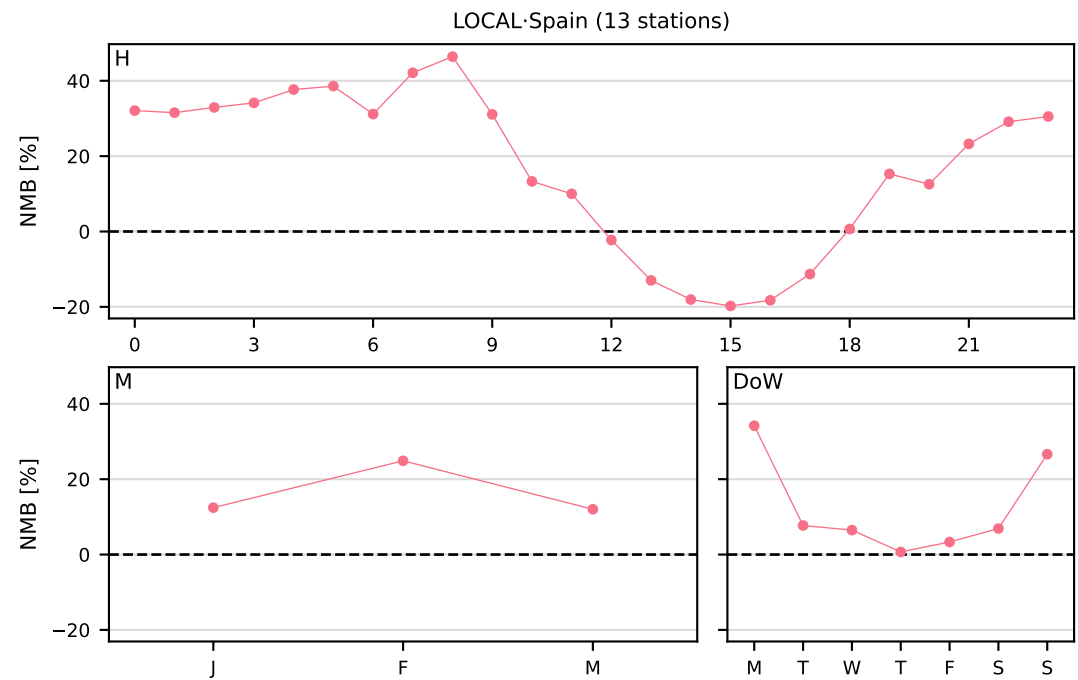
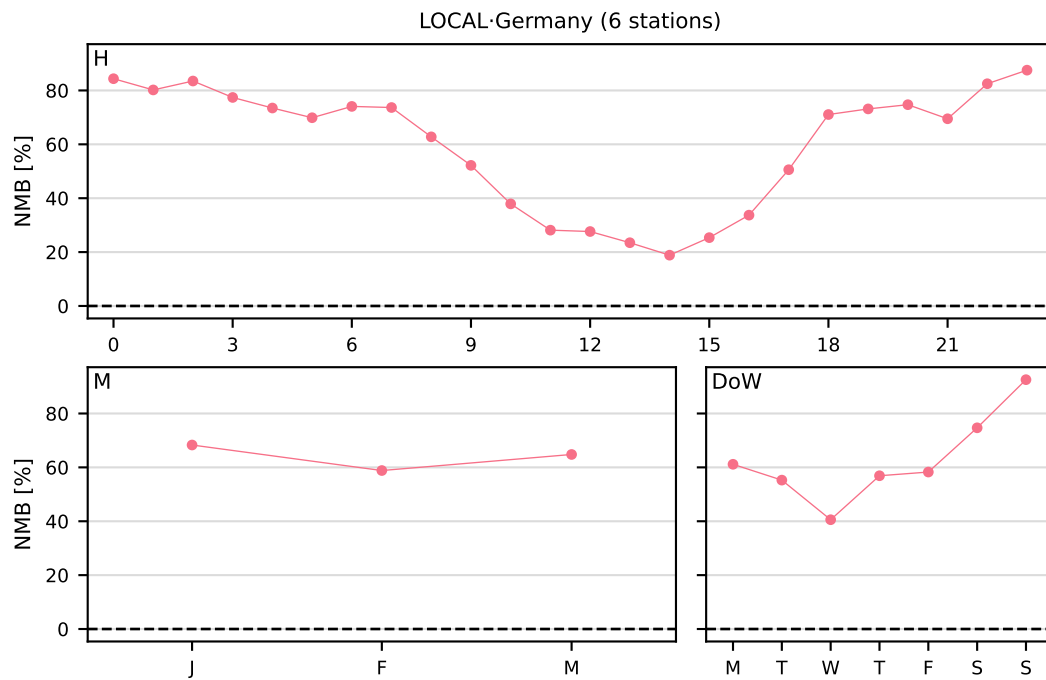
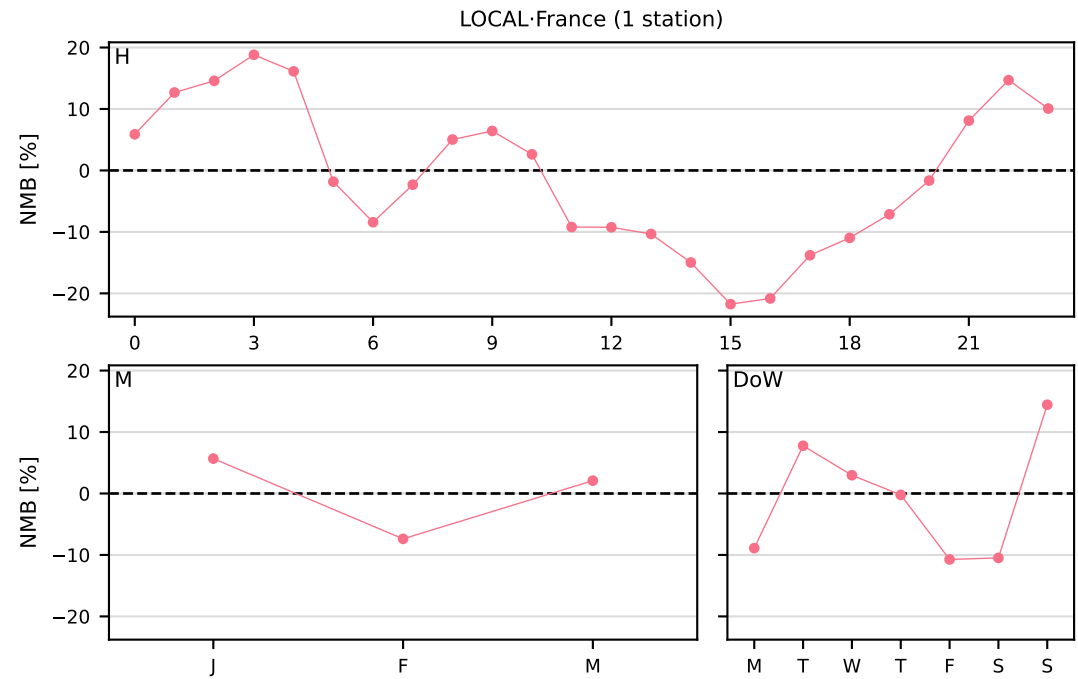
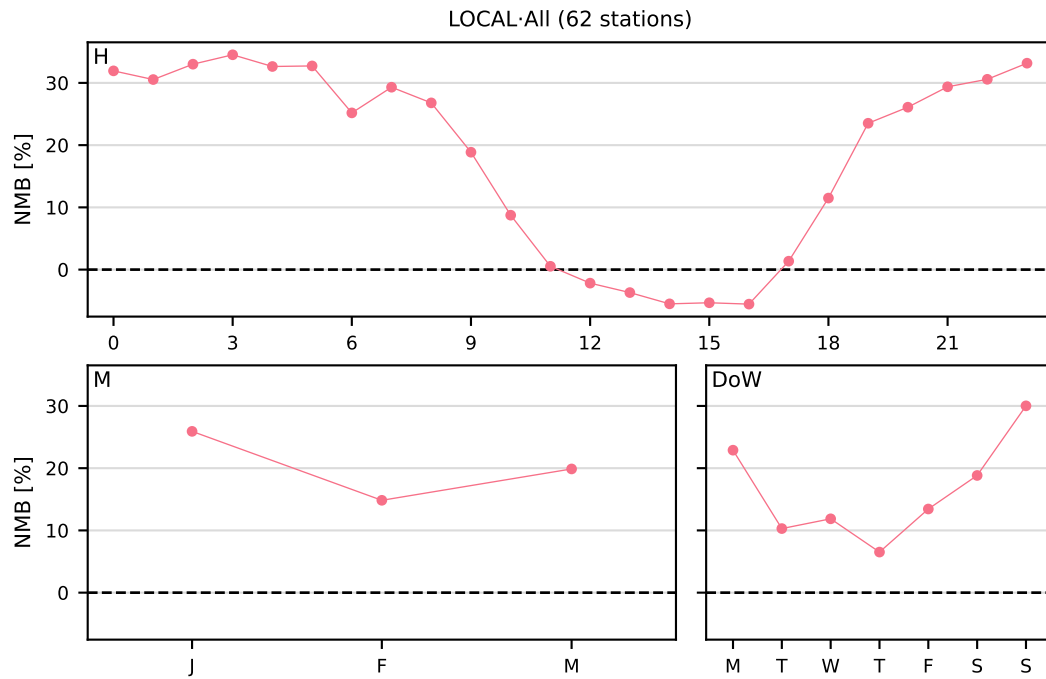
● MONARCH



Periodic NMB (Summary)

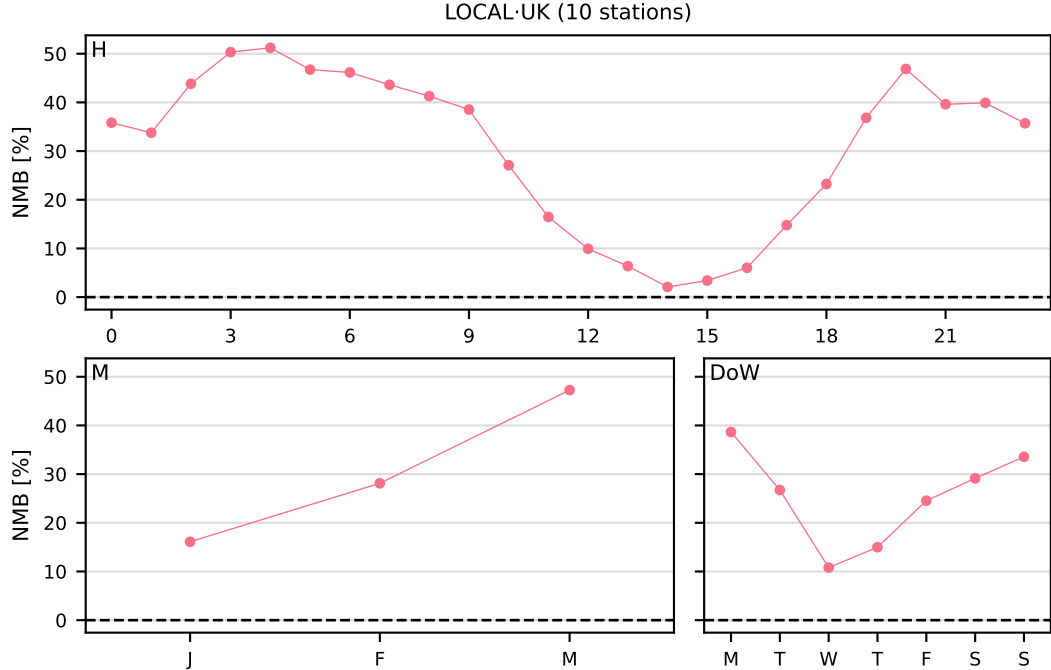
EBAS|sconcn02

MONARCH



Periodic NMB (Summary)
EBAS|sconcn02

● MONARCH

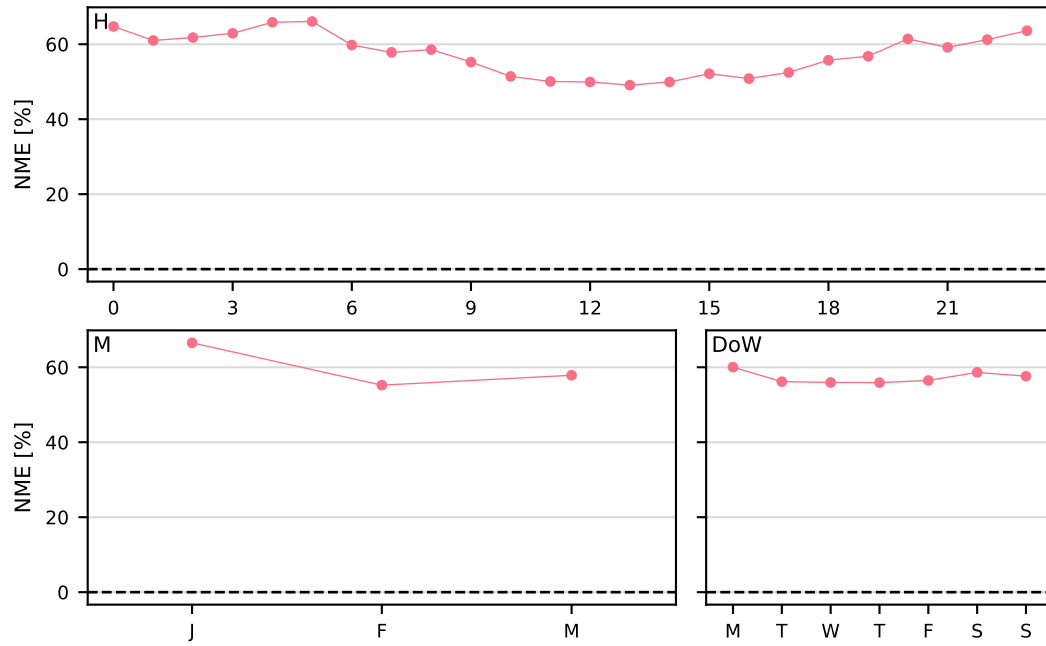


Periodic NME (Summary)

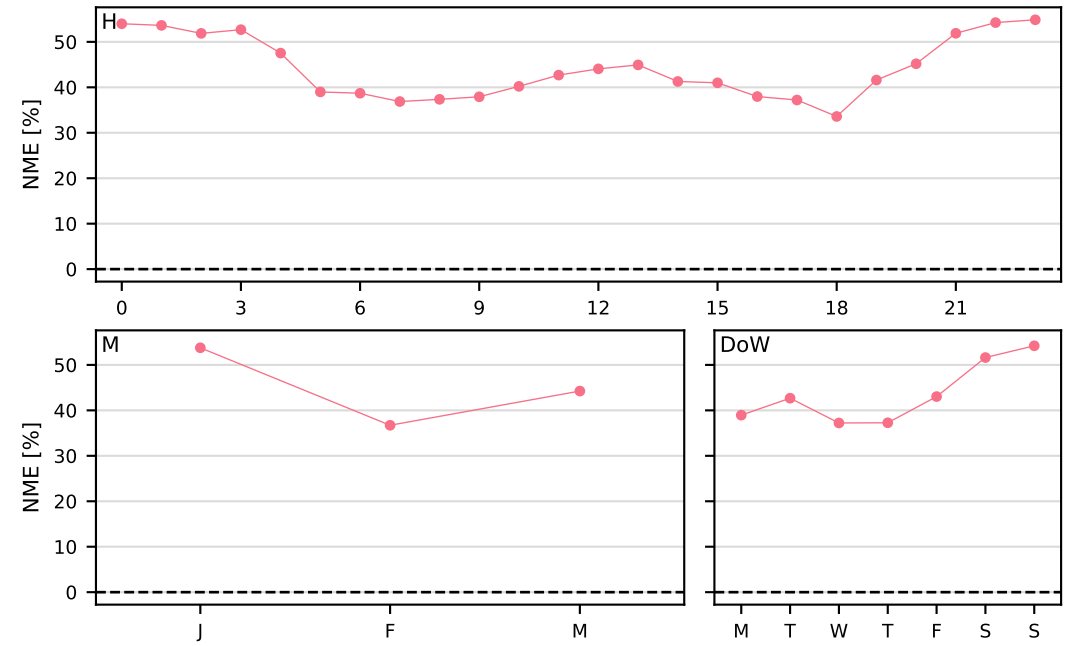
EBAS|sconcn02

MONARCH

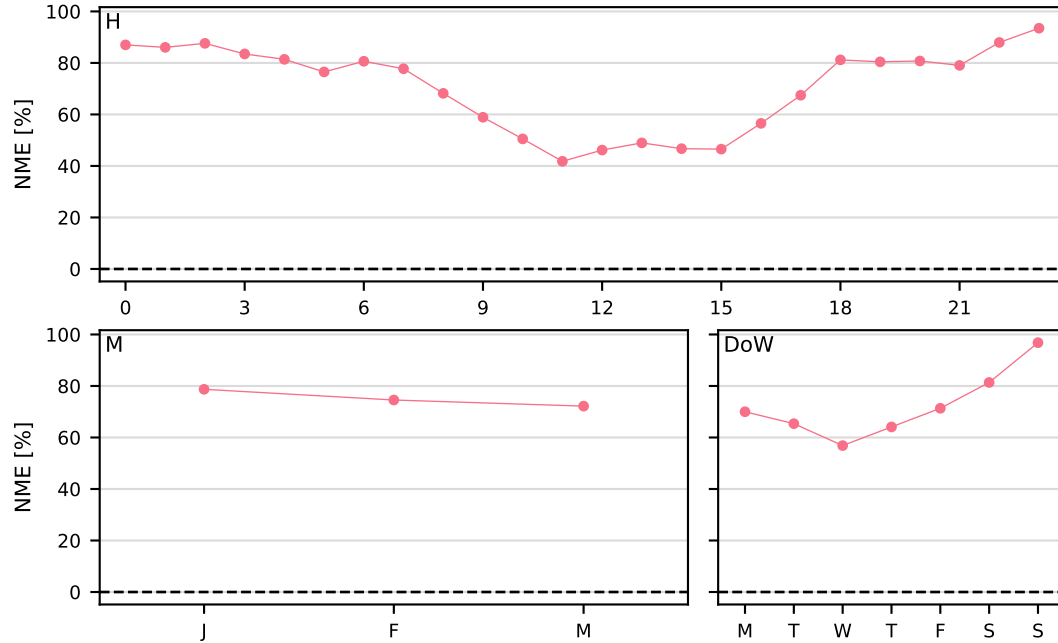
LOCAL·All (62 stations)



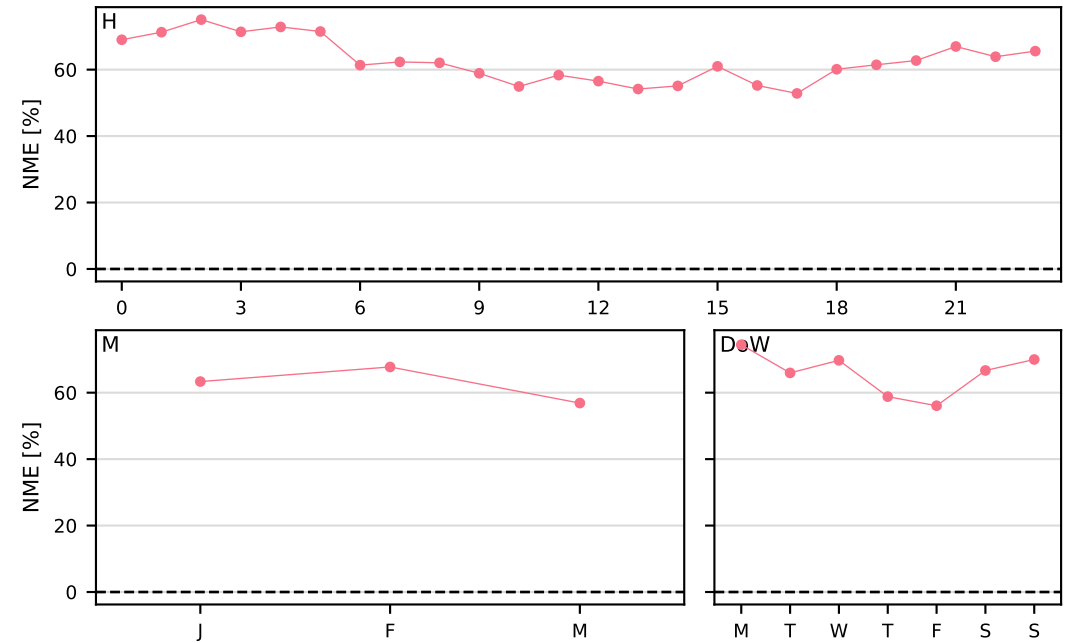
LOCAL·France (1 station)

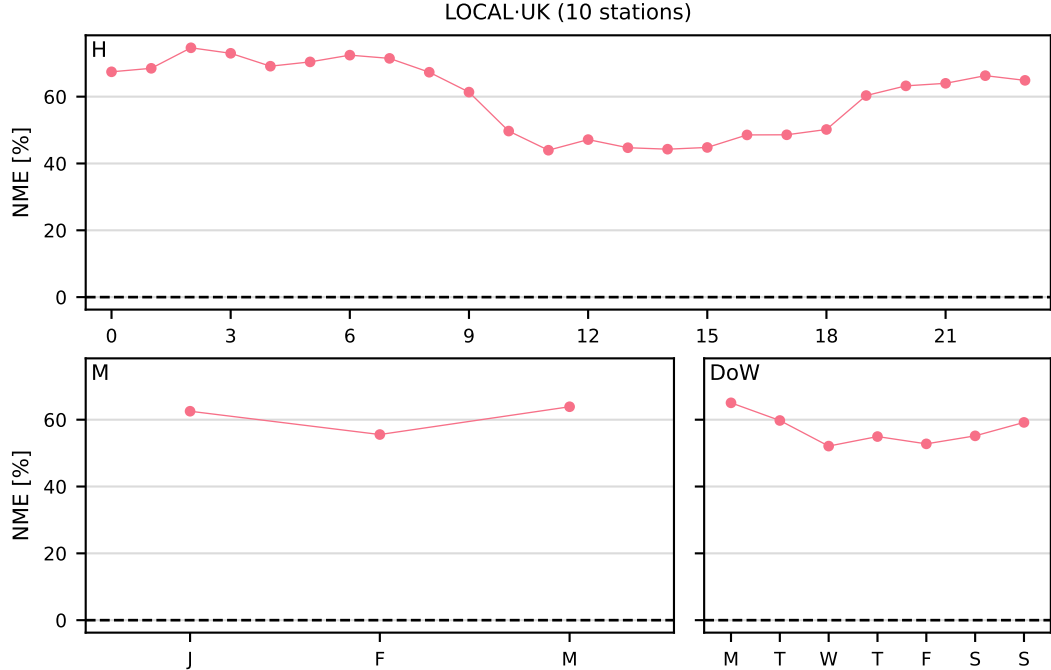


LOCAL·Germany (6 stations)



LOCAL·Spain (13 stations)



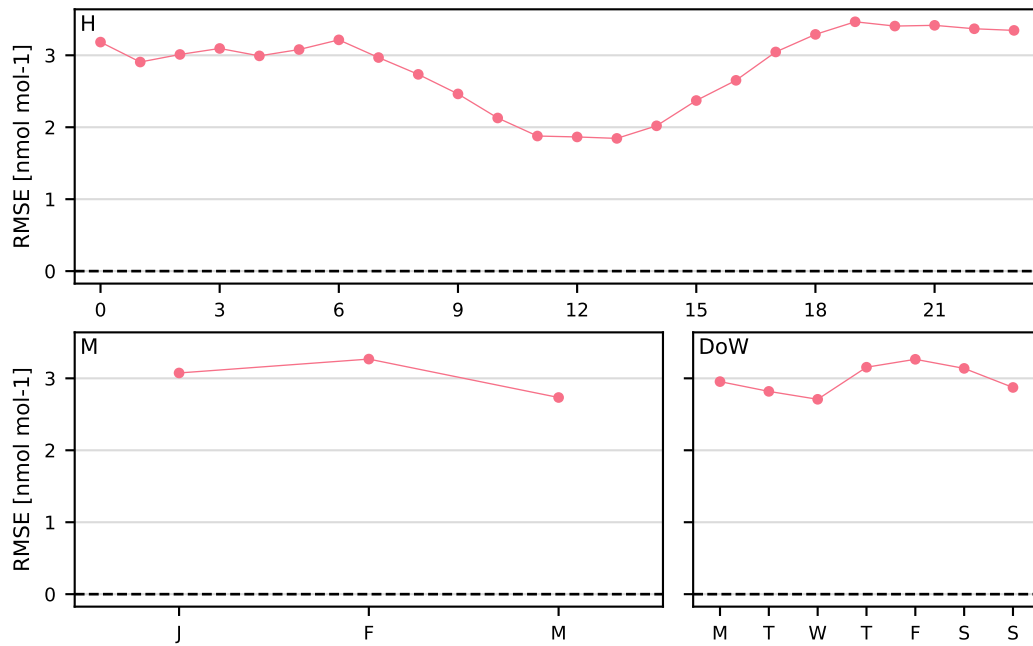


Periodic RMSE (Summary)

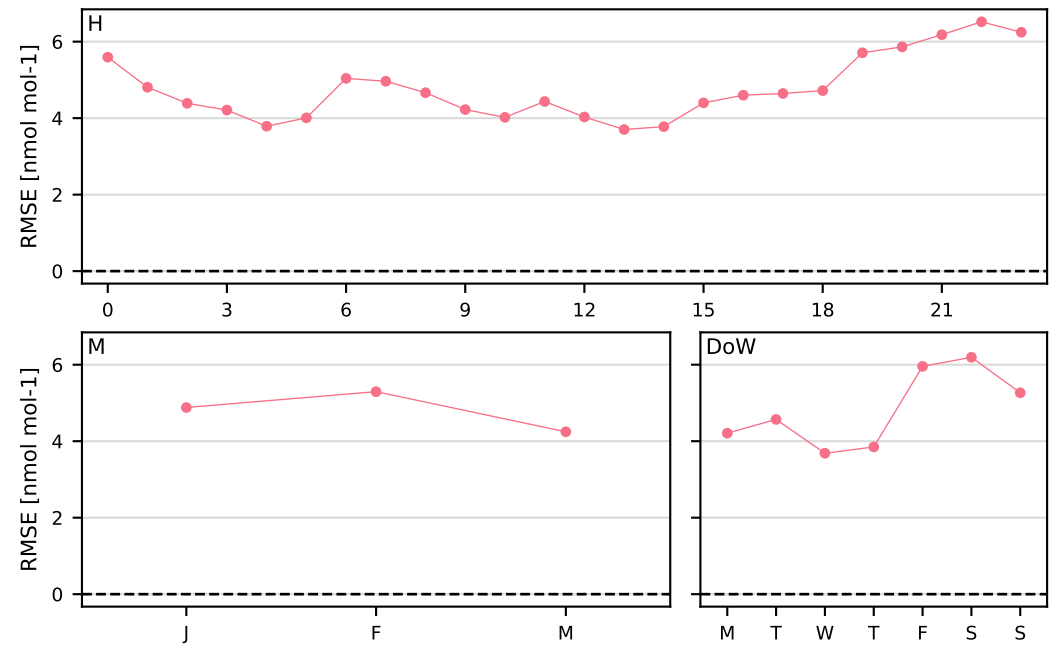
EBAS|sconcn02

MONARCH

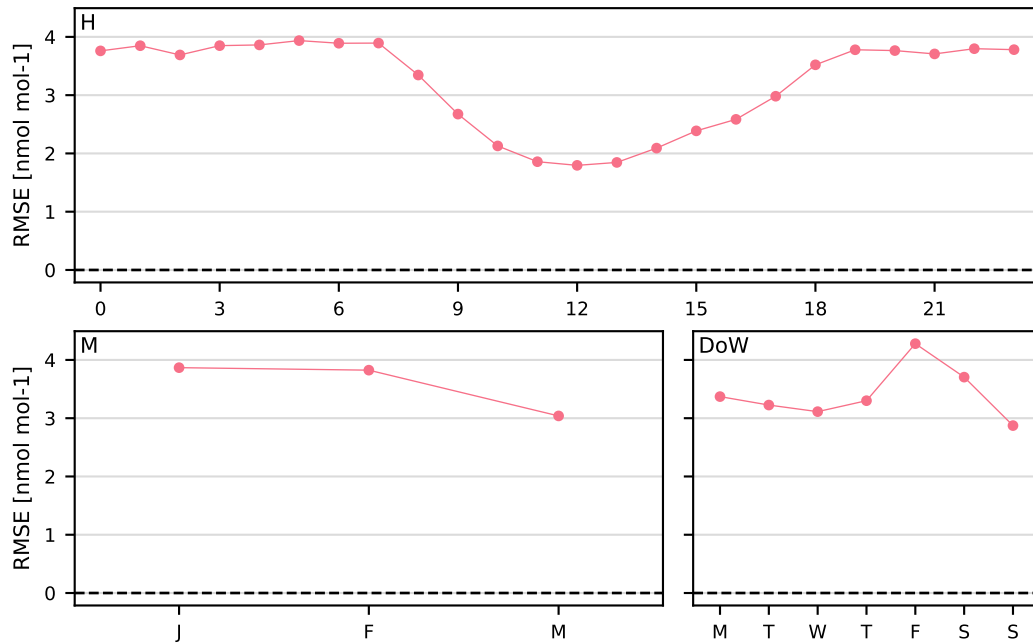
LOCAL·All (62 stations)



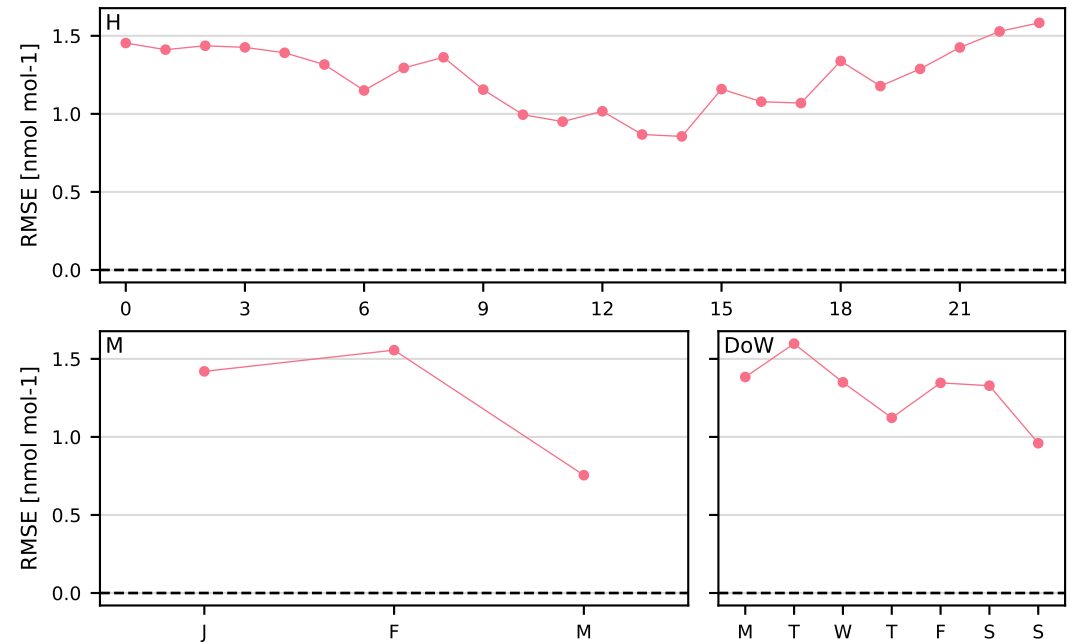
LOCAL·France (1 station)



LOCAL·Germany (6 stations)



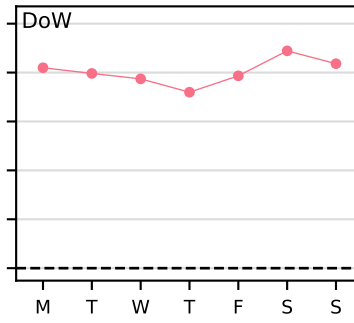
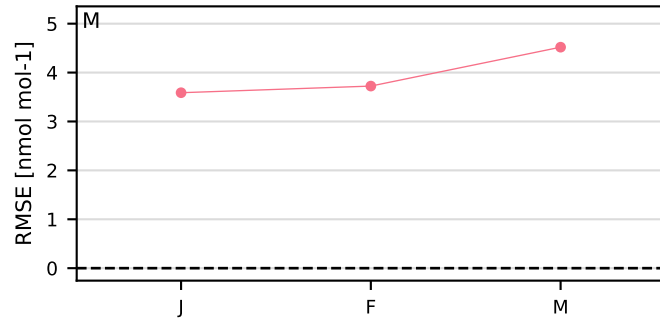
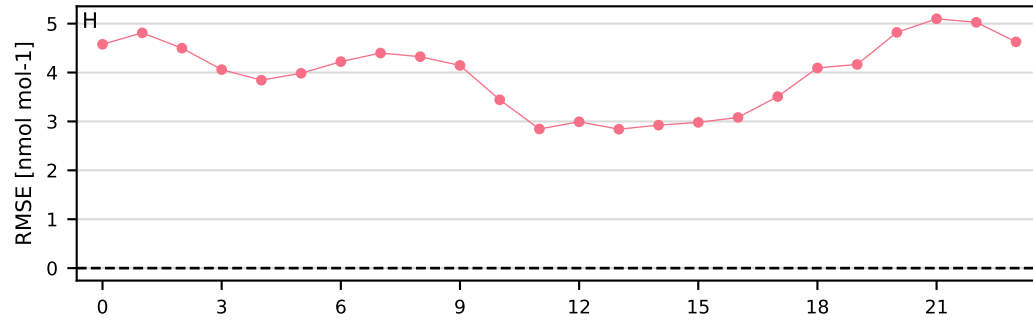
LOCAL·Spain (13 stations)



Periodic RMSE (Summary)
EBAS|sconcn02

● MONARCH

LOCAL-UK (10 stations)

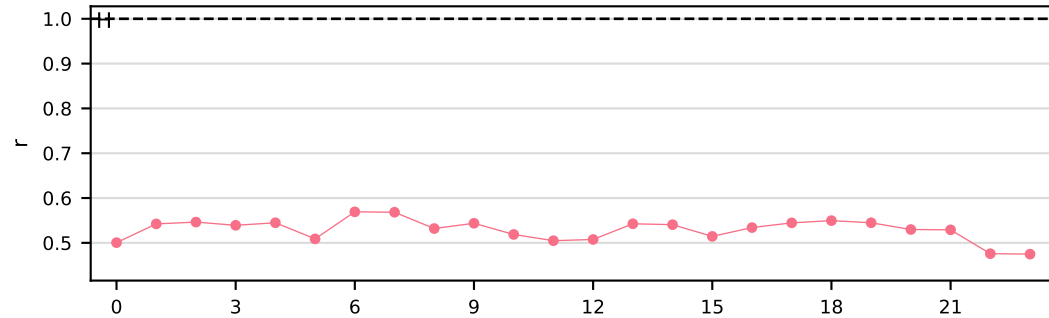


Periodic r (Summary)

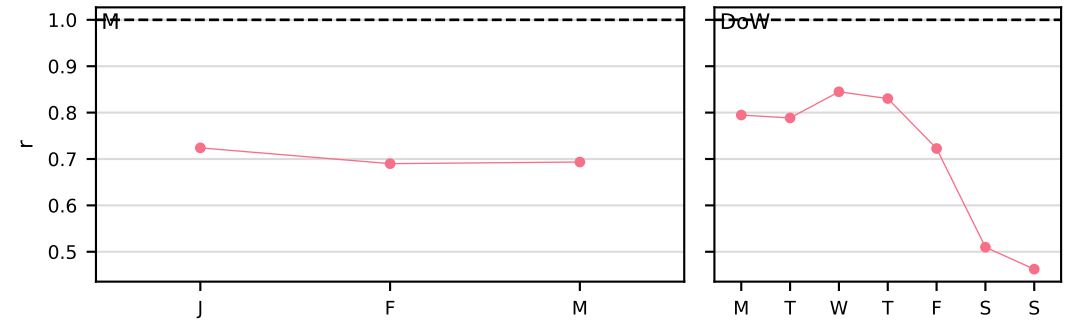
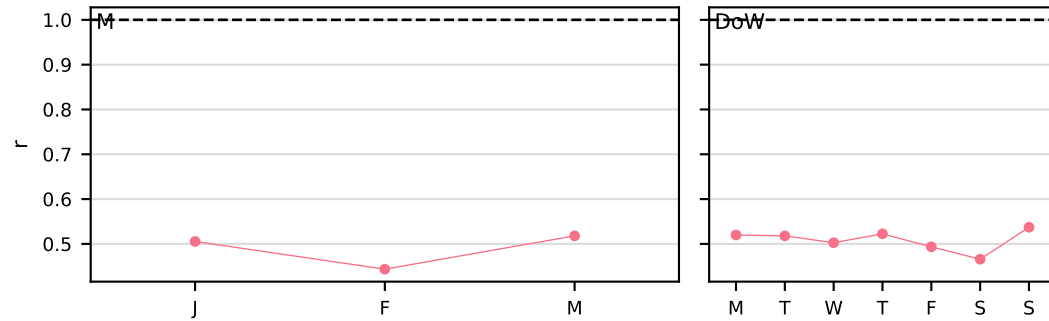
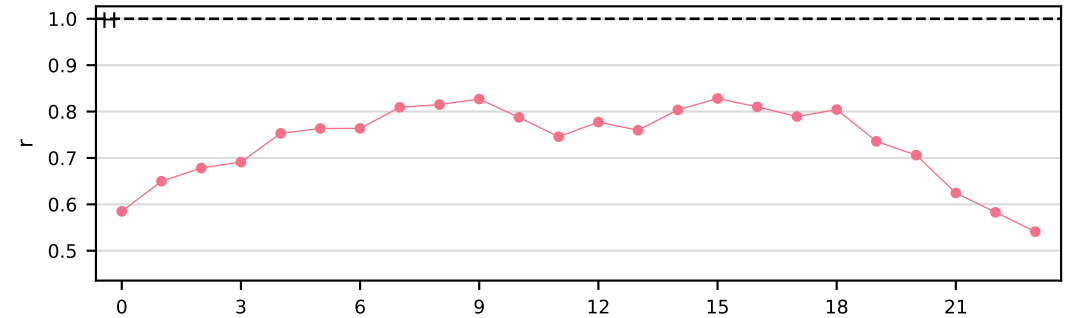
EBAS|sconcn02

MONARCH

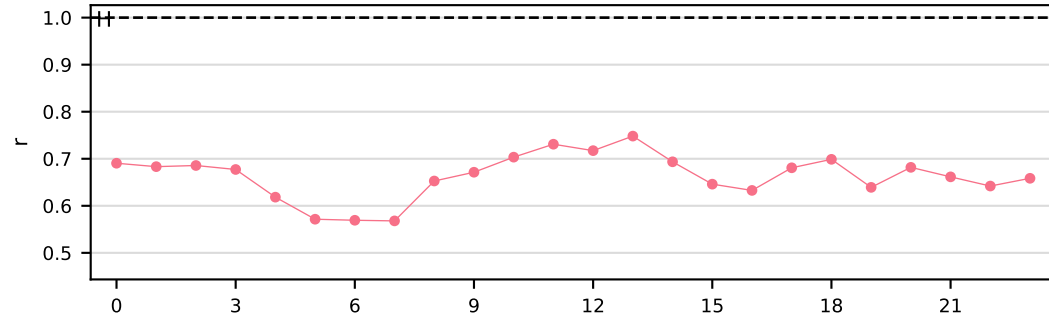
LOCAL-All (62 stations)



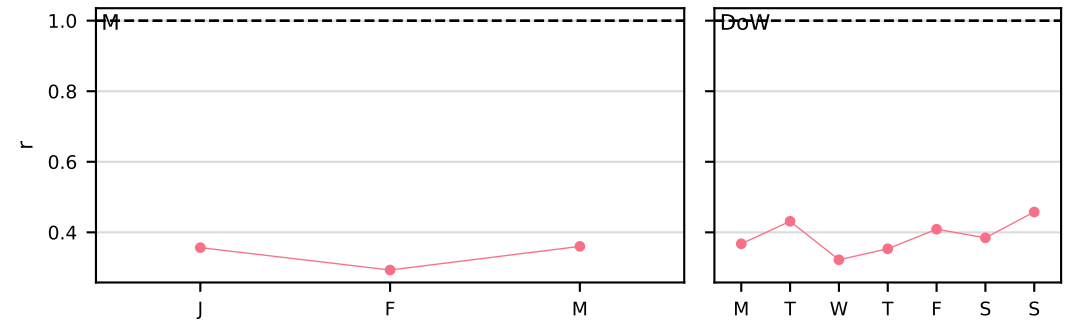
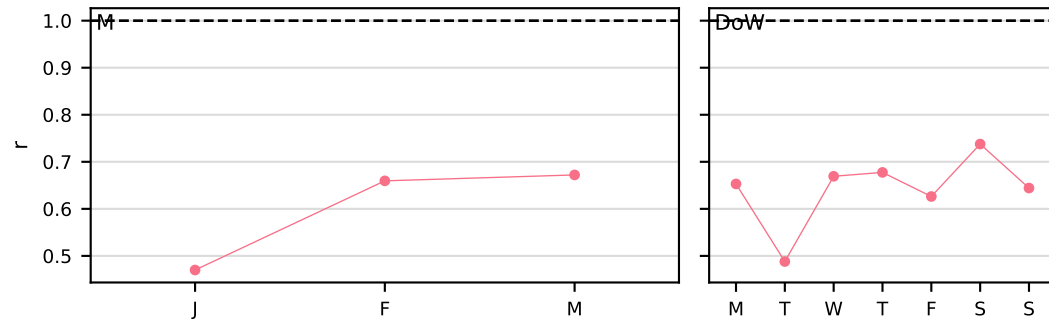
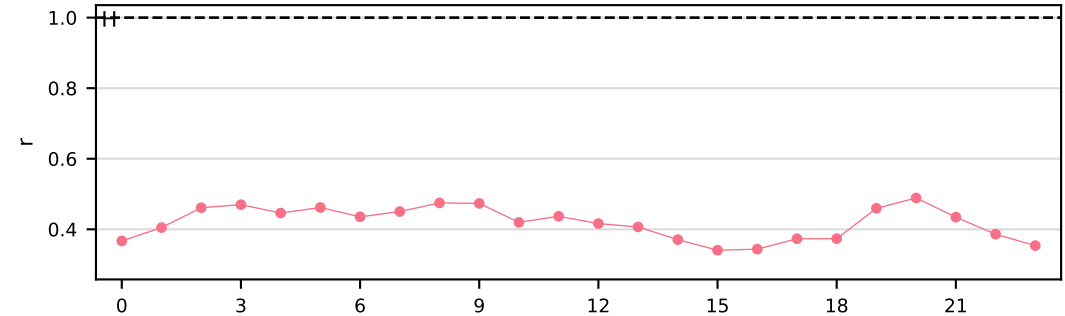
LOCAL-France (1 station)



LOCAL-Germany (6 stations)

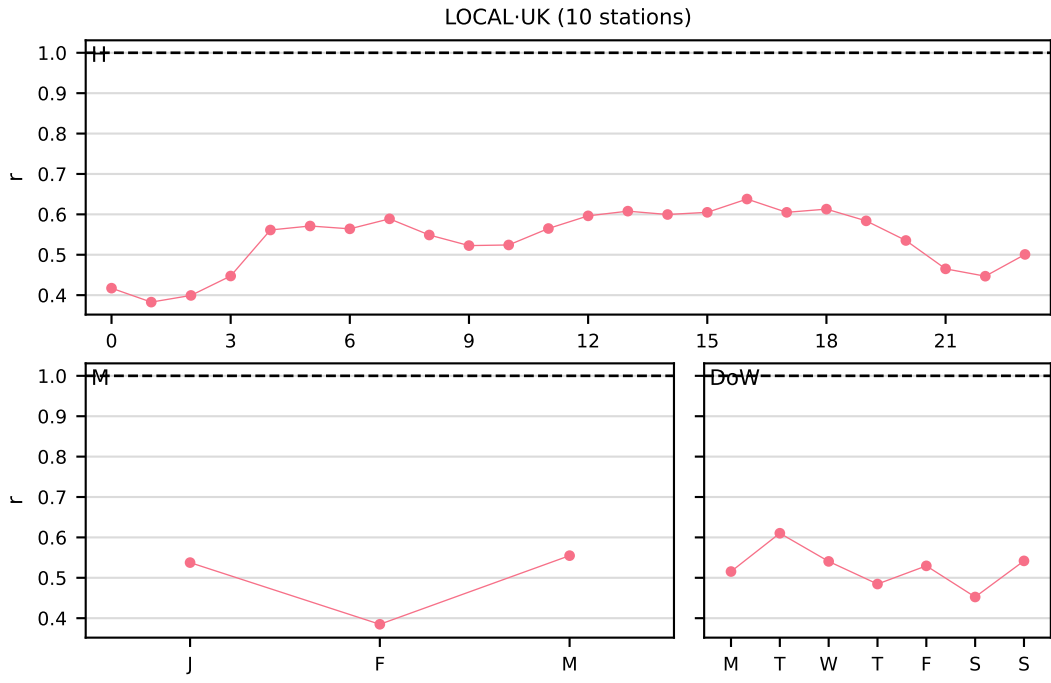


LOCAL-Spain (13 stations)



Periodic r (Summary)
EBAS|sconcn02

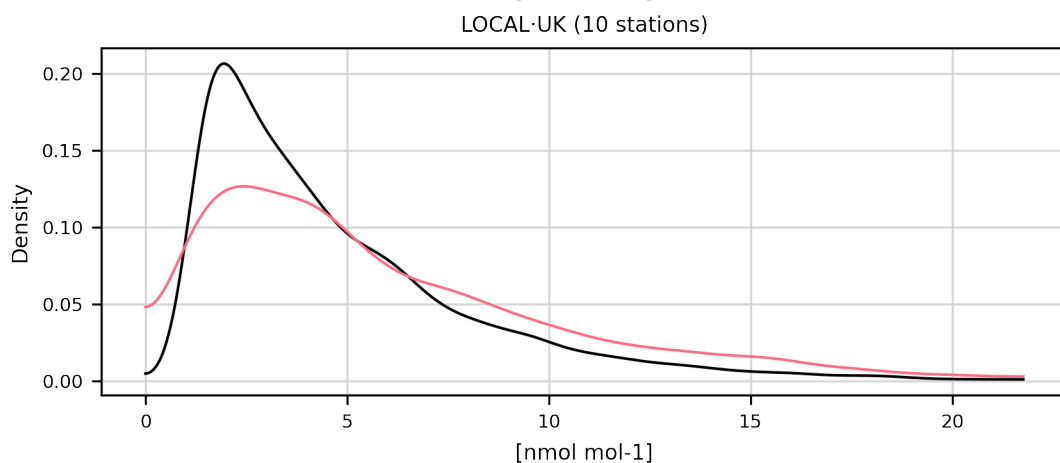
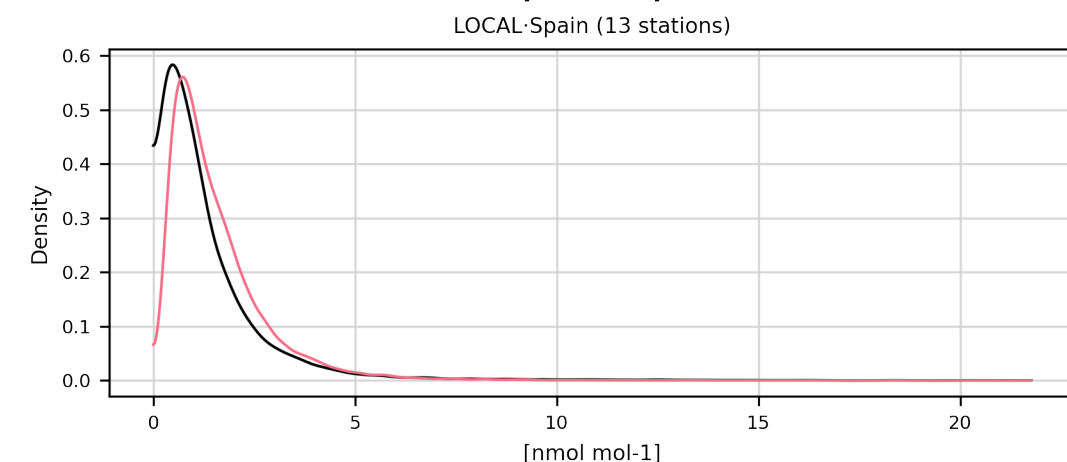
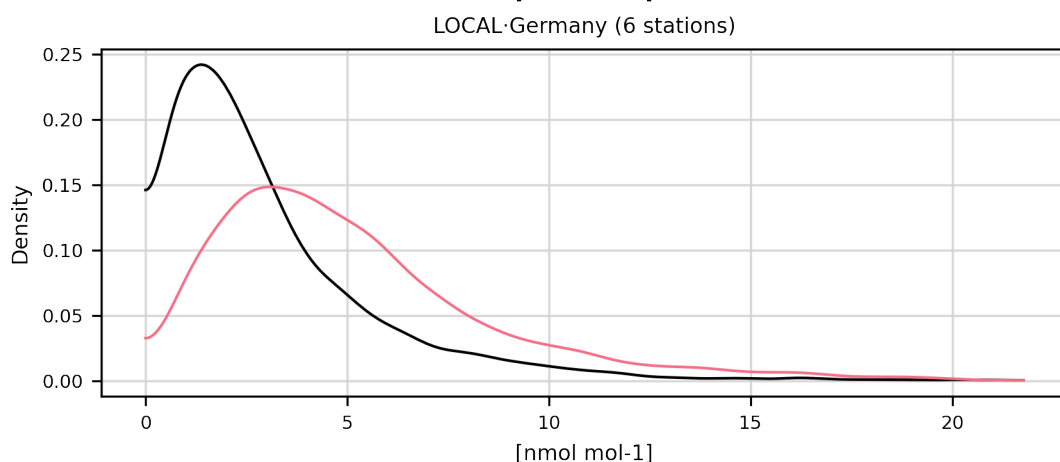
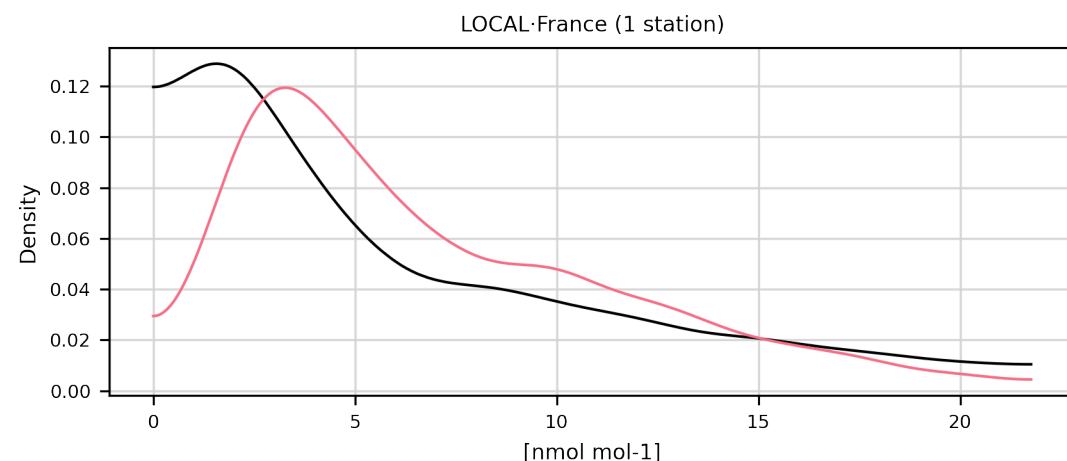
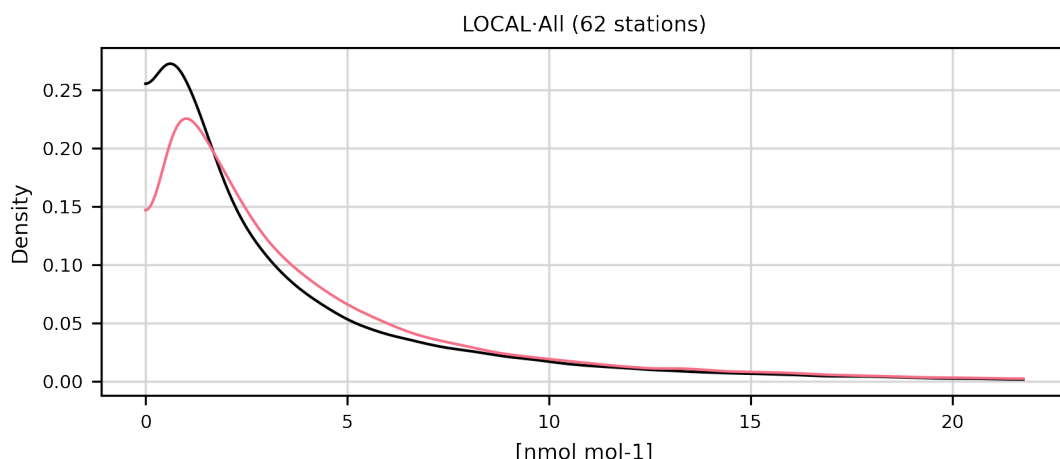
● MONARCH



Distribution (Summary)

EBAS|sconcn02

● observations ● MONARCH

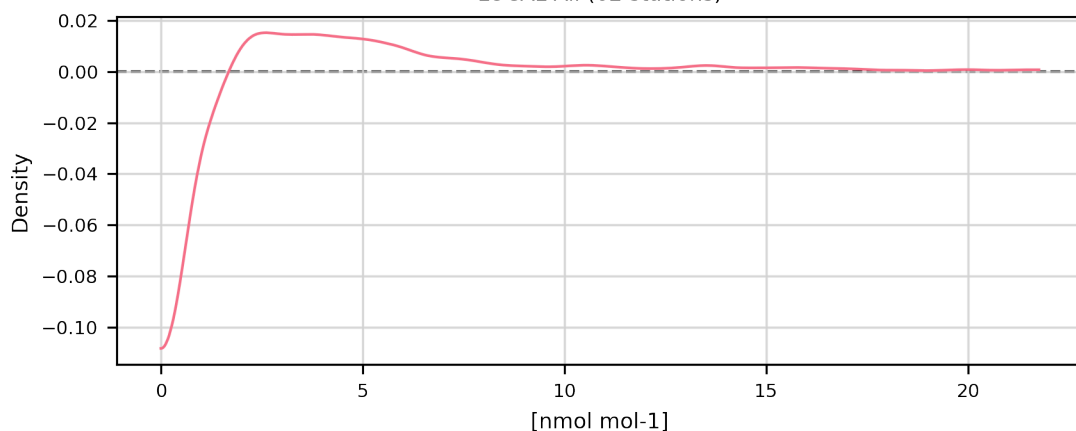


Distribution bias (Summary)

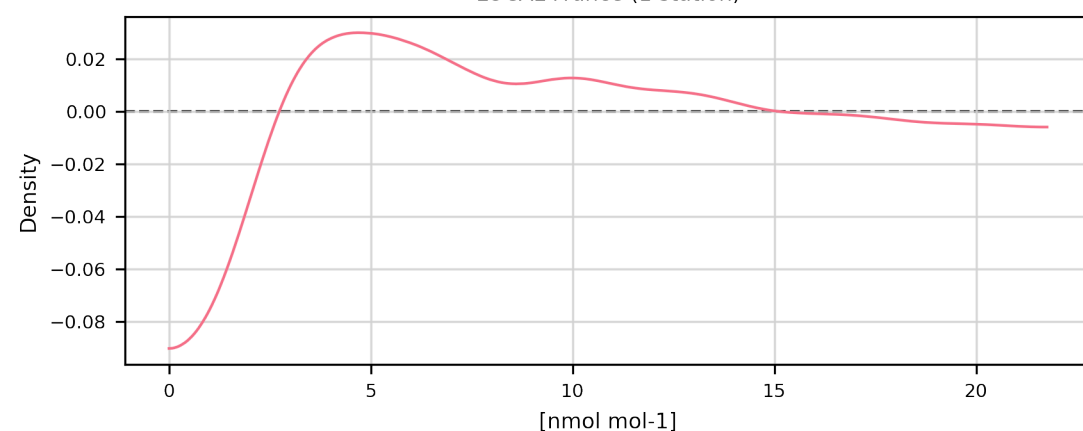
EBAS|sconcn02

● MONARCH

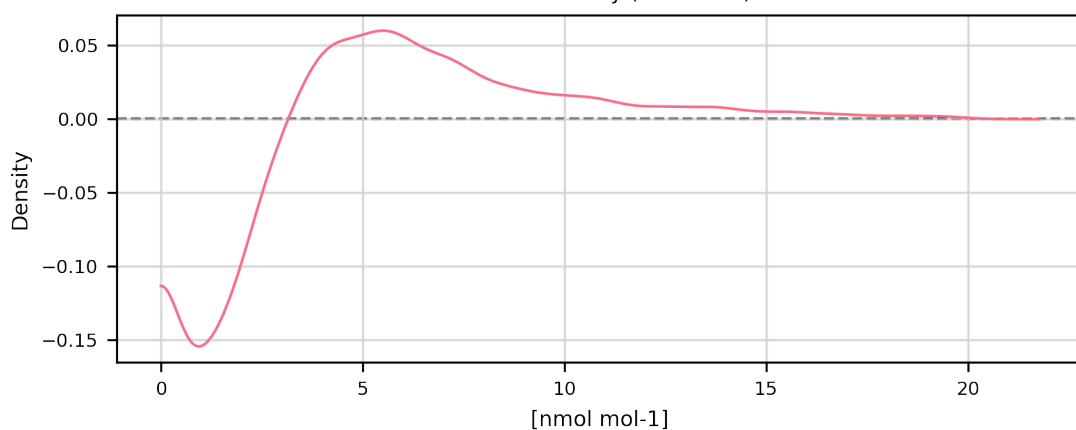
LOCAL·All (62 stations)



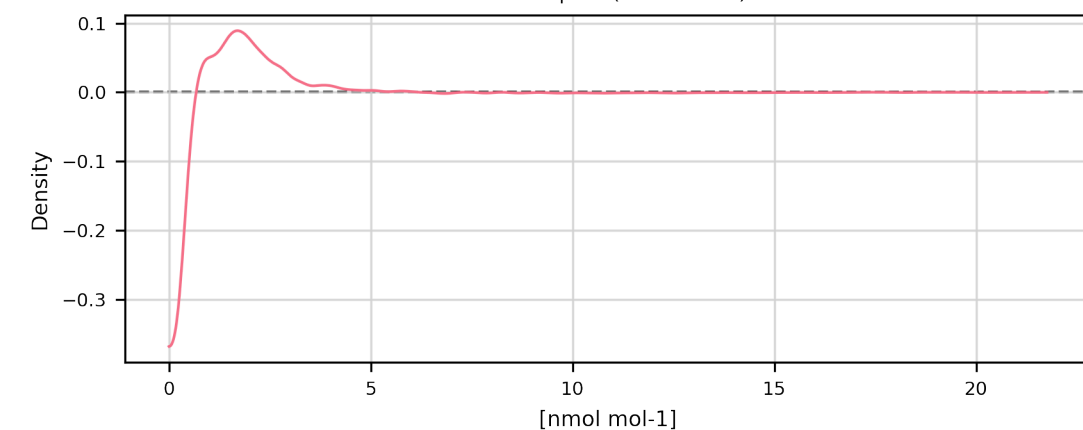
LOCAL·France (1 station)



LOCAL·Germany (6 stations)



LOCAL·Spain (13 stations)



LOCAL·UK (10 stations)

